



Northern University of Bussiness and Technology Khulna

NUBTK $\$(\hat{w})\$$

Sourav Mondal, Ayon Adhikary Arko, Devjyoti Tikader Dipto

Comming Soon

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- 1 Contest
- 2 Mathematics
- 3 Data Structures
- 4 Number Theory
- 5 Combinatorial
- 6 Numerical
- 7 Graphs
- 8 Geometry
- 9 Strings
- 10 Various

Contest (1)

739397.cpp	18 lines
#include <bits/stdc++.h> using namespace std; template<class T> using V = vector<T>; typedef long long ll; typedef pair<int, int> pii; #define f first #define s second #define sz(x) (int)(x).size() #define all(x) begin(x), end(x) #define each(a, x) for(auto& a : x) #define FOR(i, a, b) for(int i = (a); i < (b); ++i) int main() { cin.sync_with_stdio(0); cin.tie(0); cin.exceptions(cin.failbit); } .bashrc	
alias clr="printf '\33c'" co() { g++ -std=c++17 -O2 -Wall -Wextra -Wshadow -Wconversion - ↪o \$1 \$1.cpp; } run() { co \$1 && ./\$1; }	
hash.sh	3 lines
# Hash file ignoring whitespace and comments. Verifies that # code was correctly typed. Usage: 'sh hash.sh < A.cpp' cpp -dD -P -fpreprocessed tr -d '[:space:]' md5sum cut -c-6	
gen.cpp	179 lines
#include "bits/stdc++.h" #include "assert.h" using namespace std;	

1	////////// DISTRIBUTIONS
2	mt19937 rng(chrono::steady_clock::now().time_since_epoch().
3	↪count());
4	// return int in [L,R] inclusive
5	int rng_int(int L, int R) {
6	assert(L <= R);
7	return uniform_int_distribution<int>(L, R)(rng);
8	}
9	int64_t rng_int64(int64_t L, int64_t R) {
10	assert(L <= R);
11	return uniform_int_distribution<int64_t>(L, R)(rng);
12	}
13	// return double in [L,R] inclusive
14	double rng_db(double L, double R) { assert(L <= R);
15	return uniform_real_distribution<double>(L, R)(rng);
16	}
17	// http://cplusplus.com/reference/random/geometric_distribution
18	↪geometric_distribution/
19	// flip a coin which is heads with probability p until you flip
20	↪ heads
21	// mean value of c is 1/p-1
22	int rng_geo(double p) {
23	assert(0 < p && p <= 1); // p large -> closer to 0
24	return geometric_distribution<int>(p)(rng);
25	}
26	////////// VECTORS + PERMS
27	// shuffle a vector
28	template<class T> void randShuffle(vector<T>& v) {
29	shuffle(begin(v), end(v), rng);
30	}
31	// generate random permutation of [0,N-1]
32	vector<int> randPermutation(int N) {
33	vector<int> v(N); iota(begin(v), end(v), 0);
34	randShuffle(v); return v;
35	}
36	// random permutation of [0,N-1] with first element 0
37	vector<int> randPermutationZero(int N) {
38	vector<int> v(N-1); iota(begin(v), end(v), 1);
39	randShuffle(v);
40	v.insert(begin(v), 0); return v;
41	}
42	// shuffle permutation of [0,N-1]
43	vi shufPerm(vi v) {
44	int N = sz(v); vi key = randPerm(N);
45	vi res(N); FOR(i,N) res[key[i]] = key[v[i]];
46	return res;
47	}
48	// vector with all entries in [L,R]
49	vector<int> randVector(int N, int L, int R) {
50	vector<int> v(N);
51	for (int i = 0; i < N; i++) {
52	v[i] = rng_int(L, R);
53	}
54	return v;
55	}
56	// vector with all entries in [L,R], unique
57	vector<int> rngDistinctVector(int N, int L, int R) {
58	set<int> used; vector<int> v;
59	while ((int)size(v) < N) {

60	int x = rng_int(L, R);
61	if (!used.count(x)) {
62	used.insert(x);
63	v.push_back(x);
64	}
65	} return v;
66	}
67	// generate random prime in [L,R]
68	int randPrime(int L, int R) {
69	while(1) {
70	int x = rng_int(L, R);
71	bool bad = 0;
72	for (int i = 2; i*i <= x; i++) if (x%i == 0) bad = 1;
73	if (!bad) return x;
74	}
75	}
76	////////// GRAPHS
77	// relabel edges ed according to perm, shuffle
78	vpi relabelAndShuffle(vpi ed, vi perm) {
79	each(t,ed) {
80	t.f = perm[t.f], t.s = perm[t.s];
81	if (rng()&1) swap(t.f,t.s);
82	}
83	shuf(ed); return ed;
84	}
85	// shuffle graph with vertices [0,N-1]
86	vpi shufGraph(int N, vpi ed) { // randomly swap endpoints,
87	↪rearrange labels
88	return relabelAndShuffle(ed,randPerm(N)); }
89	vpi shufGraphZero(int N, vpi ed) {
90	return relabelAndShuffle(ed,randPermZero(N)); }
91	// shuffle tree given N-1 edges
92	vpi shufTree(vpi ed) { return shufGraph(sz(ed)+1,ed); }
93	// randomly swap endpoints, rearrange labels
94	vpi shufRootedTree(vpi ed) {
95	return relabelAndShuffle(ed,randPermZero(sz(ed)+1)); }
96	void pgraphOne(int N, vpi ed) {
97	ps(N,sz(ed));
98	each(e,ed) ps(1+e.f,1+e.s);
99	}
100	////////// GENERATING TREES
101	// for generating tall tree
102	pi geoEdge(int i, db p) { assert(i > 0);
103	return {i,max(0,i-1-rng_geo(p))}; }
104	// generate edges of tree with verts [0,N-1]
105	// smaller back -> taller tree
106	vpi treeRand(int N, int back) {
107	assert(N >= 1 && back >= 0); vpi ed;
108	FOR(i,1,N) ed.eb(i,i-1-rng_int(0,min(back,i-1)));
109	return ed; }
110	// generate path
111	vpi path(int N) { return treeRand(N,0); }
112	// generate tall tree (large diameter)
113	// the higher the p the taller the tree
114	vpi treeTall(int N, db p) { assert(N >= 1);
115	vpi ed; FOR(i,1,N) ed.pb(geoEdge(i,p));
116	return ed; }

```
// generate tall tree, then add rand at end
vpi treeTallShort(int N, db p) {
    assert(N >= 1); int mid = (N+1)/2;
    vpi ed = treeTall(mid,p);
    FOR(i,mid,N) ed.pb(i, rng_int(0,i-1));
    return ed; }

// lots of stuff connected to either heavy1 or heavy2
vpi treeTallHeavy(int N, db p) {
    assert(N >= 1); // + bunch of rand
    vpi ed; int heavy1 = 0, heavy2 = N/2;
    FOR(i,1,N) {
        if(i < N/4) ed.pb(i,heavy1);
        else if (i > heavy2 && i < 3*N/4) ed.pb(i,heavy2);
        else ed.pb(geoEdge(i,p));
    }
    return ed;
}

// heavy tall tree + random
// lots of verts connected to heavy1 or heavy2
vpi treeTallHeavyShort(int N, db p) {
    assert(N >= 1); // + almost-path + rand
    vpi ed; int heavy1 = 0, heavy2 = N/2;
    FOR(i,1,N) {
        if(i < N/4) ed.pb(i,heavy1);
        else if (i <= heavy2) ed.pb(geoEdge(i,p)); // tall ->
            ↪heavy1
        else if (i > heavy2 && i < 3*N/4) ed.pb(i,heavy2);
        else ed.pb(i, rng_int(0,i-1));
    }
    return ed;
}

int main(int, char* argv[]) {
    int tc; sscanf(argv[1], "%d", &tc); // test case #
    rng.seed(tc);
    // sscanf(argv[2], "%d", &b); // random seed
    // sscanf(argv[3], "%d", &c); // also random seed

}
```

stress.sh10 lines

```
# A and B are executables you want to compare, gen takes int
# as command line arg. Usage: 'sh stress.sh'
for((i = 1; ; ++i)); do
    echo $i
    ./gen $i > in
    ./sol < in > outSol
    ./slow < in > outSlow
    diff -w outSol outSlow || break
    # diff -w < ./A < in > ./B < in || break
done
```

troubleshoot.txt75 lines

```
General:
Write down most of your thoughts, even if you're not sure
whether they're useful.
Give your variables (and files) meaningful names.
Stay organized and don't leave papers all over the place!
You should know what your code is doing ...

Pre-submit:
Write a few simple test cases if sample is not enough.
Are time limits close? If so, generate max cases.
Is the memory usage fine?
Could anything overflow?
```

```
Remove debug output.
Make sure to submit the right file.

Wrong answer:
Print your solution! Print debug output as well.
Read the full problem statement again.
Have you understood the problem correctly?
Are you sure your algorithm works?
Try writing a slow (but correct) solution.
Can your algorithm handle the whole range of input?
Did you consider corner cases (ex. n=1)?
Is your output format correct? (including whitespace)
Are you clearing all data structures between test cases?
Any uninitialized variables?
Any undefined behavior (array out of bounds)?
Any overflows or NaNs (or shifting ll by >=64 bits)?
Confusing N and M, i and j, etc.?
Confusing ++i and i++?
Return vs continue vs break?
Are you sure the STL functions you use work as you think?
Add some assertions, maybe resubmit.
Create some test cases to run your algorithm on.
Go through the algorithm for a simple case.
Go through this list again.
Explain your algorithm to a teammate.
Ask the teammate to look at your code.
Go for a small walk, e.g. to the toilet.
Rewrite your solution from the start or let a teammate do it.

Geometry:
Work with ints if possible.
Correctly account for numbers close to (but not) zero. Related:
for functions like acos make sure absolute val of input is not
(slightly) greater than one.
Correctly deal with vertices that are collinear, concyclic,
coplanar (in 3D), etc.
Subtracting a point from every other (but not itself)?

Runtime error:
Have you tested all corner cases locally?
Any uninitialized variables?
Are you reading or writing outside the range of any vector?
Any assertions that might fail?
Any possible division by 0? (mod 0 for example)
Any possible infinite recursion?
Invalidated pointers or iterators?
Are you using too much memory?
Debug with resubmits (e.g. remapped signals, see Various).

Time limit exceeded:
Do you have any possible infinite loops?
What's your complexity? Large TL does not mean that something
simple (like NlogN) isn't intended.
Are you copying a lot of unnecessary data? (References)
Avoid vector, map. (use arrays/unordered_map)
How big is the input and output? (consider FastIO)
What do your teammates think about your algorithm?
Calling count() on multiset?

Memory limit exceeded:
What is the max amount of memory your algorithm should need?
Are you clearing all data structures between test cases?
If using pointers try BumpAllocator.
```

Mathematics (2)

2.1 Trigonometry

$$\sin(v + w) = \sin v \cos w + \cos v \sin w$$
$$\cos(v + w) = \cos v \cos w - \sin v \sin w$$
$$\tan(v + w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}$$
$$\sin v + \sin w = 2 \sin \frac{v + w}{2} \cos \frac{v - w}{2}$$
$$\cos v + \cos w = 2 \cos \frac{v + w}{2} \cos \frac{v - w}{2}$$
$$a \cos x + b \sin x = r \cos(x - \phi)$$
$$a \sin x + b \cos x = r \sin(x + \phi)$$

where $r = \sqrt{a^2 + b^2}$, $\phi = \operatorname{atan2}(b, a)$.

2.2 Geometry

2.2.1 Triangles

Side lengths: a, b, c
Semiperimeter: $s = \frac{a + b + c}{2}$
Area: $A = \sqrt{s(s - a)(s - b)(s - c)}$
Circumradius: $R = \frac{abc}{4A}$
Inradius: $r = \frac{A}{p}$
Length of median (divides triangle into two equal-area triangles):
 $m_a = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}$
Length of bisector (divides angles in two):

$$s_a = \sqrt{bc \left[1 - \left(\frac{a}{b + c} \right)^2 \right]}$$
$$\text{Law of sines: } \frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c} = \frac{1}{2R}$$
$$\text{Law of cosines: } a^2 = b^2 + c^2 - 2bc \cos \alpha$$
$$\text{Law of tangents: } \frac{a + b}{a - b} = \frac{\tan \frac{\alpha + \beta}{2}}{\tan \frac{\alpha - \beta}{2}}$$

2.3 Derivatives/Integrals

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1 - x^2}} \qquad \frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1 - x^2}}$$
$$\frac{d}{dx} \tan x = 1 + \tan^2 x \qquad \frac{d}{dx} \arctan x = \frac{1}{1 + x^2}$$
$$\int \tan ax = -\frac{\ln |\cos ax|}{a} \qquad \int x \sin ax = \frac{\sin ax - ax \cos ax}{a^2}$$
$$\int e^{-x^2} = \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) \qquad \int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1)$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

2.4 Sums/Series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \quad (-1 < x \leq 1)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, \quad (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, \quad (-\infty < x < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, \quad (-\infty < x < \infty)$$

Data Structures (3)

3.1 STL

MapComparator.h

Description: example of function object (functor) for map or set
Usage: set<int,cmp> s; map<int,int,cmp> m;

struct cmp{bool operator()(int l,int r)const{return l>r;}};

HashMap.h

Description: Hash map with similar API as unordered_map. Initial capacity must be a power of 2 if provided.

Usage: gp_hash_table<int, int, chash> h({},{},{},{},{1<<16});

Memory: ~1.5x unordered map

Time: ~3x faster than unordered map

<ext/pb_ds/assoc_container.hpp>

```
using namespace __gnu_pbds;
struct chash {
    const uint64_t C = 4e18*acos(0)+71;
    const int R = chrono::steady_clock::now().time_since_epoch().count();
    size_t operator()(uint64_t x) const {
        return __builtin_bswap64((x^R)*C);
    }
};
template <class K, class V> using HT = gp_hash_table<K, V, chash>;
template <class K, class V> V get(HT<K, V> &u, K x) {
    auto it = u.find(x); return it == u.end() ? 0 : it->second;
}
```

OrderStatisticTree.h

Description: A set (not multiset!) with support for finding the *n*'th element, and finding the index of an element. Change null_type to get a map.

Time: $\mathcal{O}(\log N)$

<ext/pb_ds/assoc_container.hpp>

```
using namespace __gnu_pbds;
tcT> using Tree = tree<T, null_type, less<T>,
    rb_tree_tag, tree_order_statistics_node_update>;
#define ook order_of_key
#define fbo find_by_order
```

```
void treeExample() {
    Tree<int> t, t2; t.insert(8);
    auto it = t.insert(10).f; assert(it == t.lb(9));
    assert(t.ook(10) == 1 && t.ook(11) == 2 && *t.fbo(0) == 8);
}
```

```
t.join(t2); // assuming T < T2 or T > T2, merge t2 into t
}
```

LineContainer.h

Description: Add lines of the form $ax + b$, query maximum y -coordinate for any x .

Time: $\mathcal{O}(\log N)$

df7386, 29 lines

```
using T = ll; const T INF = LLONG_MAX; // a/b rounded down
// ll fdiv(ll a, ll b) { return a/b-((a^b)<0&&a%b); }
```

```
bool _Q = 0;
struct Line {
    T a, b; mutable T lst;
    T eval(T x) const { return a*x+b; }
    bool operator<(const Line&o)const{return _Q?lst<o.lst:a<o.a;}
    T last_gre(const Line&o) const { assert(a <= o.a);
        // greatest x s.t. a*x+b >= o.a*x+o.b
        return lst=(a==o.a?(b>=o.b?INF:-INF):fdiv(b-o.b,o.a-a)); }
};
```

```
struct LineContainer: multiset<Line> {
    bool isect(iterator it) { auto n_it = next(it);
        if (n_it == end()) return it->lst = INF, 0;
        return it->last_gre(*n_it) >= n_it->lst; }
    void add(T a, T b) {
        auto it = ins({a,b,0}); while (isect(it)) erase(next(it));
        if (it == begin()) return;
        if (isect(--it)) erase(next(it)), isect(it);
        while (it != begin()) {
            --it; if (it->lst < next(it)->lst) break;
            erase(next(it)); isect(it); }
    }
    T qmax(T x) { assert(!empty());
        _Q = 1; T res = lb({0,0,x})->eval(x); _Q = 0;
        return res; }
};
```

3.2 1D Range Queries

RMQ.h

Description: 1D range minimum query. If TL is an issue, use arrays instead of vectors and store values instead of indices.

Memory: $\mathcal{O}(N \log N)$

Time: $\mathcal{O}(1)$

a3f881, 19 lines

```
tcT> struct RMQ { // floor(log_2(x))
    int level(int x) { return 31-__builtin_clz(x); }
    V<T> v; V<vi> jmp;
    int cmb(int a, int b) {
        return v[a]==v[b]?min(a,b):(v[a]<v[b]?a:b); }
    void init(const V<T>& _v) {
        v = _v; jmp = {vi(sz(v))};
        iota(all(jmp[0]),0);
        for (int j = 1; 1<=j <= sz(v); ++j) {
            jmp.pb(vi(sz(v)-(1<=j)+1));
            F0R(i,sz(jmp[j])) jmp[j][i] = cmb(jmp[j-1][i],
                jmp[j-1][i+(1<=(j-1))]);
        }
    }
    int index(int l, int r) {
        assert(l <= r); int d = level(r-l+1);
        return cmb(jmp[d][l],jmp[d][r-(1<=d)+1]); }
    T query(int l, int r) { return v[index(l,r)]; }
};
```

BIT.h

Description: Computes partial sums $a[0] + a[1] + \dots + a[\text{pos} - 1]$, and updates single elements $a[i]$, taking the difference between the old and new value. 0-Indexed.

Time: Both operations are $\mathcal{O}(\log N)$.

29f859, 26 lines

```
template<class T> struct BIT {
    int N; vector<T> bit;
    BIT(int _N) {
        N = _N;
        bit.resize(N);
    }
    void add(int p, T x) {
        for (++p; p <= N; p += p&-p) bit[p - 1] += x;
    }
    T sum(int l, int r) { return sum(r+1)-sum(l); }
    T sum(int r) {
        T res = 0;
        for (; r; r -= r&-r) res += bit[r - 1];
        return res;
    }
    int lower_bound(T sum) {
        if (sum <= 0) return -1;
        int pos = 0;
        for (int pw = 1 << 25; pw; pw >= 1) {
            int npos = pos + pw;
            if (npos <= N && bit[npos - 1] < sum)
                pos = npos, sum -= bit[pos - 1];
        }
        return pos;
    }
};
```

SegmentTree.h

Description: 1D point update and range query where cmb is any associative operation. seg[1]==query(0,N-1).

Time: $\mathcal{O}(\log N)$

a09d9c, 36 lines

```
template<class T> struct SegmentTree {
    const T ID{};
    T cmb(T a, T b) { return a + b; }
    int N = 1; vector<T> seg;
    SegmentTree(int _N) {
        while (N < _N) N *= 2;
        seg.assign(2*N, ID);
    }
    void pull(int p) { seg[p] = cmb(seg[2*p],seg[2*p+1]); }
    void update(int p, T val) { // set val at position p
        seg[p += N] = val;
        for (p /= 2; p > 0; p /= 2) pull(p);
    }
    T query(int l, int r) { // zero-indexed, inclusive
        T lf = ID, rf = ID;
        for (l += N, r += N + 1; l < r; l /= 2, r /= 2) {
            if (l & 1) lf = cmb(lf, seg[l++]);
            if (r & 1) rf = cmb(seg[--r], rf);
        }
        return cmb(lf, rf);
    }
    // int first_at_least(int lo, int val, int ind, int l, int r)
    //     ↪ { // if seg stores max across range
    //         if (r < lo || val > seg[ind]) return -1;
    //         if (l == r) return l;
    //         int m = (l+r)/2;
    //         int res = first_at_least(lo,val,2*ind,l,m); if (res !=
    //             ↪-1) return res;
    //         return first_at_least(lo,val,2*ind+1,m+1,r);
    //     }
    //     int k_th_one(int k, int ind, int l, int r) {
    //         if (l == r) return l;
    //         int m = (l + r) / 2;
    //         int lc = 2*ind, rc = 2*ind+1;
    //         if (seg[lc] >= k) return k_th_one(k, lc, l, m);
    //         else return k_th_one(k-seg[lc], rc, m+1, r);
    //     }
};
```

```
// }
};

LazySegmentTree.h
Description: 1D range increment and sum query.
Time:  $\mathcal{O}(\log N)$ 
```

```
78a06d, 26 lines

tcT, int SZ> struct LazySeg {
    static_assert(pct(SZ) == 1); // SZ must be power of 2
    const T ID{}; T cmb(T a, T b) { return a+b; }
    T seg[2*SZ], lazy[2*SZ];
    LazySeg() { F0R(i,2*SZ) seg[i] = lazy[i] = ID; }
    void push(int ind, int L, int R) {
        seg[ind] += (R-L+1)*lazy[ind]; // dependent on operation
        if (L != R) F0R(i,2) lazy[2*ind+i] += lazy[ind];
        lazy[ind] = 0;
    } // recalc values for current node
    void pull(int ind){seg[ind]=cmb(seg[2*ind],seg[2*ind+1]);}
    void build() { ROF(i,1,SZ) pull(i); }
    void upd(int lo,int hi,T inc,int ind=1,int L=0, int R=SZ-1) {
        push(ind,L,R); if (hi < L || R < lo) return;
        if (lo <= L && R <= hi) {
            lazy[ind] = inc; push(ind,L,R); return; }
        int M = (L+R)/2; upd(lo,hi,inc,2*ind,L,M);
        upd(lo,hi,inc,2*ind+1,M+1,R); pull(ind);
    }
    T query(int lo, int hi, int ind=1, int L=0, int R=SZ-1) {
        push(ind,L,R); if (lo > R || L > hi) return ID;
        if (lo <= L && R <= hi) return seg[ind];
        int M = (L+R)/2; return cmb(query(lo,hi,2*ind,L,M),
            query(lo,hi,2*ind+1,M+1,R));
    }
};
```

PSeg.h

Description: Persistent min segtree with lazy updates, no propagation. If making d a vector then save the results of upd and build in local variables first to avoid issues when vector resizes in C++14 or lower.

Memory: $\mathcal{O}(N + Q \log N)$

```
8f37fa, 46 lines

tcT, int SZ> struct pseg {
    static const int LIM = 2e7;
    struct node {
        int l, r; T val = 0, lazy = 0;
        void inc(T x) { lazy += x; }
        T get() { return val+lazy; }
    };
    node d[LIM]; int nex = 0;
    int copy(int c) { d[nex] = d[c]; return nex++; }
    T cmb(T a, T b) { return min(a,b); }
    void pull(int c) { d[c].val =
        cmb(d[d[c].l].get(), d[d[c].r].get()); }
    //// MAIN FUNCTIONS
    T query(int c, int lo, int hi, int L, int R) {
        if (lo <= L && R <= hi) return d[c].get();
        if (R < lo || hi < L) return MOD;
        int M = (L+R)/2;
        return d[c].lazy+cmb(query(d[c].l,lo,hi,L,M),
            query(d[c].r,lo,hi,M+1,R));
    }
    int upd(int c, int lo, int hi, T v, int L, int R) {
        if (R < lo || hi < L) return c;
        int x = copy(c);
        if (lo <= L && R <= hi) { d[x].inc(v); return x; }
        int M = (L+R)/2;
        d[x].l = upd(d[x].l,lo,hi,v,L,M);
        d[x].r = upd(d[x].r,lo,hi,v,M+1,R);
        pull(x); return x;
    }
};
```

```
int build(const V<T&& arr, int L, int R) {
    int c = nex++;
    if (L == R) {
        if (L < sz(arr)) d[c].val = arr[L];
        return c;
    }
    int M = (L+R)/2;
    d[c].l = build(arr,L,M), d[c].r = build(arr,M+1,R);
    pull(c); return c;
}

vi loc; //// PUBLIC
void upd(int lo, int hi, T v) {
    loc.pb(upd(loc.bk,lo,hi,v,0,SZ-1)); }
T query(int ti, int lo, int hi) {
    return query(loc[ti],lo,hi,0,SZ-1); }
void build(const V<T&&arr) {loc.pb(build(arr,0,SZ-1));}
};
```

Treap.h

Description: Easy BBST. Use split and merge to implement insert and delete.

Time: $\mathcal{O}(\log N)$

```
bdb758, 65 lines

using pt = struct tnode*;
struct tnode {
    int pri, val; pt c[2]; // essential
    int sz; ll sum; // for range queries
    bool flip = 0; // lazy update
    tnode(int _val) {
        pri = rng(); sum = val = _val;
        sz = 1; c[0] = c[1] = nullptr;
    }
    ~tnode() { F0R(i,2) delete c[i]; }
};
int getsz(pt x) { return x?x->sz:0; }
ll getsum(pt x) { return x?x->sum:0; }
pt prop(pt x) { // lazy propagation
    if (!x || !x->flip) return x;
    swap(x->c[0],x->c[1]);
    x->flip = 0; F0R(i,2) if (x->c[i]) x->c[i]->flip ^= 1;
    return x;
}
pt calc(pt x) {
    pt a = x->c[0], b = x->c[1];
    assert(!x->flip); prop(a), prop(b);
    x->sz = 1+getsz(a)+getsz(b);
    x->sum = x->val+getsum(a)+getsum(b);
    return x;
}
void tour(pt x, vi& v) { // print values of nodes,
    if (!x) return; // inorder traversal
    prop(x); tour(x->c[0],v); v.pb(x->val); tour(x->c[1],v);
}
pair<pt,pt> split(pt t, int v) { // >= v goes to the right
    if (!t) return {t,t};
    prop(t);
    if (t->val >= v) {
        auto p = split(t->c[0], v); t->c[0] = p.s;
        return {p.f,calc(t)};
    } else {
        auto p = split(t->c[1], v); t->c[1] = p.f;
        return {calc(t),p.s};
    }
}
pair<pt,pt> splitsz(pt t, int sz) { // sz nodes go to left
    if (!t) return {t,t};
    prop(t);
    if (getsz(t->c[0]) >= sz) {
        auto p = splitsz(t->c[0],sz); t->c[0] = p.s;
```

```
        return {p.f,calc(t)};
    } else {
        auto p=splitsz(t->c[1],sz-getsz(t->c[0])-1); t->c[1]=p.f;
        return {calc(t),p.s};
    }
}
pt merge(pt l, pt r) { // keys in l < keys in r
    if (!l || !r) return l?:r;
    prop(l), prop(r); pt t;
    if (l->pri > r->pri) l->c[1] = merge(l->c[1],r), t = l;
    else r->c[0] = merge(l,r->c[0]), t = r;
    return calc(t);
}
pt ins(pt x, int v) { // insert v
    auto a = split(x,v), b = split(a.s,v+1);
    return merge(a.f,merge(new tnode(v),b.s)); }
pt del(pt x, int v) { // delete v
    auto a = split(x,v), b = split(a.s,v+1);
    return merge(a.f,b.s); }
```

Number Theory (4)

4.1 Modular Arithmetic

ModIntShort.h

Description: Modular arithmetic. Assumes MOD is prime.

Usage: `mi a = MOD+5; inv(a);` // 400000003

```
2672f9, 24 lines

template<int MOD, int RT> struct mint {
    static const int mod = MOD;
    static constexpr mint rt() { return RT; } // primitive root
    int v;
    explicit operator int() const { return v; }
    mint():v(0) {}
    mint(ll _v):v(int(_v%MOD)) { v += (v<0)*MOD; }
    mint& operator+=(mint o) {
        if ((v += o.v) >= MOD) v -= MOD;
        return *this; }
    mint& operator-=(mint o) {
        if ((v -= o.v) < 0) v += MOD;
        return *this; }
    mint& operator*=(mint o) {
        v = int((ll)v*o.v%MOD); return *this; }
    friend mint pow(mint a, ll p) { assert(p >= 0);
        return p==0?1:pow(a*a,p/2)*(p&1?a:1); }
    friend mint inv(mint a) { assert(a.v != 0); return pow(a,MOD
        ↪-2); }
    friend mint operator+(mint a, mint b) { return a += b; }
    friend mint operator-(mint a, mint b) { return a -= b; }
    friend mint operator*(mint a, mint b) { return a *= b; }
};
using mi = mint<(int)1e9+7, 5>;
using vmi = V<mi>;
```

ModFact.h

Description: Combinations modulo a prime MOD . Assumes $2 \leq N \leq MOD$.

Usage: `F.init(10); F.C(6, 4);` // 15

Time: $\mathcal{O}(N)$

```
364271, 13 lines

"ModInt.h"

struct {
    vmi invs, fac, ifac;
    void init(int N) { // idempotent
        invs.rsz(N), fac.rsz(N), ifac.rsz(N);
        invs[1] = fac[0] = ifac[0] = 1;
        FOR(i,2,N) invs[i] = mi(-(ll)MOD/i*(int)invs[MOD%i]);
        FOR(i,1,N) fac[i] = fac[i-1]*i, ifac[i] = ifac[i-1]*invs[i
            ↪);
    }
};
```

```
    }
    mi C(int a, int b) {
        if (a < b || b < 0) return 0;
        return fac[a]*ifac[b]*ifac[a-b];
    }
} F;
```

ModMulLL.h
Description: Multiply two 64-bit integers mod another if 128-bit is not available. modMul is equivalent to (ul) ((__int128(a)*b%mod)). Works for $0 \leq a, b < mod < 2^{63}$. NOTE : Only works if system supports 80-bit floating point.

```

d22eee, 12 lines
using ul = uint64_t;
// ul modMul(ul a, ul b, const ul mod) {
//     ll ret = a*b-mod*(ul)((db)a*b/mod);
//     return ret+((ret<0)-(ret>=(ll)mod))*mod; }
ul modMul(ul a, ul b, const ul mod){
    return __int128(a) * __int128(b) % mod;
}
ul modPow(ul a, ul b, const ul mod) {
    if (b == 0) return 1;
    ul res = modPow(a,b/2,mod); res = modMul(res,res,mod);
    return b&1 ? modMul(res,a,mod) : res;
}
```

FastMod.h
Description: Barrett reduction computes $a\%b$ about 4 times faster than usual where $b > 1$ is constant but not known at compile time. Division by b is replaced by multiplication by m and shifting right 64 bits.

```

aa19c9, 7 lines
using ul = uint64_t; using L = __uint128_t;
struct FastMod {
    ul b, m; FastMod(ul b) : b(b), m(-1ULL / b) {}
    ul reduce(ul a) {
        ul q = (ul)((__uint128_t(m) * a) >> 64), r = a - q * b;
        return r - (r >= b) * b; }
};
```

ModSqrt.h
Description: Tonelli-Shanks algorithm for square roots mod a prime. -1 if doesn't exist.
Usage: sqrt(mi((ll)1e10)); // 100000
Time: $\mathcal{O}(\log^2(MOD))$

```

bcfa63, 14 lines
"ModInt.h"
using T = int;
T sqrt(mi a) {
    mi p = pow(a, (MOD-1)/2);
    if (p.v != 1) return p.v == 0 ? 0 : -1;
    T s = MOD-1; int r = 0; while (s%2 == 0) s /= 2, ++r;
    mi n = 2; while (pow(n, (MOD-1)/2).v == 1) n = T(n)+1;
    // n non-square, ord(g)=2^r, ord(b)=2^m, ord(g)=2^r, m<r
    for (mi x = pow(a, (s+1)/2), b = pow(a,s), g = pow(n,s);) {
        if (b.v == 1) return min(x.v,MOD-x.v); // x^2=ab
        int m = 0; for (mi t = b; t.v != 1; t *= t) ++m;
        rep(r-m-1) g *= g; // ord(g)=2^(m+1)
        x *= g, g *= g, b *= g, r = m; // ord(g)=2^m, ord(b)<2^m
    }
}
```

ModSum.h
Description: Counts # of lattice points (x,y) in the triangle $1 \leq x, 1 \leq y, ax+by \leq s \pmod{2^{64}}$ and related quantities.
Time: $\mathcal{O}(\log ab)$

```

23cbf6, 20 lines
using ul = uint64_t;
ul sum2(ul n) { return n/2*((n-1)|1); } // sum(0..n-1)
// \return |\{(x,y) | 1 <= x, 1 <= y, a*x+b*y <= S\}|
// = sum_{i=1}^n \{qs\} (S-a*i)/b
```

```

ul triSum(ul a, ul b, ul s) { assert(a > 0 && b > 0);
    ul qs = s/a, rs = s%a; // ans = sum_{i=0}^{qs-1} (i*a+rs)/b
    ul ad = a/b*sum2(qs)+rs/b*qs; a %= b, rs %= b;
    return ad+(a?triSum(b,a,a*qs+rs):0); // reduce if a >= b
} // then swap x and y axes and recurse
```

```

// \return sum_{x=0}^{n-1} (a*x+b)/m
// = |\{(x,y) | 0 < m*y <= a*x+b < a*n+b\}|
// assuming a*n+b does not overflow
ul divSum(ul n, ul a, ul b, ul m) { assert(m > 0);
    ul extra = b/m*n; b %= m;
    return extra+(a?triSum(m,a,a*n+b):0); }
// \return sum_{x=0}^{n-1} (a*x+b)%m
ul modSum(ul n, ll a, ll b, ul m) { assert(m > 0);
    a = (a%m+m)%m, b = (b%m+m)%m;
    return a*sum2(n)+b*n-m*divSum(n,a,b,m); }
```

4.2 Primality

4.2.1 Primes

$p = 962592769$ is such that $2^{21} \mid p-1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power p^a , except for $p = 2, a > 2$, and there are $\phi(\phi(p^a))$ many. For $p = 2, a > 2$, the group $\mathbb{Z}_{2^a}^\times$ is instead isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$.

4.2.2 Divisors

$$\sum_{d \mid n} d = O(n \log \log n).$$

The number of divisors of n is at most around 100 for $n < 5e4$, 500 for $n < 1e7$, 2000 for $n < 1e10$, 200 000 for $n < 1e19$.

Let $n = \prod_i p_i^{e_i}$ be the prime factorization of n .

Number of Divisors: $d(n) = \prod_i (e_i + 1)$.

Sum of Divisors: $\sigma(n) = \prod_i \frac{p_i^{e_i+1}-1}{p_i-1}$.

Product of Divisors:

Case 1: n is a perfect square. Then $\prod_{d \mid n} d = \sqrt{n}^{d(n)}$.

Case 2: n is not a perfect square. Then $\prod_{d \mid n} d = n^{d(n)/2}$.

Euler's Totient Function:
 $\phi(n) = n \prod_i (1 - \frac{1}{p_i}) = \prod_i (p_i^{a_i} - p_i^{a_i-1})$.

Dirichlet Convolution: Given a function $f(x)$, let

$$(f * g)(x) = \sum_{d \mid x} g(d) f(x/d).$$

If the partial sums $s_{f * g}(n), s_g(n)$ can be computed in $O(1)$ and $s_f(1 \dots n^{2/3})$ can be computed in $O\left(n^{2/3}\right)$ then all $s_f\left(\frac{n}{d}\right)$ can as well. Use

$$s_{f * g}(n) = \sum_{d=1}^n g(d) s_f(n/d).$$

If $f(x) = \mu(x)$ then $g(x) = 1, (f * g)(x) = (x == 1)$, and $s_f(n) = 1 - \sum_{i=2}^n s_f(n/i)$.

If $f(x) = \phi(x)$ then $g(x) = 1, (f * g)(x) = x$, and $s_f(n) = \frac{n(n+1)}{2} - \sum_{i=2}^n s_f(n/i)$.

Let $s(x) = \sum_{i=1}^x \phi(i)$. Then

$$s(n) = \frac{n(n+1)}{2} - \sum_{i=2}^n s\left(\left\lfloor \frac{n}{i} \right\rfloor\right)$$

can be computed in faster than $\Theta(n)$.

FactorBasic.h
Description: Factors integers.
Time: $\mathcal{O}(\sqrt{N})$

```

8b0341, 53 lines
inline namespace factorBasic {
template<class T> vector<pair<T, int>> factor(T x) {
    vector<pair<T, int>> primes;
    for (T i = 2; i*i <= x; ++i) if (x%i == 0) {
        int t = 0;
        while (x%i == 0) x /= i, t++;
        primes.push_back({i, t});
    }
    if (x > 1) primes.push_back({x, 1});
    return primes;
}
```

```

/* Note:
 * number of operations needed s.t.
 *     phi(phi(...phi(n)...))=1
 * is O(log n).
 * Euler's theorem: a^{\phi(p)} \equiv 1 (mod p), gcd(a,p)=1
 */
template<class T> T phi(T x) {
    for (auto& [p, exp]: factor(x)) x -= x/p;
    return x;
}
```

```

template<class T> void tour(vector<pair<T, int>> &v,
vector<T> &V, int ind, T cur) {
    if (ind == (int)v.size()) V.push_back(cur);
    else {
        T mul = 1;
        for (int i = 0; i <= v[ind].second; i++) {
            tour(v, V, ind+1, cur*mul);
            mul *= v[ind].first;
        }
    }
}
```

```

template<class T> vector<T> getDiv(T x) {
    auto v = factor(x);
    vector<T> V;
    tour(v, V, 0, (T)1);
    ranges::sort(V);
    return V;
}
```

```

template<class T> vector<T> getDiv(T x) {
    vector<T> V;
    for (T i = 1; i*i <= x; ++i) if (x%i == 0) {
        V.push_back(i);
        if (x/i != i) V.push_back(x/i);
    }
    ranges::sort(V);
}
```

```
    return V;
}
}
```

Sieve.h
Description: Tests primality up to N . Runs faster if only odd indices are stored.
Time: $\mathcal{O}(N \log \log N)$ or $\mathcal{O}(N)$

```
template<int N> struct Sieve {
    bitset<N> is_prime; vector<int> primes;
    Sieve() {
        is_prime.set(); is_prime[0] = is_prime[1] = 0;
        for (int i = 4; i < N; i += 2) is_prime[i] = 0;
        for (int i = 3; i*i < N; i += 2) if (is_prime[i])
            for (int j = i*i; j < N; j += i*2) is_prime[j] = 0;
        for (int i = 0; i < N; i++) if (is_prime[i]) primes.
            ↪push_back(i);
    }
    // array<int, N> spf{} // smallest prime that divides
    // Sieve() { // above is faster
    //     for (int i = 2; i < N; i++) {
    //         if (spf[i] == 0) spf[i] = i, primes.push_back(i);
    //         for (int p: primes) {
    //             if (p > spf[i] || i*p >= N) break;
    //             spf[i*p] = p;
    //         }
    //     }
    // }
```

MultiplicativePrefixSums.h
Description: $\sum_{i=1}^N f(i)$ where $f(i) = \prod \text{val}[e]$ for each p^e in the factorization of i . Must satisfy $\text{val}[1] = 1$. Generalizes to any multiplicative function with $f(p) = p^{\text{fixed power}}$.
Time: $\mathcal{O}(\sqrt{N})$

```
"Sieve.h" 3151ea, 12 lines

vmi val;
mi get_prefix(ll N, int p = 0) {
    mi ans = N;
    for (; S.primes.at(p) <= N / S.primes.at(p); ++p) {
        ll new_N = N / S.primes.at(p) / S.primes.at(p);
        for (int idx = 2; new_N; ++idx, new_N /= S.primes.at(p)) {
            ans += (val.at(idx) - val.at(idx - 1))
                * get_prefix(new_N, p + 1);
        }
    }
    return ans;
}
```

PrimeCnt.h
Description: Counts number of primes up to N . Can also count sum of primes.
Time: $\mathcal{O}(N^{3/4}/\log N)$, 60ms for $N = 10^{11}$, 2.5s for $N = 10^{13}$

```
ll count_primes(ll N) { // count_primes(1e13) == 346065536839
    if (N <= 1) return 0;
    int sq = (int)sqrt(N);
    vl big_ans((sq+1)/2), small_ans(sq+1);
    FOR(i,1,sq+1) small_ans[i] = (i-1)/2;
    F0R(i,sz(big_ans)) big_ans[i] = (N/(2*i+1)-1)/2;
    vb skip(sq+1); int prime_cnt = 0;
    for (int p = 3; p <= sq; p += 2) if (!skip[p]) { // primes
        for (int j = p; j <= sq; j += 2*p) skip[j] = 1;
        F0R(j,min((ll)sz(big_ans), (N/p/p+1)/2)) {
            ll prod = (ll)(2*j+1)*p;
            big_ans[j] -= (prod > sq ? small_ans[(double)N/prod]
```

```
                : big_ans[prod/2])-prime_cnt;
        }
        for (int j = sq, q = sq/p; q >= p; --q) for (;j >= q*p;--j)
            small_ans[j] -= small_ans[q]-prime_cnt;
        ++prime_cnt;
    }
    return big_ans[0]+1;
}
```

MillerRabin.h
Description: Deterministic primality test, works up to 2^{64} . For larger numbers, extend A randomly.

```
"ModMulLL.h" f5b0fe, 11 lines

bool prime(ul n) { // not ll!
    if (n < 2 || n % 6 % 4 != 1) return n - 2 < 2;
    ul s = __builtin_ctzll(n - 1), d = n >> s;
    ul A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022};
    for (auto a : A) {
        ul p = modPow(a, d, n), i = s;
        while (p != 1 && p != n-1 && a % n && i--) p = modMul(p, p,
            ↪n);
        if (p != n-1 && i != s) return false;
    }
    return true;
}
```

FactorFast.h
Description: Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}).

```
Time:  $\mathcal{O}(N^{1/4})$ , less for numbers with small factors

"MillerRabin.h", "ModMulLL.h" 99cf33, 16 lines

ul pollard(ul n) { // return some nontrivial factor of n
    auto f = [n](ul x) { return modMul(x, x, n) + 1; };
    ul x = 0, y = 0, t = 30, prd = 2, i = 1, q;
    while (t++ % 40 || gcd(prd, n) == 1) {
        if (x == y) x = ++i, y = f(x);
        if ((q = modMul(prd, max(x,y)-min(x,y), n)) prd = q;
            x = f(x), y = f(f(y));
        }
    }
    return gcd(prd, n);
}

void factor_rec(ul n, map<ul,int>& cnt) {
    if (n == 1) return;
    if (prime(n)) { ++cnt[n]; return; }
    ul u = pollard(n);
    factor_rec(u,cnt), factor_rec(n/u,cnt);
}
```

4.3 Euclidean Algorithm

4.3.1 Bézout’s identity

For $a \neq 0, b \neq 0$, then $d = \gcd(a, b)$ is the smallest positive integer for which there are integer solutions to

$$ax + by = d$$

If (x, y) is one solution, then all solutions are given by

$$\left(x + \frac{kb}{\gcd(a,b)}, y - \frac{ka}{\gcd(a,b)}\right), \quad k \in \mathbb{Z}$$

FracInterval.h
Description: Given fractions $a < b$ with non-negative numerators and denominators, finds fraction f with lowest denominator such that $a < f < b$. Should work with all numbers less than 2^{62} .

```
pl bet(pl a, pl b) {
    ll num = a.f/a.s; a.f -= num*a.s, b.f -= num*b.s;
    if (b.f > b.s) return {1+num,1};
    auto x = bet({b.s,b.f},{a.s,a.f});
    return {x.s+num*x.f,x.f};
}
```

Euclid.h
Description: Generalized Euclidean algorithm. euclid and invGeneral work for $A, B < 2^{62}$.
Time: $\mathcal{O}(\log AB)$

```
c7e528, 9 lines

// ceil(a/b)
// ll cdiv(ll a, ll b) { return a/b+((a^b)>0&&a%b); }
pl euclid(ll A, ll B) { // For A,B>=0, finds (x,y) s.t.
    // Ax+By=gcd(A,B), |Ax|,|By|<=AB/gcd(A,B)
    if (!B) return {1,0};
    pl p = euclid(B,A%B); return {p.s,p.f-A/B*p.s}; }
ll invGeneral(ll A, ll B) { // find x in [0,B) such that Ax=1
    ↪mod B
    pl p = euclid(A,B); assert(p.f*A+p.s*B == 1);
    return p.f+(p.f<0)*B; } // must have gcd(A,B)=1
```

CRT.h
Description: Chinese Remainder Theorem. $a.f \pmod{a.s}, b.f \pmod{b.s} \implies ? \pmod{\text{lcm}(a.s,b.s)}$. Should work for $ab < 2^{62}$.

```
"Euclid.h" 23df64, 10 lines

pl CRT(pl a, pl b) { assert(0 <= a.f && a.f < a.s && 0 <= b.f
    ↪&& b.f < b.s);
    if (a.s < b.s) swap(a,b); // will overflow if b.s^2 > 2^{62}
    ll x,y; tie(x,y) = euclid(a.s,b.s);
    ll g = a.s*x+b.s*y, l = a.s/g*b.s;
    if ((b.f-a.f)%g) return {-1,-1}; // no solution
    // ?*a.s+a.f \equiv b.f \pmod{b.s}
    // ?=(b.f-a.f)/g*(a.s/g)^{-1} \pmod{b.s/g}
    x = (b.f-a.f)%b.s*x*b.s/g*a.s+a.f;
    return {x+(x<0)*1,l};
}
```

ModArith.h
Description: Statistics on mod’ed arithmetic series. minBetween and minRemainder both assume that $0 \leq L \leq R < B, AB < 2^{62}$

```
f68a6d, 40 lines

ll minBetween(ll A, ll B, ll L, ll R) {
    // min x s.t. exists y s.t. L <= A*x-B*y <= R
    A %= B;
    if (L == 0) return 0;
    if (A == 0) return -1;
    ll k = cdiv(L,A); if (A*k <= R) return k;
    ll x = minBetween(B,A,A-R%A,A-L%A); // min x s.t. exists y
    // s.t. -R <= Bx-Ay <= -L
    return x == -1 ? x : cdiv(B*x+L,A); // solve for y
}

// find min((Ax+C)%B) for 0 <= x <= M
// aka find minimum non-negative value of A*x-B*y+C
// where 0 <= x <= M, 0 <= y
ll minRemainder(ll A, ll B, ll C, ll M) {
    assert(A >= 0 && B > 0 && C >= 0 && M >= 0);
    A %= B, C %= B; ckmin(M,B-1);
    if (A == 0) return C;
    if (C >= A) { // make sure C<A
        ll ad = cdiv(B-C,A);
        M -= ad; if (M < 0) return C;
        C += ad*A-B;
    }
    ll q = B/A, new_B = B%A; // new_B < A
    if (new_B == 0) return C; // B=q*A
```

```
// now minimize A*x-new_B*y+C
// where 0 <= x,y and x+q*y <= M, 0 <= C < new_B < A
// q*y -> C-new_B*y
if (C/new_B > M/q) return C-M/q*new_B;
M -= C/new_B*q; C %= new_B; // now C < new_B

// given y, we can compute x = ceil(((B-q*A)*y-C)/A)
// so x+q*y = ceil((B*y-C)/A) <= M
ll max_Y = (M+A*C)/B; // must have y <= max_Y
ll max_X = cdiv(new_B*max_Y-C,A); // must have x <= max_X
if (max_X*A-new_B*max_Y+C >= new_B) --max_X;
// now we can remove upper bound on y
return minRemainder(A,new_B,C,max_X);
}
```

4.4 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$a = k \cdot (m^2 - n^2), \quad b = k \cdot (2mn), \quad c = k \cdot (m^2 + n^2),$

with $m > n > 0, k > 0, m \perp n$, and either m or n even.

4.5 Lifting the Exponent

For $n > 0, p$ prime, and ints x, y s.t. $p \nmid x, y$ and $p | x - y$:

- $p \neq 2$ or $p = 2, 4 | x - y \implies v_p(x^n - y^n) = v_p(x - y) + v_p(n).$
- $p = 2, 2 | n \implies v_2(x^n - y^n) = v_2((x^2)^{n/2} - (y^2)^{n/2}).$

Combinatorial (5)

5.1 Permutations

5.1.1 Cycles

Let $g_S(n)$ be the number of n -permutations whose cycle lengths all belong to the set S . Then

$$\sum_{n=0}^\infty g_S(n) \frac{x^n}{n!} = \exp\left(\sum_{n \in S} \frac{x^n}{n}\right)$$

5.1.2 Burnside’s lemma

Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts “configurations” (of some sort) of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k | n} f(k) \phi(n/k).$$

5.2 Partitions and subsets

5.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k - 1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	$\sim 2e5$	$\sim 2e8$

5.2.2 Lucas’ Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + \dots + n_1 p + n_0$ and $m = m_k p^k + \dots + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod p$.

5.3 General purpose numbers

5.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t - 1}$ (FFT-able).
 $B[0, \dots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \dots]$

Sums of powers:

$$\sum_{i=1}^n i^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^\infty f(i) = \int_m^\infty f(x) dx - \sum_{k=1}^\infty \frac{B_k}{k!} f^{(k-1)}(m)$$

$$\approx \int_m^\infty f(x) dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

5.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n, k) = c(n - 1, k - 1) + (n - 1)c(n - 1, k), \quad c(0, 0) = 1$$

$$\sum_{k=0}^n c(n, k) x^k = x(x + 1) \dots (x + n - 1)$$

$c(8, k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$
 $c(n, 2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \dots$

5.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j :s s.t. $\pi(j) > \pi(j + 1)$, $k + 1$ j :s s.t. $\pi(j) \geq j$, k j :s s.t. $\pi(j) > j$.

$$E(n, k) = (n - k)E(n - 1, k - 1) + (k + 1)E(n - 1, k)$$

$$E(n, 0) = E(n, n - 1) = 1$$

$$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n + 1}{j} (k + 1 - j)^n$$

5.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n, k) = S(n - 1, k - 1) + kS(n - 1, k)$$

$$S(n, 1) = S(n, n) = 1$$

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

5.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$. For p prime,

$$B(p^m + n) \equiv mB(n) + B(n + 1) \pmod p$$

5.3.6 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i : $n_1 n_2 \dots n_k n^{k-2}$
with degrees d_i : $(n - 2)! / ((d_1 - 1)! \dots (d_n - 1)!)$

5.3.7 Catalan numbers

$$C_n = \frac{1}{n + 1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n + 1} = \frac{(2n)!}{(n + 1)!n!}$$

$$C_0 = 1, \quad C_{n+1} = \frac{2(2n + 1)}{n + 2} C_n, \quad C_{n+1} = \sum C_i C_{n-i}$$

$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \dots$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n + 1$ leaves (0 or 2 children).
- ordered trees with $n + 1$ vertices.
- ways a convex polygon with $n + 2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

5.4 Young Tableaux

Let a **Young diagram** have shape $\lambda = (\lambda_1 \geq \dots \geq \lambda_k)$, where λ_i equals the number of cells in the i -th (left-justified) row from the top. A **Young tableau** of shape λ is a filling of the $n = \sum \lambda_i$ cells with a permutation of $1 \dots n$ such that each row and column is increasing.

Hook-Length Formula: For the cell in position (i, j) , let $h_\lambda(i, j) = |\{(I, J) | i \leq I, j \leq J, (I = i \text{ or } J = j)\}|$. The number of Young tableaux of shape λ is equal to $f^\lambda = \frac{n!}{\prod h_\lambda(i, j)}$.

Schensted’s Algorithm: converts a permutation σ of length n into a pair of Young Tableaux $(S(\sigma), T(\sigma))$ of the same shape. When inserting $x = \sigma_i$,

1. Add x to the first row of S by inserting x in place of the largest y with $x < y$. If y doesn't exist, push x to the end of the row, set the value of T at that position to be i , and stop.
2. Add y to the second row using the same rule, keep repeating as necessary.

All pairs $(S(\sigma),T(\sigma))$ of the same shape correspond to a unique σ , so $n! = \sum (f^\lambda)^2$. Also, $S(\sigma^R) = S(\sigma)^T$.

Let $d_k(\sigma), a_k(\sigma)$ be the lengths of the longest subseqs which are a union of k decreasing/ascending subseqs, respectively. Then $a_k(\sigma) = \sum_{i=1}^k \lambda_i, d_k(\sigma) = \sum_{i=1}^k \lambda_i^*$, where λ_i^* is size of the i -th column.

5.5 Other

DeBruijnSequence.h

Description: Given alphabet $[0, k)$ constructs a cyclic string of length k^n that contains every length n string as substring.

dfecad, 14 lines

```
vector<int> deBruijnSequence(int k, int n) {
    if (k == 1) return {0};
    vector<int> seq, aux(n+1);
    auto gen = [&](this auto&& self, int t, int p)->void {
        if (t > n && n%p == 0) for (int i = 1; i <= p; i++) {
            seq.push_back(aux[i]);
        }
        else if (t <= n) {
            aux[t] = aux[t-p]; self(t+1, p);
            while (++aux[t] < k) self(t+1, t);
        }
    };
    gen(1, 1); return seq;
}
```

NimProduct.h

Description: Product of numbers is associative, commutative, and distributive over addition (xor). Forms finite field of size 2^{2^k} . Defined by $ab = \text{mex}(\{a'b + ab' + a'b' : a' < a, b' < b\})$. Application: Given 1D coin turning games G_1, G_2 $G_1 \times G_2$ is the 2D coin turning game defined as follows. If turning coins at $x_1, x_2, \dots, x_m$ is legal in G_1 and $y_1, y_2, \dots, y_n$ is legal in G_2 , then turning coins at all positions (x_i, y_j) is legal assuming that the coin at (x_m, y_n) goes from heads to tails. Then the Grundy function $g(x, y)$ of $G_1 \times G_2$ is $g_1(x) \times g_2(y)$.
Time: 64^2 xors per multiplication, memorize to speed up.

5afe17, 46 lines

```
using ul = uint64_t;
struct Precalc {
    ul tmp[64][64], y[8][8][256];
    unsigned char x[256][256];
    Precalc() { // small nim products, all < 256
        F0R(i,256) F0R(j,256) x[i][j] = mult<8>(i,j);
        F0R(i,8) F0R(j,i+1) F0R(k,256)
            y[i][j][k] = mult<64>(prod2(8*i,8*j),k);
    }
    ul prod2(int i, int j) { // nim prod of 2^i, 2^j
        ul& u = tmp[i][j]; if (u) return u;
        if (!(i&j)) return u = 1ULL<<(i|j);
        int a = (i&j)&~(i&j); // a=2^k, consider 2^{2^k}
        return u=prod2(i^a,j)^prod2((i^a)|(a-1),(j^a)|(i&(a-1)));
        // 2^{2^k}*2^{2^k} = 2^{2^k}*2^{2^k-1}
    } // 2^{2^i}*2^{2^j} = 2^{2^i+2^j} if i<j
    template<int L> ul mult(ul a, ul b) {
        ul c = 0; F0R(i,L) if (a>i&i)
            F0R(j,L) if (b>>j&1) c ^= prod2(i,j);
    }
}
```

```
        return c;
    }
    // 2^{8*i}*(a>>(8*i)&255) * 2^{8*j}*(b>>(8*j)&255)
    // -> (2^{8*i}*2^{8*j})*(a>>(8*i)&255)*(b>>(8*j)&255))
    ul multFast(ul a, ul b) const { // faster nim product
        ul res = 0; auto f=[](ul c,int d) {return c>>(8*d)&255;};
        F0R(i,8) {
            F0R(j,i) res ^= y[i][j][x[f(a,i)][f(b,j)]
                ^x[f(a,j)][f(b,i)]];
            res ^= y[i][i][x[f(a,i)][f(b,i)]];
        }
        return res;
    }
};
const Precalc P;
```

struct nb { // number
 ul x; nb() { x = 0; }
 nb(ul _x): x(_x) {}
 explicit operator ul() { return x; }
 nb operator+(nb y) { return nb(x^y.x); }
 nb operator*(nb y) { return nb(P.multFast(x,y.x)); }
 friend nb pow(nb b, ul p) {
 nb res = 1; for (;p;p/=2,b=b*b) if (p&1) res = res*b;
 return res; } // b^{2^{2^A}-1}=1 where 2^{2^A} > b
 friend nb inv(nb b) { return pow(b,-2); }
};

MatroidIsect.h

Description: Computes a set of maximum size which is independent in both graphic and colorful matroids, aka a spanning forest where no two edges are of the same color. In general, construct the exchange graph and find a short-est path. Can apply similar concept to partition matroid.
Usage: MatroidIsect<Gmat, Cmat> M(sz(ed), Gmat(ed), Cmat(col))
Time: $\mathcal{O}(GI^{1.5})$ calls to oracles, where G is size of ground set and I is size of independent set.

"DSU.h" d0051c, 51 lines

```
struct Gmat { // graphic matroid
    int V = 0; vpi ed; DSU D;
    Gmat(vpi _ed):ed(_ed) {
        map<int,int> m; each(t,ed) m[t.f] = m[t.s] = 0;
        each(t,m) t.s = V++;
        each(t,ed) t.f = m[t.f], t.s = m[t.s];
    }
    void clear() { D.init(V); }
    void ins(int i) { assert(D.unite(ed[i].f,ed[i].s)); }
    bool indep(int i) { return !D.sameSet(ed[i].f,ed[i].s); }
};
struct Cmat { // colorful matroid
    int C = 0; vi col; V<bool> used;
    Cmat(vi col):col(col) {each(t,col) ckmax(C,t+1); }
    void clear() { used.assign(C,0); }
    void ins(int i) { used[col[i]] = 1; }
    bool indep(int i) { return !used[col[i]]; }
};
template<class M1, class M2> struct MatroidIsect {
    int n; V<bool> iset; M1 m1; M2 m2;
    bool augment() {
        vi pre(n+1,-1); queue<int> q({n});
        while (sz(q)) {
            int x = q.ft; q.pop();
            if (iset[x]) {
                m1.clear(); F0R(i,n) if (iset[i] && i != x) m1.ins(i);
                F0R(i,n) if (!iset[i] && pre[i] == -1 && m1.indep(i))
                    pre[i] = x, q.push(i);
            } else {
                auto backE = [&]() { // back edge
                    m2.clear();
                    F0R(c,2)F0R(i,n)if((x==i||iset[i])&&(pre[i]==-1)==c){
```

```
                        if (!m2.indep(i))return c?pre[i]=x,q.push(i),i:-1;
                        m2.ins(i); }
                    return n;
                };
            }
            for (int y; (y = backE()) != -1;) if (y == n) {
                for(; x != n; x = pre[x]) iset[x] = !iset[x];
                return 1; }
        }
    }
    return 0;
}
MatroidIsect(int n, M1 m1, M2 m2):n(n), m1(m1), m2(m2) {
    iset.assign(n+1,0); iset[n] = 1;
    m1.clear(); m2.clear(); // greedily add to basis
    R0F(i,n) if (m1.indep(i) && m2.indep(i))
        iset[i] = 1, m1.ins(i), m2.ins(i);
    while (augment());
}
};
```

Numerical (6)

6.1 Matrix

Matrix.h

Description: 2D matrix operations.

"ModInt.h" b18e29, 21 lines

```
using T = mi;
using Mat = V<V<T>>; // use array instead if tight TL
```

Mat makeMat(int r, int c) { return Mat(r,V<T>(c)); }
Mat makeId(int n) {
 Mat m = makeMat(n,n); F0R(i,n) m[i][i] = 1;
 return m;
}
Mat operator*(const Mat& a, const Mat& b) {
 int x = sz(a), y = sz(a[0]), z = sz(b[0]);
 assert(y == sz(b)); Mat c = makeMat(x,z);
 F0R(i,x) F0R(j,y) F0R(k,z) c[i][k] += a[i][j]*b[j][k];
 return c;
}
Mat& operator*=(Mat& a, const Mat& b) { return a = a*b; }
Mat pow(Mat m, ll p) {
 int n = sz(m); assert(n == sz(m[0]) && p >= 0);
 Mat res = makeId(n);
 for (; p; p /= 2, m *= m) if (p&1) res *= m;
 return res;
}

MatrixInv.h

Description: Uses gaussian elimination to convert into reduced row echelon form and calculates determinant. For determinant via arbitrary modulus, use a modified form of the Euclidean algorithm because modular inverse may not exist. If you have computed $A^{-1} \pmod{p^k}$, then the inverse $\pmod{p^{2k}}$ is $A^{-1}(2I - AA^{-1})$.
Time: $\mathcal{O}(N^3)$, determinant of 1000×1000 matrix of modints in 1 second if you reduce # of operations by half

73ec43, 38 lines

```
const db EPS = 1e-9; // adjust?
int getRow(V<V<db>>& m, int R, int i, int nex) {
    pair<db,int> bes{0,-1}; // find row with max abs value
    F0R(j,nex,R) ckmax(bes,{abs(m[j][i]),j});
    return bes.f < EPS ? -1 : bes.s; }
int getRow(V<vmi>& m, int R, int i, int nex) {
    F0R(j,nex,R) if (m[j][i] != 0) return j;
    return -1; }
pair<T,int> gauss(Mat& m) { // convert to reduced row echelon
    ↪form
```

```
if (!sz(m)) return {1,0};
int R = sz(m), C = sz(m[0]), rank = 0, nex = 0;
T prod = 1; // determinant
F0R(i,C) {
    int row = getRow(m,R,i,nex);
    if (row == -1) { prod = 0; continue; }
    if (row != nex) prod *= -1, swap(m[row],m[nex]);
    prod *= m[nex][i]; rank++;
    T x = 1/m[nex][i]; FOR(k,i,C) m[nex][k] *= x;
    F0R(j,R) if (j != nex) {
        T v = m[j][i]; if (v == 0) continue;
        FOR(k,i,C) m[j][k] -= v*m[nex][k];
    }
    nex++;
}
return {prod,rank};
}
Mat inv(Mat m) {
    int R = sz(m); assert(R == sz(m[0]));
    Mat x = makeMat(R,2*R);
    F0R(i,R) {
        x[i][i+R] = 1;
        F0R(j,R) x[i][j] = m[i][j];
    }
    if (gauss(x).s != R) return Mat();
    Mat res = makeMat(R,R);
    F0R(i,R) F0R(j,R) res[i][j] = x[i][j+R];
    return res;
}
```

MatrixTree.h
Description: Kirchhoff's Matrix Tree Theorem. Given adjacency matrix, calculates # of spanning trees.

"MatrixInv.h"	48363d, 11 lines
---------------	------------------

```
T numSpan(const Mat& m) {
    int n = sz(m); Mat res = makeMat(n-1,n-1);
    F0R(i,n) FOR(j,i+1,n) {
        mi ed = m[i][j]; res[i][i] += ed;
        if (j != n-1) {
            res[j][j] += ed;
            res[i][j] -= ed, res[j][i] -= ed;
        }
    }
    return gauss(res).f;
}
```

ShermanMorrison.h
Description: Calculates $(A + uv^T)^{-1}$ given $B = A^{-1}$. Not invertible if sum=0.

"MatrixInv.h"	3a3f34, 7 lines
---------------	-----------------

```
void ad(Mat& B, const V<T>& u, const V<T>& v) {
    int n = sz(A); V<T> x(n), y(n);
    F0R(i,n) F0R(j,n)
        x[i] += B[i][j]*u[j], y[j] += v[i]*B[i][j];
    T sum = 1; F0R(i,n) F0R(j,n) sum += v[i]*B[i][j]*u[j];
    F0R(i,n) F0R(j,n) B[i][j] -= x[i]*y[j]/sum;
}
```

6.2 Polynomials

Poly.h
Description: Basic poly ops including division. Can replace T with double, complex.

"ModInt.h"	cd218a, 73 lines
------------	------------------

```
using T = mi; using poly = V<T>;
void remz(poly& p) { while (sz(p)&&p.bk==T(0)) p.pop_back(); }
poly REMZ(poly p) { remz(p); return p; }
poly rev(poly p) { reverse(all(p)); return p; }
```

```
poly shift(poly p, int x) {
    if (x >= 0) p.insert(begin(p),x,0);
    else assert(sz(p)+x >= 0), p.erase(begin(p),begin(p)-x);
    return p;
}
poly RSZ(const poly& p, int x) {
    if (x <= sz(p)) return poly(begin(p),begin(p)+x);
    poly q = p; q.rsz(x); return q; }
T eval(const poly& p, T x) { // evaluate at point x
    T res = 0; R0F(i,sz(p)) res = x*res+p[i];
    return res; }
poly dif(const poly& p) { // differentiate
    poly res; FOR(i,1,sz(p)) res.pb(T(i)*p[i]);
    return res; }
poly integ(const poly& p) { // integrate
    static poly invs{0,1};
    for (int i = sz(invs); i <= sz(p); ++i)
        invs.pb(-MOD/i*invs[MOD%i]);
    poly res(sz(p)+1); F0R(i,sz(p)) res[i+1] = p[i]*invs[i+1];
    return res;
}

poly& operator+=(poly& l, const poly& r) {
    l.rsz(max(sz(l),sz(r))); F0R(i,sz(r)) l[i] += r[i];
    return l; }
poly& operator-=(poly& l, const poly& r) {
    l.rsz(max(sz(l),sz(r))); F0R(i,sz(r)) l[i] -= r[i];
    return l; }
poly& operator*=(poly& l, const T& r) { each(t,l) t *= r;
    return l; }
poly& operator/=(poly& l, const T& r) { each(t,l) t /= r;
    return l; }
poly operator+(poly l, const poly& r) { return l += r; }
poly operator-(poly l, const poly& r) { return l -= r; }
poly operator*(poly l, const T& r) { return l *= r; }
poly operator*(const T& r, const poly& l) { return l*r; }
poly operator/(poly l, const T& r) { return l /= r; }
poly operator*(const poly& l, const poly& r) {
    if (!min(sz(l),sz(r))) return {};
    poly x(sz(l)+sz(r)-1);
    F0R(i,sz(l)) F0R(j,sz(r)) x[i+j] += l[i]*r[j];
    return x;
}
poly& operator*=(poly& l, const poly& r) { return l = l*r; }
```

```
pair<poly,poly> quoRemSlow(poly a, poly b) {
    remz(a); remz(b); assert(sz(b));
    T lst = b.bk, B = T(1)/lst; each(t,a) t *= B;
    each(t,b) t *= B;
    poly q(max(sz(a)-sz(b)+1,0));
    for (int dif; (dif=sz(a)-sz(b)) >= 0; remz(a)) {
        q[dif] = a.bk; F0R(i,sz(b)) a[i+dif] -= q[dif]*b[i]; }
    each(t,a) t *= lst;
    return {q,a}; // quotient, remainder
}
poly operator%(const poly& a, const poly& b) {
    return quoRemSlow(a,b).s; }
T resultant(poly a, poly b) { // R(A,B)
    // =b_m^n*prod_{j=1}^mA(mu_j)
    // =b_m^na_n^m*prod_{i=1}^nprod_{j=1}^m(mu_j-lambda_i)
    // =(-1) ^{mn}a_n^m*prod_{i=1}^nB(lambda_i)
    // =(-1) ^{nm}R(B,A)
    // Also, R(A,B)=b_m^{deg(A)-deg(A-CB)}R(A-CB,B)
    int ad = sz(a)-1, bd = sz(b)-1;
    if (bd <= 0) return bd < 0 ? 0 : pow(b.bk,ad);
    int pw = ad; a = a&b; pw -= (ad = sz(a)-1);
    return resultant(b,a)*pow(b.bk,pw)*T((bd&ad&1)?-1:1);
}
```

PolyInterpolate.h
Description: n points determine unique polynomial of degree $\leq n-1$. For numerical precision pick $v[k].f = c * \cos(k/(n-1) * \pi), k = 0 \dots n-1$.
Time: $\mathcal{O}(n^2)$

"Poly.h"	aada3a, 8 lines
----------	-----------------

```
poly interpolate(V<pair<T,T>> v) {
    poly res, tmp{1};
    F0R(i,sz(v)) { T prod = 1; // add one point at a time
        F0R(j,i) v[i].s -= prod*v[j].s, prod *= v[i].f-v[j].f;
        v[i].s /= prod; res += v[i].s*tmp; tmp *= poly{-v[i].f,1};
    } // add multiple of (x-v[0].f)*(x-v[1].f)*...*(x-v[i-1].f)
    return res;
}
```

FFT.h
Description: Multiply polynomials of ints for any modulus $< 2^{31}$. For XOR convolution ignore m within fft.
Time: $\mathcal{O}(N \log N)$. For $N = 10^6$, conv ~0.13ms, conv-general ~320ms.

"ModInt.h"	19ab20, 39 lines
------------	------------------

```
// const int MOD = 998244353;
tcT> void fft(V<T>& A, bool invert = 0) { // NTT
    int n = sz(A); assert((T::mod-1)%n == 0); V<T> B(n);
    for(int b = n/2; b /= 2, swap(A,B)) { // w = n/b'th root
        T w = pow(T::rt(), (T::mod-1)/n*b), m = 1;
        for(int i = 0; i < n; i += b*2, m *= w) F0R(j,b) {
            T u = A[i+j], v = A[i+j+b]*m;
            B[i/2+j] = u+v; B[i/2+j+n/2] = u-v;
        }
    }
    if (invert) { reverse(l+all(A));
        T z = inv(T(n)); each(t,A) t *= z; }
} // for NTT-able moduli
tcT> V<T> conv(V<T> A, V<T> B) {
    if (!min(sz(A),sz(B))) return {};
    int s = sz(A)+sz(B)-1, n = 1; for (; n < s; n *= 2);
    A.rsz(n), fft(A); B.rsz(n), fft(B);
    F0R(i,n) A[i] *= B[i];
    fft(A,1); A.rsz(s); return A;
}
template<class M, class T> V<M> mulMod(const V<T>& x, const V<T
    <=>y) {
    auto con = [] (const V<T>& v) {
        V<M> w(sz(v)); F0R(i,sz(v)) w[i] = (int)v[i];
        return w; };
    return conv(con(x), con(y));
} // arbitrary moduli
tcT> V<T> conv_general(const V<T>& A, const V<T>& B) {
    using m0 = mint<(119<<23)+1,62>; auto c0 = mulMod<m0>(A,B);
    using m1 = mint<(5<<25)+1, 62>; auto c1 = mulMod<m1>(A,B);
    using m2 = mint<(7<<26)+1, 62>; auto c2 = mulMod<m2>(A,B);
    int n = sz(c0); V<T> res(n); m1 r01 = inv(m1(m0::mod));
    m2 r02 = inv(m2(m0::mod)), r12 = inv(m2(m1::mod));
    F0R(i,n) { // a=remainder mod m0::mod, b fixes it mod m1::mod
        int a = c0[i].v, b = ((c1[i]-a)*r01).v,
            c = (((c2[i]-a)*r02-b)*r12).v;
        res[i] = (T(c)*m1::mod+b)*m0::mod+a; // c fixes m2::mod
    }
    return res;
}
```

PolyInvSimpler.h
Description: computes A^{-1} such that $AA^{-1} \equiv 1 \pmod{x^n}$. Newton's method: If you want $F(x) = 0$ and $F(Q_k) \equiv 0 \pmod{x^a}$ then $Q_{k+1} = Q_k - \frac{F(Q_k)}{F'(Q_k)} \pmod{x^{2a}}$ satisfies $F(Q_{k+1}) \equiv 0 \pmod{x^{2a}}$. Application: if $f(n), g(n)$ are the #s of forests and trees on n nodes then $\sum_{n=0}^{\infty} f(n)x^n = \exp\left(\sum_{n=1}^{\infty} \frac{g(n)}{n!}\right)$.

```
Usage: vmi v{1,5,2,3,4}; ps(exp(2*log(v,9),9)); // squares v
Time:  $\mathcal{O}(N \log N)$ . For  $N = 5 \cdot 10^9$ , inv~270ms, log ~350ms, exp~550ms
"FFT.h", "Poly.h" 6e5362, 30 lines

poly inv(poly A, int n) { //  $Q-(1/Q-A)/(-Q^{-2})$ 
    poly B(inv(A[0]));
    for (int x = 2; x/2 < n; x *= 2)
        B = 2*B-RSZ(conv(RSZ(A,x),conv(B,B)),x);
    return RSZ(B,n);
}

poly sqrt(const poly& A, int n) { //  $Q-(Q^2-A)/(2Q)$ 
    assert(A[0].v == 1); poly B{1};
    for (int x = 2; x/2 < n; x *= 2)
        B = inv(T(2))*RSZ(B+conv(RSZ(A,x),inv(B,x)),x);
    return RSZ(B,n);
}

// return {quotient, remainder}
pair<poly,poly> quoRem(const poly& f, const poly& g) {
    if (sz(f) < sz(g)) return {{},f};
    poly q = conv(inv(rev(g),sz(f)-sz(g)+1),rev(f));
    q = rev(RSZ(q,sz(f)-sz(g)+1));
    poly r = RSZ(f-conv(q,g),sz(g)-1); return {q,r};
}

poly log(poly A, int n) { assert(A[0].v == 1); //  $(\ln A)' = A'/\hookrightarrow A$ 
    A.rsz(n); return integ(RSZ(conv(dif(A),inv(A,n-1)),n-1)); }
poly exp(poly A, int n) { assert(A[0].v == 0);
    poly B{1}, IB{1}; // inverse of B
    for (int x = 1; x < n; x *= 2) {
        IB = 2*IB-RSZ(conv(B,conv(IB,IB)),x);
        poly Q = dif(RSZ(A,x)); Q += RSZ(conv(IB,dif(B)-conv(B,Q)
             $\hookrightarrow, 2*x-1$ );
        B = B+RSZ(conv(B,RSZ(A,2*x)-integ(Q)),2*x);
    }
    return RSZ(B,n);
}
```

6.3 Misc

```
LinearRecurrence.h
Description: Berlekamp-Massey. Computes linear recurrence C of order
N for sequence s of 2N terms.  $C[0] = 1$  and for all  $i \geq sz(C) - 1$ ,
 $\sum_{j=0}^{sz(C)-1} C[j]s[i-j] = 0$ .
Usage: LinRec L; L.init({0,1,1,2,3}); L.eval(5); L.eval(6); //
5, 8
Time: init  $\Rightarrow \mathcal{O}(N|C|)$ , eval  $\Rightarrow \mathcal{O}(|C|^2 \log p)$  or faster with FFT
"Poly.h" 39ea71, 29 lines

struct LinRec {
    poly s, C, rC;
    void BM() {
        int x = 0; T b = 1;
        poly B; B = C = {1}; // B is fail vector
        F0R(i,sz(s)) { // update C after adding a term of s
            ++x; int L = sz(C), M = i+3-L;
            T d = 0; F0R(j,L) d += C[j]*s[i-j]; //  $[D^{\wedge}i]C*s$ 
            if (d.v == 0) continue; //  $[D^{\wedge}i]C*s=0$ 
            poly _C = C; T coef = d*inv(b);
            C.rsz(max(L,M)); F0R(j,sz(B)) C[j+x] -= coef*B[j];
            if (L < M) B = _C, b = d, x = 0;
        }
    }
    void init(const poly& _s) {
        s = _s; BM();
        s = C; reverse(all(rC));
        C.erase(begin(C)); each(t,C) t *= -1;
    } // now  $s[i]=\sum_{j=0}^i\{sz(C)-1\}C[j]*s[i-j-1]$ 
    poly getPow(ll p) { // get  $x^p$  mod rC
        if (p == 0) return {1};
        poly r = getPow(p/2); r = (r*r)%rC;
```

```
        return p&1?(r*poly{0,1})%rC:r;
    }
    T dot(poly v) { // dot product with s
        T ans = 0; F0R(i,sz(v)) ans += v[i]*s[i];
        return ans; } // get p-th term of rec
    T eval(ll p) { assert(p >= 0); return dot(getPow(p)); }
};
```

```
Integrate.h
Description: Integration of a function over an interval using Simpson's rule,
exact for polynomials of degree up to 3. The error should be proportional to
 $dif^4$ , although in practice you will want to verify that the result is stable to
desired precision when epsilon changes.
Usage: quad({}(db x) { return x*x+3*x+1; }, 2, 3) // 14.833,
3ebaab, 6 lines

template<class F> db quad(F f, db a, db b) {
    const int n = 1000;
    db dif = (b-a)/2/n, tot = f(a)+f(b);
    FOR(i,1,2*n) tot += f(a+i*dif)*(i&1?4:2);
    return tot*dif/3;
}
```

```
IntegrateAdaptive.h
Description: Unused. Fast integration using adaptive Simpson's rule, exact
for polynomials of degree up to 5.
Usage: db z, y;
db h(db x) { return x*x + y*y + z*z <= 1; }
db g(db y) { ::y = y; return quad(h, -1, 1); }
db f(db z) { ::z = z; return quad(g, -1, 1); }
db sphereVol = quad(f,-1,1), pi = sphereVol*3/4; 3b316e, 10 lines

template<class F> db simpson(F f, db a, db b) {
    db c = (a+b)/2; return (f(a)+4*f(c)+f(b))*(b-a)/6; }
template<class F> db rec(F& f, db a, db b, db eps, db S) {
    db c = (a+b)/2;
    db S1 = simpson(f,a,c), S2 = simpson(f,c,b), T = S1+S2;
    if (abs(T-S)<=15*eps || b-a<1e-10) return T+(T-S)/15;
    return rec(f,a,c,eps/2,S1)+rec(f,c,b,eps/2,S2);
}
template<class F> db quad(F f, db a, db b, db eps = 1e-8) {
    return rec(f,a,b,eps,simpson(f,a,b)); }
```

```
Simplex.h
Description: Solves a general linear maximization problem: maximize  $c^T x$ 
subject to  $Ax \leq b, x \geq 0$ . Returns -inf if there is no solution, inf if there
are arbitrarily good solutions, or the maximum value of  $c^T x$  otherwise. The
input vector is set to an optimal  $x$  (or in the unbounded case, an arbitrary
solution fulfilling the constraints). Numerical stability is not guaranteed. For
better performance, define variables such that  $x = 0$  is viable.
Usage: vvd A{{1,-1}, {-1,1}, {-1,-2}};
vd b{1,1,-4}, c{-1,-1}, x;
T val = LPSolver(A, b, c).solve(x);
Time:  $\mathcal{O}(NM \cdot \#pivots)$ , where a pivot may be e.g. an edge relaxation.
 $\mathcal{O}(2^N)$  in the general case. c99f9c, 67 lines

using T = db; // double probably suffices
using vd = V<T>; using vvd = V<vd>;
const T eps = 1e-8, inf = 1./0;

#define ltj(X) if (s==-1 || mp(X[j],N[j])<mp(X[s],N[s])) s=j
struct LPSolver {
    int m, n; // # m = constraints, # n = variables
    vi N, B; // N[j] = non-basic variable (j-th column), = 0
    vvd D; // B[j] = basic variable (j-th row)
    LPSolver(const vvd& A, const vd& b, const vd& c) :
        m(sz(b)), n(sz(c)), N(n+1), B(m), D(m+2, vd(n+2)) {
        F0R(i,m) F0R(j,n) D[i][j] = A[i][j];
        F0R(i,m) B[i] = n+i, D[i][n] = -1, D[i][n+1] = b[i];
        // B[i]: basic variable for each constraint
```

```
        // D[i][n]: artificial variable for testing feasibility
        F0R(j,n) N[j] = j, D[m][j] = -c[j];
        // D[m] stores negation of objective,
        // which we want to minimize
        N[n] = -1; D[m+1][n] = 1; // to find initial feasible
    } // solution, minimize artificial variable
    void pivot(int r, int s) { // swap B[r] (row)
        T inv = 1/D[r][s]; // with N[r] (column)
        F0R(i,m+2) if (i != r && abs(D[i][s]) > eps) {
            T binv = D[i][s]*inv;
            F0R(j,n+2) if (j != s) D[i][j] -= D[r][j]*binv;
            D[i][s] = -binv;
        }
        D[r][s] = 1; F0R(j,n+2) D[r][j] *= inv; // scale r-th row
        swap(B[r],N[s]);
    }
    bool simplex(int phase) {
        int x = m+phase-1;
        while (1) { // if phase=1, ignore artificial variable
            int s = -1; F0R(j,n+1) if (N[j] != -phase) ltj(D[x]);
            // find most negative col for nonbasic (NB) variable
            if (D[x][s] >= -eps) return 1;
            // can't get better sol by increasing NB variable
            int r = -1;
            F0R(i,m) {
                if (D[i][s] <= eps) continue;
                if (r == -1 || mp(D[i][n+1] / D[i][s], B[i])
                    < mp(D[r][n+1] / D[r][s], B[r])) r = i;
                // find smallest positive ratio
            } // -> max increase in NB variable
            if (r == -1) return 0; // objective is unbounded
            pivot(r,s);
        }
    }
    T solve(vd& x) { // 1. check if x=0 feasible
        int r = 0; FOR(i,1,m) if (D[i][n+1] < D[r][n+1]) r = i;
        if (D[r][n+1] < -eps) { // if not, find feasible start
            pivot(r,n); // make artificial variable basic
            assert(simplex(2)); // I think this will always be true??
            if (D[m+1][n+1] < -eps) return -inf;
            // D[m+1][n+1] is max possible value of the negation of
            // artificial variable, optimal value should be zero
            // if exists feasible solution
            F0R(i,m) if (B[i] == -1) { // artificial var basic
                int s = 0; FOR(j,1,n+1) ltj(D[i]); // -> nonbasic
                pivot(i,s);
            }
        }
        bool ok = simplex(1); x = vd(n);
        F0R(i,m) if (B[i] < n) x[B[i]] = D[i][n+1];
        return ok ? D[m][n+1] : inf;
    }
};
```

Graphs (7)

Erdos-Gallai: $d_1 \geq \dots \geq d_n$ can be degree sequence of simple graph on n vertices iff their sum is even and $\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k), \forall 1 \leq k \leq n$.

7.1 Basics

```
DirectedCycle.h
Description: stack c746ee, 28 lines

struct DirectedCycle {
    int N;
```

```
vector<bool> inStk, vis;
vector<int> stk, cyc;
vector<vector<int>>> adj;
DirectedCycle(int _N) {
    N = _N;
    adj.resize(N);
    inStk.resize(N), vis.resize(N);
}
void ae(int u, int v) { adj[u].push_back(v); }
vector<int> gen() {
    for (int u = 0; u < N; u++) {
        if (!vis[u] && empty(cyc)) dfs(u);
    }
    return cyc;
}
void dfs(int u) {
    stk.push_back(u);
    inStk[u] = vis[u] = 1;
    for (auto& v: adj[u]) {
        if (inStk[v]) cyc = {ranges::find(stk, v), end(stk)};
        else if (!vis[v]) dfs(v);
        if (size(cyc)) return;
    }
    stk.pop_back(); inStk[u] = 0;
}
};
```

GridBFS.h

Description: BFS through grid with fixed xdir and ydir arrays

4931b7, 20 lines

```
const int dx[4] = {1, 0, -1, 0};
const int dy[4] = {0, 1, 0, -1};

bool valid(int x, int y) {
    return !(x < 0 || x >= n || y < 0 || y >= m);
}

void bfs(pair<int, int> src) {
    dist[src.first][src.second] = 0;
    queue<pair<int, int>>> todo; todo.push(src);
    while (!todo.empty()) {
        auto [uX, uY] = todo.front(); todo.pop();
        for (int i = 0; i < 4; i++) {
            int vX = uX+dx[i], vY = uY+dy[i];
            if (!valid(vX, vY) || dist[vX][vY] != -1) continue;
            dist[vX][vY] = dist[uX][uY]+1;
            todo.push({vX, vY});
        }
    }
}
};
```

Bipartite.h

Description: Checks if an undirected graph is bipartite. Colors nodes with 1/2. If not bipartite, returns false.

Time: $\mathcal{O}(N + M)$

4f7998, 28 lines

```
struct Bipartite {
    int N;
    vector<int> color;
    vector<vector<int>>> adj;
    Bipartite(int _N) {
        N = _N;
        adj.resize(N); color.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    bool check() {
```

```
    queue<int> todo;
    for (int src = 0; src < N; src++) {
        if (color[src]) continue;
        color[src] = 1; todo.push(src);
        while (!todo.empty()) {
            int u = todo.front(); todo.pop();
            for (auto &v: adj[u]) {
                if (color[v] == color[u]) return false;
                if (!color[v]) color[v] = 3-color[u], todo.push(v);
            }
        }
        return true;
    }
};
```

TopoSort.h

Description: sorts vertices such that if there exists an edge u->v, then u goes before v

7b10d8, 20 lines

```
struct TopoSort {
    int N;
    vector<int> in, res;
    vector<vector<int>>> adj;
    TopoSort(int _N) {
        N = _N; in.resize(N); adj.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v), ++in[v];
    }
    bool sort() {
        queue<int> todo;
        for (int u = 0; u < N; ++u) if (!in[u]) todo.push(u);
        while (size(todo)) {
            int u = todo.front(); todo.pop(); res.push_back(u);
            for (auto &v: adj[u]) if (--in[v]) todo.push(v);
        }
        return (int)size(res) == N;
    }
};
```

Dijkstra.h

Description: shortest path

Time: $\mathcal{O}(N \log N + M)$

8cfac5, 27 lines

```
template<class T, bool directed> struct Dijkstra {
    int N;
    vector<T> dist;
    vector<vector<pair<int, T>>> adj;
    Dijkstra(int _N) {
        N = _N;
        adj.resize(N);
    }
    void ae(int u, int v, T w) {
        adj[u].push_back({v, w});
        if (!directed) adj[v].push_back({u, w});
    }
    void gen(int st) {
        dist.assign(N, numeric_limits<T>::max());
        priority_queue<pair<T, int>>> PQ;
        auto relax = [&](int v, T d) {
            if (dist[v] <= d) return;
            dist[v] = d;
            PQ.push({-dist[v], v});
        }; relax(st, 0);
        while (!PQ.empty()) {
            auto [d, u] = PQ.top(); PQ.pop(); d = -d;
            if (dist[u] < d) continue;
            for (auto& [v, w]: adj[u]) relax(v, w+d);
        }
    }
};
```

NegativeCycle.h

Description: use Bellman-Ford (make sure no underflow)

e3180d, 22 lines

```
vector<int> NegativeCycle(int N, vector<array<int, 3>> &ed) {
    vector<int64_t> dist(N);
    vector<int> p(N);
    int x = -1;
    for (int i = 0; i < N; i++) {
        x = -1;
        for (auto &[u, v, w] : ed) {
            if (dist[v] > dist[u]+w) {
                dist[v] = dist[u]+w;
                p[v] = u, x = v;
            }
        }
        if (x == -1) return {};
    }

    for (int i = 0; i < N; i++) x = p[x]; // enter cycle
    vector<int> cyc{x};
    while (p[cyc.back()] != x) cyc.push_back(p[cyc.back()]);
    cyc.push_back(x); // complete cycle
    ranges::reverse(cyc);
    return cyc;
}
```

BellmanFord.h

Description: Shortest Path w/ negative edge weights

c2a89c, 33 lines

```
struct BellmanFord {
    int N;
    vector<int64_t> dist;
    vector<vector<int>>> adj;
    vector<array<int, 3>> ed;
    BellmanFord(int _N) {
        N = _N;
        adj.resize(N);
        dist.resize(N, INF);
    }
    void ae(int u, int v, int w) {
        adj[u].push_back(v);
        ed.push_back({u, v, w});
    }
    void genBad(int u) {
        if (dist[u] == -INF) return;
        dist[u] = -INF;
        for (auto &v : adj[u]) genBad(v);
    }
    void genDist(int src) {
        dist[src] = 0;
        for (int i = 0; i < N; i++) {
            for (auto &[u, v, w] : ed) {
                if (dist[u] < INF) {
                    dist[v] = min(dist[v], dist[u]+w);
                }
            }
        }
        for (auto &[u, v, w] : ed) {
            if (dist[u] < INF && dist[v] > dist[u]+w) genBad(v);
        }
    }
};
```

7.2 Trees

LCAjump.h

Description: Calculates least common ancestor in tree with verts $0 \dots N-1$ and root R using binary jumping.

Memory: $\mathcal{O}(N \log N)$

Time: $\mathcal{O}(N \log N)$ build, $\mathcal{O}(\log N)$ query

6b0ee9, 28 lines

```
struct LCA {
    int N; V<vi> par, adj; vi depth;
    void init(int _N) { N = _N;
        int d = 1; while ((1<<d) < N) ++d;
        par.assign(d,vi(N)); adj.rsz(N); depth.rsz(N);
    }
    void ae(int x, int y) { adj[x].pb(y), adj[y].pb(x); }
    void gen(int R = 0) { par[0][R] = R; dfs(R); }
    void dfs(int x = 0) {
        FOR(i,1,sz(par)) par[i][x] = par[i-1][par[i-1][x]];
        each(y,adj[x]) if (y != par[0][x])
            depth[y] = depth[par[0][y]=x]+1, dfs(y);
    }
    int jmp(int x, int d) {
        FOR(i,sz(par)) if ((d>>i)&1) x = par[i][x];
        return x; }
    int lca(int x, int y) {
        if (depth[x] < depth[y]) swap(x,y);
        x = jmp(x,depth[x]-depth[y]); if (x == y) return x;
        R0F(i,sz(par)) {
            int X = par[i][x], Y = par[i][y];
            if (X != Y) x = X, y = Y;
        }
        return par[0][x];
    }
    int dist(int x, int y) { // # edges on path
        return depth[x]+depth[y]-2*depth[lca(x,y)]; }
};
```

LCArm q.h

Description: Euler Tour LCA. Compress takes a subset S of nodes and computes the minimal subtree that contains all the nodes pairwise LCAs and compressing edges. Returns a list of (par, orig_index) representing a tree rooted at 0. The root points to itself.

Time: $\mathcal{O}(N \log N)$ build, $\mathcal{O}(1)$ LCA, $\mathcal{O}(|S| \log |S|)$ compress

"RMQ.h" e5a035, 28 lines

```
struct LCA {
    int N; V<vi> adj;
    vi depth, pos, par, rev; // rev is for compress
    vpi tmp; RMQ<pi> r;
    void init(int _N) { N = _N; adj.rsz(N);
        depth = pos = par = rev = vi(N); }
    void ae(int x, int y) { adj[x].pb(y), adj[y].pb(x); }
    void dfs(int x) {
        pos[x] = sz(tmp); tmp.eb(depth[x],x);
        each(y,adj[x]) if (y != par[x]) {
            depth[y] = depth[par[y]=x]+1, dfs(y);
            tmp.eb(depth[x],x); }
    }
    void gen(int R = 0) { par[R] = R; dfs(R); r.init(tmp); }
    int lca(int u, int v){
        u = pos[u], v = pos[v]; if (u > v) swap(u,v);
        return r.query(u,v).s; }
    int dist(int u, int v) {
        return depth[u]+depth[v]-2*depth[lca(u,v)]; }
    vpi compress(vi S) {
        auto cmp = [&](int a, int b) { return pos[a] < pos[b]; };
        sort(all(S),cmp); R0F(i,sz(S)-1) S.pb(lca(S[i],S[i+1]));
        sort(all(S),cmp); S.erase(unique(all(S)),end(S));
        vpi ret{{0,S[0]}}; F0R(i,sz(S)) rev[S[i]] = i;
        FOR(i,1,sz(S)) ret.eb(rev[lca(S[i-1],S[i])],S[i]);
    }
};
```

```
        return ret;
    }
};
```

HLD.h

Description: Heavy-Light Decomposition, add val to verts and query sum in path/subtree.

Time: any tree path is split into $\mathcal{O}(\log N)$ parts

"LazySeq.h" 1802e2, 48 lines

```
template<int SZ, bool VALS_IN_EDGES> struct HLD {
    int N; vi adj[SZ];
    int par[SZ], root[SZ], depth[SZ], sz[SZ], ti;
    int pos[SZ]; vi rpos; // rpos not used but could be useful
    void ae(int x, int y) { adj[x].pb(y), adj[y].pb(x); }
    void dfsSz(int x) {
        sz[x] = 1;
        each(y,adj[x]) {
            par[y] = x; depth[y] = depth[x]+1;
            adj[y].erase(find(all(adj[y]),x));
            dfsSz(y); sz[x] += sz[y];
            if (sz[y] > sz[adj[x][0]]) swap(y,adj[x][0]);
        }
    }
    void dfsHld(int x) {
        pos[x] = ti++; rpos.pb(x);
        each(y,adj[x]) {
            root[y] = (y == adj[x][0] ? root[x] : y);
            dfsHld(y); }
    }
    void init(int _N, int R = 0) { N = _N;
        par[R] = depth[R] = ti = 0; dfsSz(R);
        root[R] = R; dfsHld(R);
    }
    int lca(int x, int y) {
        for (; root[x] != root[y]; y = par[root[y]])
            if (depth[root[x]] > depth[root[y]]) swap(x,y);
        return depth[x] < depth[y] ? x : y;
    }
    LazySeq<ll,SZ> tree; // segtree for sum
    template <class BinaryOp>
    void processPath(int x, int y, BinaryOp op) {
        for (; root[x] != root[y]; y = par[root[y]]) {
            if (depth[root[x]] > depth[root[y]]) swap(x,y);
            op(pos[root[y]],pos[y]); }
        if (depth[x] > depth[y]) swap(x,y);
        op(pos[x]+VALS_IN_EDGES,pos[y]);
    }
    void modifyPath(int x, int y, int v) {
        processPath(x,y,[this,&v])(int l, int r) {
            tree.upd(l,r,v); }; }
    ll queryPath(int x, int y) {
        ll res = 0; processPath(x,y,[this,&res])(int l, int r) {
            res += tree.query(l,r); };
        return res; }
    void modifySubtree(int x, int v) {
        tree.upd(pos[x]+VALS_IN_EDGES,pos[x]+sz[x]-1,v); }
};
```

Centroid.h

Description: The centroid of a tree of size N is a vertex such that after removing it, all resulting subtrees have size at most $\frac{N}{2}$. Supports updates in the form “add 1 to all verts v such that $dist(x,v) \leq y$.”

Memory: $\mathcal{O}(N \log N)$

Time: $\mathcal{O}(N \log N)$ build, $\mathcal{O}(\log N)$ update and query

907e21, 54 lines

```
void ad(vi& a, int b) { ckmin(b,sz(a)-1); if (b>=0) a[b]++; }
void prop(vi& a) { R0F(i,sz(a)-1) a[i] += a[i+1]; }
template<int SZ> struct Centroid {
    vi adj[SZ]; void ae(int a,int b){adj[a].pb(b),adj[b].pb(a); }
```

```
bool done[SZ]; // processed as centroid yet
int N,sub[SZ],cen[SZ],lev[SZ]; // subtree size, centroid anc
int dist[32-__builtin_clz(SZ)][SZ]; // dists to all ancs
vi stor[SZ], STOR[SZ];
void dfs(int x, int p) { sub[x] = 1;
    each(y,adj[x]) if (!done[y] && y != p)
        dfs(y,x), sub[x] += sub[y];
}
int centroid(int x) {
    dfs(x,-1);
    for (int sz = sub[x];; ) {
        pi mx = {0,0};
        each(y,adj[x]) if (!done[y] && sub[y] < sub[x])
            ckmax(mx,{sub[y],y});
        if (mx.f*2 <= sz) return x;
        x = mx.s;
    }
}
void genDist(int x, int p, int lev) {
    dist[lev][x] = dist[lev][p]+1;
    each(y,adj[x]) if (!done[y] && y != p) genDist(y,x,lev); }
void gen(int CEN, int _x) { // CEN = centroid above x
    int x = centroid(_x); done[x] = 1; cen[x] = CEN;
    sub[x] = sub[_x]; lev[x] = (CEN == -1 ? 0 : lev[CEN]+1);
    dist[lev[x]][x] = 0;
    stor[x].rsz(sub[x]),STOR[x].rsz(sub[x]+1);
    each(y,adj[x]) if (!done[y]) genDist(y,x,lev[x]);
    each(y,adj[x]) if (!done[y]) gen(x,y);
}
void init(int _N) { N = _N; FOR(i,1,N+1) done[i] = 0;
    gen(-1,1); } // start at vert 1
void upd(int x, int y) {
    int cur = x, pre = -1;
    R0F(i,lev[x]+1) {
        ad(stor[cur],y-dist[i][x]);
        if (pre != -1) ad(STOR[pre],y-dist[i][x]);
        if (i > 0) pre = cur, cur = cen[cur];
    }
}
} // call propAll() after all updates
void propAll() { FOR(i,1,N+1) prop(stor[i]), prop(STOR[i]); }
int query(int x) { // get value at vertex x
    int cur = x, pre = -1, ans = 0;
    R0F(i,lev[x]+1) { // if pre != -1, subtract those from
        ans += stor[cur][dist[i][x]]; // same subtree
        if (pre != -1) ans -= STOR[pre][dist[i][x]];
        if (i > 0) pre = cur, cur = cen[cur];
    }
    return ans;
}
```

7.2.1 SqrtDecompton

HLD generally suffices. If not, here are some common strategies:

- Rebuild the tree after every $\sqrt{N}$ queries.
- Consider vertices with $>$ or $< \sqrt{N}$ degree separately.
- For subtree updates, note that there are $\mathcal{O}(\sqrt{N})$ distinct sizes among child subtrees of any node.

Block Tree: Use a DFS to split edges into contiguous groups of size $\sqrt{N}$ to $2\sqrt{N}$.

Mo’s Algorithm for Tree Paths: Maintain an array of vertices where each one appears twice, once when a DFS enters the vertex (st) and one when the DFS exists (en). For a tree path $u \leftrightarrow v$ such that $st[u] < st[v]$,

- If u is an ancestor of v , query $[st[u], st[v]]$.
- Otherwise, query $[en[u], st[v]]$ and consider $LCA(u, v)$ separately.

Solutions with worse complexities can be faster if you optimize the operations that are performed most frequently. Use arrays instead of vectors whenever possible. Iterating over an array in order is faster than iterating through the same array in some other order (ex. one given by a random permutation) or DFSing on a tree of the same size. Also, the difference between $\sqrt{N}$ and the optimal block (or buffer) size can be quite large. Try up to 5x smaller or larger (at least).

7.3 DFS Algorithms

EulerPath.h
Description: Eulerian path starting at src if it exists, visits all edges exactly once. Works for both directed and undirected. Returns vector of {vertex,label of edge to vertex}. Second element of first pair is always -1.
Time: $\mathcal{O}(N + M)$

```
template<bool directed> struct Euler {
    int N; V<vpi> adj; V<vpi::iterator> its; vb used;
    void init(int _N) { N = _N; adj.rsz(N); }
    void ae(int a, int b) {
        int M = sz(used); used.pb(0);
        adj[a].eb(b,M); if (!directed) adj[b].eb(a,M); }
    vpi solve(int src = 0) {
        its.rsz(N); F0R(i,N) its[i] = begin(adj[i]);
        vpi ans, s{{src,-1}}; // {{vert,prev vert},edge label}
        int lst = -1; // ans generated in reverse order
        while (sz(s)) {
            int x = s.bk.f; auto& it=its[x], en=end(adj[x]);
            while (it != en && used[it->s]) ++it;
            if (it == en) { // no more edges out of vertex
                if (lst != -1 && lst != x) return {};
                // not a path, no tour exists
                ans.pb(s.bk); s.pop_back(); if (sz(s)) lst=s.bk.f;
            } else s.pb(*it), used[it->s] = 1;
        } // must use all edges
        if (sz(ans) != sz(used)+1) return {};
        reverse(all(ans)); return ans;
    }
};
```

SCCT.h
Description: Tarjan’s, DFS once to generate strongly connected components in topological order. a, b in same component if both $a \rightarrow b$ and $b \rightarrow a$ exist. Uses less memory than Kosaraju b/c doesn’t store reverse edges.
Time: $\mathcal{O}(N + M)$

```
struct SCC {
    int N, ti = 0; V<vi> adj;
    vi disc, comp, stk, comps;
    void init(int _N){ N = _N, adj.rsz(N);
        disc.rsz(N), comp.rsz(N,-1); }
    void ae(int x, int y) { adj[x].pb(y); }
    int dfs(int x) {
        int low = disc[x] = ++ti; stk.pb(x);
        each(y,adj[x]) if (comp[y] == -1) // comp[y] == -1,
```

```
        ckmin(low,disc[y]?dfs(y)); // disc[y] != 0 -> in stack
    if (low == disc[x]) { // make new SCC
        // pop off stack until you find x
        comps.pb(x); for (int y = -1; y != x;)
            comp[y = stk.bk] = x, stk.pop_back();
        }
    return low;
}
void gen() {
    F0R(i,N) if (!disc[i]) dfs(i);
    reverse(all(comps));
}
};
```

TwoSAT.h
Description: Calculates a valid assignment to boolean variables a, b, c,... to a 2-SAT problem, so that an expression of the type $(a|||b)&&(a|||c)&&(d|||!b)&&...$ becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions (~x).
Usage: TwoSat ts;

ts.either(0, ~3); // Var 0 is true or var 3 is false
ts.setVal(2); // Var 2 is true
ts.atMostOne({0,~1,2}); // <= 1 of vars 0, ~1 and 2 are true
ts.solve(N); // Returns true iff it is solvable
ts.ans[0..N-1] holds the assigned values to the vars
"SCC.h" ffof3d, 31 lines

```
struct TwoSAT {
    int N = 0; vpi edges;
    void init(int _N) { N = _N; }
    int addVar() { return N++; }
    void either(int x, int y) {
        x = max(2*x,-1-2*x), y = max(2*y,-1-2*y);
        edges.eb(x,y); }
    void implies(int x, int y) { either(~x,y); }
    void must(int x) { either(x,x); }
    void atMostOne(const vi& li) {
        if (sz(li) <= 1) return;
        int cur = ~li[0];
        FOR(i,2,sz(li)) {
            int next = addVar();
            either(cur,~li[i]); either(cur,next);
            either(~li[i],next); cur = ~next;
        }
        either(cur,~li[1]);
    }
    vb solve() {
        SCC S; S.init(2*N);
        each(e,edges) S.ae(e.f^1,e.s), S.ae(e.s^1,e.f);
        S.gen(); reverse(all(S.comps)); // reverse topo order
        for (int i = 0; i < 2*N; i += 2)
            if (S.comp[i] == S.comp[i^1]) return {};
        vi tmp(2*N); each(i,S.comps) if (!tmp[i])
            tmp[i] = 1, tmp[S.comp[i^1]] = -1;
        vb ans(N); F0R(i,N) ans[i] = tmp[S.comp[2*i]] == 1;
        return ans;
    }
};
```

BCC.h
Description: Biconnected components of edges. Removing any vertex in BCC doesn’t disconnect it. To get block-cut tree, create a bipartite graph with the original vertices on the left and a vertex for each BCC on the right. Draw edge $u \leftrightarrow v$ if u is contained within the BCC for v . Self-loops are not included in any BCC while BCCS of size 1 represent bridges.
Time: $\mathcal{O}(N + M)$

```
struct BCC {
    V<vpi> adj; vpi ed;
    V<vi> edgeSets, vertSets; // edges for each bcc
```

```
int N, ti = 0; vi disc, stk;
void init(int _N) { N = _N; disc.rsz(N), adj.rsz(N); }
void ae(int x, int y) {
    adj[x].eb(y,sz(ed)), adj[y].eb(x,sz(ed)), ed.eb(x,y); }
int dfs(int x, int p = -1) { // return lowest disc
    int low = disc[x] = ++ti;
    each(e,adj[x]) if (e.s != p) {
        if (!disc[e.f]) {
            stk.pb(e.s); // disc[x] < LOW -> bridge
            int LOW = dfs(e.f,e.s); ckmin(low,LOW);
            if (disc[x] <= LOW) { // get edges in bcc
                edgeSets.eb(); vi& tmp = edgeSets.bk; // new bcc
                for (int y = -1; y != e.s; )
                    tmp.pb(y = stk.bk), stk.pop_back();
            }
        } else if (disc[e.f] < disc[x]) // back-edge
            ckmin(low,disc[e.f]), stk.pb(e.s);
    }
    return low;
}
void gen() {
    F0R(i,N) if (!disc[i]) dfs(i);
    vb in(N);
    each(c,edgeSets) { // edges contained within each BCC
        vertSets.eb(); // so you can easily create block cut tree
        auto ad = [&](int x) {
            if (!in[x]) in[x] = 1, vertSets.bk.pb(x); };
        each(e,c) ad(ed[e].f), ad(ed[e].s);
        each(e,c) in[ed[e].f] = in[ed[e].s] = 0;
    }
};
```

MaximalCliques.h
Description: Used only once. Finds all maximal cliques.
Time: $\mathcal{O}(3^{N/3})$

```
using B = bitset<128>; B adj[128];
int N;
// possibly in clique, not in clique, in clique
void cliques(B P = ~B(), B X={}, B R={}) {
    if (!P.any()) {
        if (!X.any()) // do smth with R
            return;
    }
    int q = (P|X)._Find_first();
    // clique must contain q or non-neighbor of q
    B cand = P&~adj[q];
    F0R(i,N) if (cand[i]) {
        R[i] = 1; cliques(P&adj[i],X&adj[i],R);
        R[i] = P[i] = 0; X[i] = 1;
    }
}
```

7.4 DSU

DSU.h
Description: Disjoint Set Union with path compression and union by size. Add edges and test connectivity. Use for Kruskal’s or Boruvka’s minimum spanning tree.
Time: $\mathcal{O}(\alpha(N))$

```
struct DSU {
    vector<int> e;
    DSU(int N) { e = vector<int>(N, -1); }
    int get(int u) { return e[u] < 0 ? u : e[u] = get(e[u]); }
    bool sameSet(int u, int v) { return get(u) == get(v); }
    int size(int u) { return -e[get(u)]; }
    bool unite(int u, int v) { // union by size
```

```

    u = get(u), v = get(v); if (u == v) return false;
    if (e[u] > e[v]) swap(u, v);
    e[u] += e[v]; e[v] = u; return true;
}
};

```

Kruskal.h

Description: Computes the weight of a Minimum Spanning Tree (MST) using Kruskal's algorithm. Requires DSU.

Time: $\mathcal{O}(M \log M)$, where M = number of edges

```

"DSU.h" f6c35c, 8 lines
template<class T> T Kruskal(int N, vector<array<T, 3>> eg) {
    ranges::sort(eg);
    T ans = 0; DSU D(N);
    for (auto &[w, u, v] : eg) {
        if (D.unite(u, v)) ans += w;
    }
    return ans;
};

```

7.5 Flows

Konig's Theorem: In a bipartite graph, max matching = min vertex cover.

Dilworth's Theorem: For any partially ordered set, the sizes of the max antichain and of the min chain decomposition are equal. Equivalent to Konig's theorem on the bipartite graph (U, V, E) where $U = V = S$ and (u, v) is an edge when $u < v$. Those vertices outside the min vertex cover in both U and V form a max antichain.

Dinic.h

Description: Fast flow. After computing flow, edges $\{u, v\}$ such that $lev[u] \neq -1$, $lev[v] = -1$ are part of min cut. Use reset and rcap for Gomory-Hu.

Time: $\mathcal{O}(N^2 M)$ flow, $\mathcal{O}(M\sqrt{N})$ bipartite matching

```

b7b370, 38 lines
struct Dinic {
    using F = ll; // flow type
    struct Edge { int to; F flo, cap; };
    int N; V<Edge> eds; V<vi> adj;
    void init(int _N) { N = _N; adj.rsz(N), cur.rsz(N); }
    void ae(int u, int v, F cap, F rcap = 0) { assert(min(cap,
        ↪rcap) >= 0);
        adj[u].pb(sz(eds)); eds.pb({v, 0, cap});
        adj[v].pb(sz(eds)); eds.pb({u, 0, rcap});
    }
    vi lev; V<vi::iterator> cur;
    bool bfs(int s, int t) { // level = shortest distance from
        ↪source
        lev = vi(N, -1); F0R(i, N) cur[i] = begin(adj[i]);
        queue<int> q({s}); lev[s] = 0;
        while (sz(q)) { int u = q.ft; q.pop();
            each(e, adj[u]) { const Edge& E = eds[e];
                int v = E.to; if (lev[v] < 0 && E.flo < E.cap)
                    q.push(v), lev[v] = lev[u] + 1;
            }
        }
        return lev[t] >= 0;
    }
    F dfs(int v, int t, F flo) {
        if (v == t) return flo;
        for (; cur[v] != end(adj[v]); cur[v]++) {
            Edge& E = eds[*cur[v]];
            if (lev[E.to] != lev[v] + 1 || E.flo == E.cap) continue;

```

Kruskal Dinic GomoryHu MCMF Hungarian

```

    F df = dfs(E.to, t, min(flo, E.cap - E.flo));
    if (df) { E.flo += df; eds[*cur[v]^1].flo -= df;
        return df; } // saturated >= 1 one edge
    }
    return 0;
}
}
F maxFlow(int s, int t) {
    F tot = 0; while (bfs(s, t)) while (F df =
        dfs(s, t, numeric_limits<F>::max())) tot += df;
    return tot;
}
};

```

GomoryHu.h

Description: Returns edges of Gomory-Hu tree (second element is weight). Max flow between pair of vertices of undirected graph is given by min edge weight along tree path. Uses the fact that for any i, j, k , $\lambda_{ik} \geq \min(\lambda_{ij}, \lambda_{jk})$, where λ_{ij} denotes the flow between i and j .

Time: $N - 1$ calls to Dinic

```

"Dinic.h" 0d712e, 16 lines
template<class F> V<pair<pi, F>> gomoryHu(int N,
    const V<pair<pi, F>>& ed) {
    vi par(N); Dinic<F> D; D.init(N);
    vpi ed_locs; each(t, ed) ed_locs.pb(D.ae(t.f.f, t.f.s, t.s, t.s));
    V<pair<pi, F>> ans;
    FOR(i, 1, N) {
        each(p, ed_locs) { // reset capacities
            auto& e = D.adj.at(p.f).at(p.s);
            auto& e_rev = D.adj.at(e.to).at(e.rev);
            e.cap = e_rev.cap = (e.cap + e_rev.cap) / 2;
        }
        ans.pb({{i, par[i]}, D.maxFlow(i, par[i])});
        FOR(j, i + 1, N) if (par[j] == par[i] && D.lev[j]) par[j] = i;
    }
    return ans;
}

```

MCMF.h

Description: Minimum-cost maximum flow, assumes no negative cycles. It is possible to choose negative edge costs such that the first run of Dijkstra is slow, but this hasn't been an issue in the past. Edge weights ≥ 0 for every subsequent run. To get flow through original edges, assign ID's during ae.

Time: Ignoring first run of Dijkstra, $\mathcal{O}(FM \log M)$ if caps are integers and F is max flow.

```

072c1b, 40 lines
struct MCMF {
    using F = ll; using C = ll; // flow type, cost type
    struct Edge { int to; F flo, cap; C cost; };
    int N; V<C> p, dist; vi pre; V<Edge> eds; V<vi> adj;
    void init(int _N) { N = _N;
        p.rsz(N), dist.rsz(N), pre.rsz(N), adj.rsz(N); }
    void ae(int u, int v, F cap, C cost) { assert(cap >= 0);
        adj[u].pb(sz(eds)); eds.pb({v, 0, cap, cost});
        adj[v].pb(sz(eds)); eds.pb({u, 0, 0, -cost});
    } // use asserts, don't try smth dumb
    bool path(int s, int t) { // find lowest cost path to send
        ↪flow through
        const C inf = numeric_limits<C>::max(); F0R(i, N) dist[i] =
            ↪inf;
        using T = pair<C, int>; priority_queue<T, vector<T>, greater<T>
            ↪>> todo;
        todo.push({dist[s] = 0, s});
        while (sz(todo)) { // Dijkstra
            T x = todo.top(); todo.pop(); if (x.f > dist[x.s])
                ↪continue;
            each(e, adj[x.s]) { const Edge& E = eds[e]; // all weights
                ↪should be non-negative
                if (E.flo < E.cap && ckmin(dist[E.to], x.f + E.cost + p[x.s]
                    ↪-p[E.to]))

```

```

        pre[E.to] = e, todo.push({dist[E.to], E.to});
    }
} // if costs are doubles, add some EPS so you
// don't traverse ~0-weight cycle repeatedly
return dist[t] != inf; // return flow
}
pair<F, C> calc(int s, int t) { assert(s != t);
    F0R(_, N) F0R(e, sz(eds)) { const Edge& E = eds[e]; //
        ↪Bellman-Ford
        if (E.cap) ckmin(p[E.to], p[eds[e^1].to] + E.cost); }
    F totFlow = 0; C totCost = 0;
    while (path(s, t)) { // p -> potentials for Dijkstra
        F0R(i, N) p[i] += dist[i]; // don't matter for unreachable
            ↪nodes
        F df = numeric_limits<F>::max();
        for (int x = t; x != s; x = eds[pre[x]^1].to) {
            const Edge& E = eds[pre[x]]; ckmin(df, E.cap - E.flo); }
        totFlow += df; totCost += (p[t] - p[s]) * df;
        for (int x = t; x != s; x = eds[pre[x]^1].to)
            eds[pre[x]].flo += df, eds[pre[x]^1].flo -= df;
    } // get max flow you can send along path
    return {totFlow, totCost};
}
};

```

7.6 Matching

Hungarian.h

Description: Given J jobs and W workers ($J \leq W$), computes the minimum cost to assign each prefix of jobs to distinct workers.

@tparam T a type large enough to represent integers on the order of $J * \max(\text{---C})$ @param C a matrix of dimensions $J \times W$ such that $C[j][w]$ = cost to assign j -th job to w -th worker (possibly negative)

@return a vector of length J , with the j -th entry equaling the minimum cost to assign the first $(j+1)$ jobs to distinct workers

Time: $\mathcal{O}(J^2 W)$

```

f382f7, 36 lines
template <class T> vector<T> hungarian(const vector<vector<T>>
    ↪&C) {
    const int J = (int)size(C), W = (int)size(C[0]);
    assert(J <= W);
    vector<int> job(W + 1, -1);
    vector<T> ys(J), yt(W + 1);
    vector<T> answers;
    const T inf = numeric_limits<T>::max();
    for (int j_cur = 0; j_cur < J; ++j_cur) {
        int w_cur = W;
        job[w_cur] = j_cur;
        vector<T> min_to(W + 1, inf);
        vector<int> prv(W + 1, -1);
        vector<bool> in_Z(W + 1);
        while (job[w_cur] != -1) {
            in_Z[w_cur] = true;
            const int j = job[w_cur];
            T delta = inf;
            int w_next;
            for (int w = 0; w < W; ++w) {
                if (!in_Z[w]) {
                    if (ckmin(min_to[w], C[j][w] - ys[j] - yt[w]))
                        prv[w] = w_cur;
                    if (ckmin(delta, min_to[w])) w_next = w;
                }
            }
            for (int w = 0; w <= W; ++w) {
                if (in_Z[w]) ys[job[w]] += delta, yt[w] -= delta;
                else min_to[w] -= delta;
            }
            w_cur = w_next;
        }
    }
}

```

```

    for (int w; w_cur != -1; w_cur = w) job[w_cur] = job[w =
        ↪prv[w_cur]];
    answers.push_back(-yt[W]);
}
return answers;
}

```

GeneralMatchBlossom.h

Description: Variant on Gabow's Impl of Edmond's Blossom Algorithm. General unweighted max matching with 1-based indexing. If white[v] = 0 after solve() returns, v is part of every max matching.

Time: $\mathcal{O}(NM)$, faster in practice

fd5cc7, 50 lines

```

struct MaxMatching {
    int N; V<vi> adj;
    V<int> mate, first; vb white; vpi label;
    void init(int _N) { N = _N; adj = V<vi>(N+1);
        mate = first = vi(N+1); label = vpi(N+1); white = vb(N+1);
        ↪}
    void ae(int u, int v) { adj.at(u).pb(v), adj.at(v).pb(u); }
    int group(int x) { if (white[first[x]]) first[x] = group(
        ↪first[x]);
        return first[x]; }
    void match(int p, int b) {
        swap(b, mate[p]); if (mate[b] != p) return;
        if (!label[p].s) mate[b] = label[p].f, match(label[p].f, b);
        ↪ // vertex label
        else match(label[p].f, label[p].s), match(label[p].s, label[p]
            ↪.f); // edge label
    }
    bool augment(int st) { assert(st);
        white[st] = 1; first[st] = 0; label[st] = {0,0};
        queue<int> q; q.push(st);
        while (!q.empty()) {
            int a = q.ft; q.pop(); // outer vertex
            each(b, adj[a]) { assert(b);
                if (white[b]) { // two outer vertices, form blossom
                    int x = group(a), y = group(b), lca = 0;
                    while (x||y) {
                        if (y) swap(x,y);
                        if (label[x] == pi{a,b}) { lca = x; break; }
                        label[x] = {a,b}; x = group(label[mate[x]].first);
                    }
                }
                for (int v: {group(a), group(b)}) while (v != lca) {
                    assert(!white[v]); // make everything along path
                    ↪white
                    q.push(v); white[v] = true; first[v] = lca;
                    v = group(label[mate[v]].first);
                }
            }
            else if (!mate[b]) { // found augmenting path
                mate[b] = a; match(a,b); white = vb(N+1); // reset
                return true;
            }
            else if (!white[mate[b]]) {
                white[mate[b]] = true; first[mate[b]] = b;
                label[b] = {0,0}; label[mate[b]] = pi{a,0};
                q.push(mate[b]);
            }
        }
    }
    return false;
}

int solve() {
    int ans = 0;
    FOR(st, 1, N+1) if (!mate[st]) ans += augment(st);
    FOR(st, 1, N+1) if (!mate[st] && !white[st]) assert(!augment(
        ↪st));
    return ans;
}
};

```

GeneralMatchBlossom GeneralWeightedMatch

GeneralWeightedMatch.h

Description: General max weight max matching with 1-based indexing. Edge weights must be positive, combo of UnweightedMatch and Hungarian. **Time:** $\mathcal{O}(N^3)$?

120873, 145 lines

```

template<int SZ> struct WeightedMatch {
    struct edge { int u,v,w; }; edge g[SZ*2][SZ*2];
    void ae(int u, int v, int w) { g[u][v].w = g[v][u].w = w; }
    int N, NX, lab[SZ*2], match[SZ*2], slack[SZ*2], st[SZ*2];
    int par[SZ*2], floFrom[SZ*2][SZ], S[SZ*2], aux[SZ*2];
    vi flo[SZ*2]; queue<int> q;
    void init(int _N) { N = _N; // init all edges
        FOR(u, 1, N+1) FOR(v, 1, N+1) g[u][v] = {u,v,0}; }
    int eDelta(edge e) { // >= 0 at all times
        return lab[e.u]+lab[e.v]-g[e.u][e.v].w*2; }
    void updSlack(int u, int x) { // smallest edge -> blossom x
        if (!slack[x] || eDelta(g[u][x]) < eDelta(g[slack[x]][x]))
            slack[x] = u; }
    void setSlack(int x) {
        slack[x] = 0; FOR(u, 1, N+1) if (g[u][x].w > 0
            && st[u] != x && S[st[u]] == 0) updSlack(u, x); }
    void qPush(int x) {
        if (x <= N) q.push(x);
        else each(t, flo[x]) qPush(t); }
    void setSt(int x, int b) {
        st[x] = b; if (x > N) each(t, flo[x]) setSt(t, b); }
    int getPr(int b, int xr) { // get even position of xr
        int pr = find(all(flo[b]), xr)-begin(flo[b]);
        if (pr&1) { reverse(1+all(flo[b])); return sz(flo[b])-pr; }
        return pr; }
    void setMatch(int u, int v) { // rearrange flo[u], matches
        edge e = g[u][v]; match[u] = e.v; if (u <= N) return;
        int xr = floFrom[u][e.u], pr = getPr(u, xr);
        FOR(i, pr) setMatch(flo[u][i], flo[u][i^1]);
        setMatch(xr, v); rotate(begin(flo[u]), pr+all(flo[u])); }
    void augment(int u, int v) { // set matches including u->v
        while (1) { // and previous ones
            int xnv = st[match[u]]; setMatch(u, v);
            if (!xnv) return;
            setMatch(xnv, st[par[xnv]]);
            u = st[par[xnv]], v = xnv;
        }
    }
    int lca(int u, int v) { // same as in unweighted
        static int t = 0; // except maybe return 0
        for (++t; u||v; swap(u,v)) {
            if (!u) continue;
            if (aux[u] == t) return u;
            aux[u] = t; u = st[match[u]];
            if (u) u = st[par[u]];
        }
        return 0;
    }
    void addBlossom(int u, int anc, int v) {
        int b = N+1; while (b <= NX && st[b]) ++b;
        if (b > NX) ++NX; // new blossom
        lab[b] = S[b] = 0; match[b] = match[anc]; flo[b] = {anc};
        auto blossom = [&](int x) {
            for (int y; x != anc; x = st[par[y]])
                flo[b].pb(x), flo[b].pb(y = st[match[x]]), qPush(y);
        };
        blossom(u); reverse(1+all(flo[b])); blossom(v); setSt(b, b);
        // identify all nodes in current blossom
        FOR(x, 1, NX+1) g[b][x].w = g[x][b].w = 0;
        FOR(x, 1, N+1) floFrom[b][x] = 0;
        each(xs, flo[b]) { // find tightest constraints
            FOR(x, 1, NX+1) if (g[b][x].w == 0 || eDelta(g[xs][x]) <
                eDelta(g[b][x])) g[b][x]=g[xs][x], g[x][b]=g[x][xs];
            FOR(x, 1, N+1) if (floFrom[xs][x]) floFrom[b][x] = xs;
        }
    }
};

```

```

    } // floFrom to deconstruct blossom
    setSlack(b); // since didn't qPush everything
}
void expandBlossom(int b) {
    each(t, flo[b]) setSt(t, t); // undo setSt(b,b)
    int xr = floFrom[b][g[b][par[b]].u], pr = getPr(b, xr);
    for(int i = 0; i < pr; i += 2) {
        int xs = flo[b][i], xns = flo[b][i+1];
        par[xs] = g[xns][xs].u; S[xs] = 1; // no setSlack(xns)?
        S[xns] = slack[xs] = slack[xns] = 0; qPush(xns);
    }
    S[xr] = 1, par[xr] = par[b];
    FOR(i, pr+1, sz(flo[b])) { // matches don't change
        int xs = flo[b][i]; S[xs] = -1, setSlack(xs); }
    st[b] = 0; // blossom killed
}
bool onFoundEdge(edge e) {
    int u = st[e.u], v = st[e.v];
    if (S[v] == -1) { // v unvisited, matched with smth else
        par[v] = e.u, S[v] = 1; slack[v] = 0;
        int nu = st[match[v]]; S[nu] = slack[nu] = 0; qPush(nu);
    } else if (S[v] == 0) {
        int anc = lca(u, v); // if 0 then match found!
        if (!anc) return augment(u, v), augment(v, u), 1;
        addBlossom(u, anc, v);
    }
    return 0;
}
bool matching() {
    q = queue<int>();
    FOR(x, 1, NX+1) {
        S[x] = -1, slack[x] = 0; // all initially unvisited
        if (st[x] == x && !match[x]) par[x] = S[x] = 0, qPush(x);
    }
    if (!sz(q)) return 0;
    while (1) {
        while (sz(q)) { // unweighted matching with tight edges
            int u = q.ft; q.pop(); if (S[st[u]] == 1) continue;
            FOR(v, 1, N+1) if (g[u][v].w > 0 && st[u] != st[v]) {
                if (eDelta(g[u][v]) == 0) { // condition is strict
                    if (onFoundEdge(g[u][v])) return 1;
                } else updSlack(u, st[v]);
            }
        }
        int d = INT_MAX;
        FOR(b, N+1, NX+1) if (st[b] == b && S[b] == 1)
            ckmin(d, lab[b]/2); // decrease lab[b]
        FOR(x, 1, NX+1) if (st[x] == x && slack[x]) {
            if (S[x] == -1) ckmin(d, eDelta(g[slack[x]][x]));
            else if (S[x] == 0) ckmin(d, eDelta(g[slack[x]][x])/2);
        } // edge weights shouldn't go below 0
        FOR(u, 1, N+1) {
            if (S[st[u]] == 0) {
                if (lab[u] <= d) return 0; // why?
                lab[u] -= d;
            } else if (S[st[u]] == 1) lab[u] += d;
        } // lab has opposite meaning for verts and blossoms
        FOR(b, N+1, NX+1) if (st[b] == b && S[b] != -1)
            lab[b] += (S[b] == 0 ? 1 : -1)*d*2;
        q = queue<int>();
        FOR(x, 1, NX+1) if (st[x]==x && slack[x] // new tight edge
            && st[slack[x]] != x && eDelta(g[slack[x]][x]) == 0)
            if (onFoundEdge(g[slack[x]][x])) return 1;
        FOR(b, N+1, NX+1) if (st[b]==b && S[b]==1 && lab[b]==0)
            expandBlossom(b); // odd dist blossom taken apart
    }
    return 0;
}
pair<ll, int> calc() {

```

```

    NX = N; st[0] = 0; FOR(i,1,2*N+1) aux[i] = 0;
    FOR(i,1,N+1) match[i] = 0, st[i] = i, flo[i].clear();
    int wMax = 0;
    FOR(u,1,N+1) FOR(v,1,N+1)
        floFrom[u][v] = (u == v ? u : 0), ckmax(wMax,g[u][v].w);
    FOR(u,1,N+1) lab[u] = wMax; // start high and decrease
    int num = 0; ll wei = 0; while (matching()) ++num;
    FOR(u,1,N+1) if (match[u] && match[u] < u)
        wei += g[u][match[u]].w; // edges in matching
    return {wei,num};
}
};
```

MaxMatchFast.h

Description: Fast bipartite matching.

Time: $\mathcal{O}\left(M\sqrt{N}\right)$ ec6c96, 31 lines

```

vpi maxMatch(int L, int R, const vpi& edges) {
    V<vi> adj = V<vi>(L);
    vi nxt(L,-1), prv(R,-1), lev, ptr;
    FOR(i,sz(edges)) adj.at(edges[i].f).pb(edges[i].s);
    while (true) {
        lev = ptr = vi(L); int max_lev = 0;
        queue<int> q; FOR(i,L) if (nxt[i]==-1) lev[i]=1, q.push(i);
        while (sz(q)) {
            int x = q.ft; q.pop();
            for (int y: adj[x]) {
                int z = prv[y];
                if (z == -1) max_lev = lev[x];
                else if (!lev[z]) lev[z] = lev[x]+1, q.push(z);
            }
            if (max_lev) break;
        }
        if (!max_lev) break;
        FOR(i,L) if (lev[i] > max_lev) lev[i] = 0;
        auto dfs = [&](auto self, int x) -> bool {
            for (;ptr[x] < sz(adj[x]);++ptr[x]) {
                int y = adj[x][ptr[x]], z = prv[y];
                if (z == -1 || (lev[z] == lev[x]+1 && self(self,z)))
                    return nxt[x]=y, prv[y]=x, ptr[x]=sz(adj[x]), 1;
            }
            return 0;
        };
        FOR(i,L) if (nxt[i] == -1) dfs(dfs,i);
    }
    vpi ans; FOR(i,L) if (nxt[i] != -1) ans.pb({i,nxt[i]});
    return ans;
}
```

7.7 Advanced

ChordalGraphRecognition.h

Description: Recognizes graph where every induced cycle has length exactly 3 using maximum adjacency search.

```

int N,M;
set<int> adj[MX];
int cnt[MX];
vi ord, rord;

vi find_path(int x, int y, int z) {
    vi pre(N,-1);
    queue<int> q; q.push(x);
    while (sz(q)) {
        int t = q.ft; q.pop();
        if (adj[t].count(y)) {
            pre[y] = t; vi path = {y};
            while (path.bk != x) path.pb(pre[path.bk]);
            path.pb(z);
        }
    }
}
```

```

        return path;
    }
    each(u,adj[t]) if (u != z && !adj[u].count(z) && pre[u] ==
        ↪ -1) {
        pre[u] = t;
        q.push(u);
    }
}
assert(0);
}

int main() {
    setIO(); re(N,M);
    FOR(i,M) {
        int a,b; re(a,b);
        adj[a].insert(b), adj[b].insert(a);
    }
    rord = vi(N,-1);
    priority_queue<pi> pq;
    FOR(i,N) pq.push({0,i});
    while (sz(pq)) {
        pi p = pq.top(); pq.pop();
        if (rord[p.s] != -1) continue;
        rord[p.s] = sz(ord); ord.pb(p.s);
        each(t,adj[p.s]) pq.push({++cnt[t],t});
    }
    assert(sz(ord) == N);
    each(z,ord) {
        pi big = {-1,-1};
        each(y,adj[z]) if (rord[y] < rord[z])
            ckmax(big,mp(rord[y],y));
        if (big.f == -1) continue;
        int y = big.s;
        each(x,adj[z]) if (rord[x] < rord[y]) if (!adj[y].count(x))
            ↪ {
                ps("NO");
                vi v = find_path(x,y,z);
                ps(sz(v));
                each(t,v) pr(t,' ');
                exit(0);
            }
    }
    ps("YES");
    reverse(all(ord));
    each(z,ord) pr(z,' ');
}
```

DominatorTree.h

Description: Used only a few times. Assuming that all nodes are reachable from *root*, *a* dominates *b* iff every path from *root* to *b* passes through *a*.

Time: $\mathcal{O}(M \log N)$ 4b8836, 41 lines

```

template<int SZ> struct Dominator {
    vi adj[SZ], ans[SZ]; // input edges, edges of dominator tree
    vi radj[SZ], child[SZ], sdomChild[SZ];
    int label[SZ], rlabel[SZ], sdom[SZ], dom[SZ], co = 0;
    int par[SZ], bes[SZ];
    void ae(int a, int b) { adj[a].pb(b); }
    int get(int x) { // DSU with path compression
        // get vertex with smallest sdom on path to root
        if (par[x] != x) {
            int t = get(par[x]); par[x] = par[par[x]];
            if (sdom[t] < sdom[bes[x]]) bes[x] = t;
        }
        return bes[x];
    }
    void dfs(int x) { // create DFS tree
        label[x] = ++co; rlabel[co] = x;
        sdom[co] = par[co] = bes[co] = co;
    }
}
```

```

    each(y,adj[x]) {
        if (!label[y]) {
            dfs(y); child[label[x]].pb(label[y]); }
        radj[label[y]].pb(label[x]);
    }
}
void init(int root) {
    dfs(root);
    ROF(i,1,co+1) {
        each(j,radj[i]) ckmin(sdom[i],sdom[get(j)]);
        if (i > 1) sdomChild[sdom[i]].pb(i);
        each(j,sdomChild[i]) {
            int k = get(j);
            if (sdom[j] == sdom[k]) dom[j] = sdom[j];
            else dom[j] = k;
        }
        each(j,child[i]) par[j] = i;
    }
    FOR(i,2,co+1) {
        if (dom[i] != sdom[i]) dom[i] = dom[dom[i]];
        ans[rlabel[dom[i]]].pb(rlabel[i]);
    }
}
};
```

EdgeColor.h

Description: Used only once. Naive implementation of Misra & Gries edge coloring. By Vizing's Theorem, a simple graph with max degree *d* can be edge colored with at most *d* + 1 colors

Time: $\mathcal{O}(N^2M)$, faster in practice cc2b29, 40 lines

```

template<int SZ> struct EdgeColor {
    int N = 0, maxDeg = 0, adj[SZ][SZ], deg[SZ];
    void init(int _N) { N = _N;
        FOR(i,N) { deg[i] = 0; FOR(j,N) adj[i][j] = 0; } }
    void ae(int a, int b, int c) {
        adj[a][b] = adj[b][a] = c; }
    int delEdge(int a, int b) {
        int c = adj[a][b]; adj[a][b] = adj[b][a] = 0;
        return c; }
    V<bool> genCol(int x) {
        V<bool> col(N+1); FOR(i,N) col[adj[x][i]] = 1;
        return col; }
    int freeCol(int u) {
        auto col = genCol(u); int x = 1;
        while (col[x]) ++x; return x; }
    void invert(int x, int d, int c) {
        FOR(i,N) if (adj[x][i] == d)
            delEdge(x,i), invert(i,c,d), ae(x,i,c); }
    void ae(int u, int v) {
        // check if you can add edge w/o doing any work
        assert(N); ckmax(maxDeg,max(++deg[u],++deg[v]));
        auto a = genCol(u), b = genCol(v);
        FOR(i,1,maxDeg+2) if (!a[i] && !b[i])
            return ae(u,v,i);
        V<bool> use(N); vi fan = {v}; use[v] = 1;
        while (1) {
            auto col = genCol(fan.bk);
            if (sz(fan) > 1) col[adj[fan.bk][u]] = 0;
            int i=0; while (i<N && (use[i] || col[adj[u][i]])) i++;
            if (i < N) fan.pb(i), use[i] = 1;
            else break;
        }
        int c = freeCol(u), d = freeCol(fan.bk); invert(u,d,c);
        int i = 0; while (i < sz(fan) && genCol(fan[i])[d]
            && adj[u][fan[i]] != d) i ++;
        assert (i != sz(fan));
        FOR(j,i) ae(u,fan[j],delEdge(u,fan[j+1]));
        ae(u,fan[i],d);
    }
}
```

```

    }
};

DirectedMST.h
Description: Chu-Liu-Edmonds algorithm. Computes minimum weight di-
rected spanning tree rooted at  $r$ , edge from  $par[i] \rightarrow i$  for all  $i \neq r$ . Use DSU
with rollback if need to return edges.
Time:  $\mathcal{O}(M \log M)$ 

```

```

"DSUrb.h" 5d5c10, 61 lines

struct Edge { int a, b; ll w; };
struct Node { // lazy skew heap node
    Edge key; Node *l, *r; ll delta;
    void prop() {
        key.w += delta;
        if (l) l->delta += delta;
        if (r) r->delta += delta;
        delta = 0;
    }
    Edge top() { prop(); return key; }
};

Node *merge(Node *a, Node *b) {
    if (!a || !b) return a ? b;
    a->prop(), b->prop();
    if (a->key.w > b->key.w) swap(a, b);
    swap(a->l, a->r = merge(b, a->r));
    return a;
}

void pop(Node*& a) { a->prop(); a = merge(a->l, a->r); }

pair<ll,vi> dmst(int n, int r, const vector<Edge>& g) {
    DSUrb dsu; dsu.init(n);
    vector<Node*> heap(n); // store edges entering each vertex
    // in increasing order of weight
    each(e,g) heap[e.b] = merge(heap[e.b], new Node{e});
    ll res = 0; vi seen(n,-1); seen[r] = r;
    vpi in(n,{-1,-1}); // edge entering each vertex in MST
    vector<pair<int,vector<Edge>>> cyscs;
    FOR(s,n) {
        int u = s, w;
        vector<pair<int,Edge>> path;
        while (seen[u] < 0) {
            if (!heap[u]) return {-1,{};};
            seen[u] = s;
            Edge e = heap[u]->top(); path.pb({u,e});
            heap[u]->delta -= e.w, pop(heap[u]);
            res += e.w, u = dsu.get(e.a);
            if (seen[u] == s) { // found cycle, contract
                Node* cyc = 0; cyscs.pb({});
                do {
                    cyc = merge(cyc, heap[w = path.bk.f]);
                    cyscs.bk.s.pb(path.bk.s);
                    path.pop_back();
                } while (dsu.unite(u,w));
                u = dsu.get(u); heap[u] = cyc, seen[u] = -1;
                cyscs.bk.f = u;
            }
        }
        each(t,path) in[dsu.get(t.s.b)] = {t.s.a,t.s.b};
    } // found path from root to s, done
    while (sz(cyscs)) { // expand cyscs to restore sol
        auto c = cyscs.bk; cyscs.pop_back();
        pi inEdge = in[c.f];
        each(t,c.s) dsu.rollback();
        each(t,c.s) in[dsu.get(t.b)] = {t.a,t.b};
        in[dsu.get(inEdge.s)] = inEdge;
    }
    vi par(n); FOR(i,n) par[i] = in[i].f;
    // i == r ? in[i].s == -1 : in[i].s == i
    return {res,par};
}

```

```

}

LCT.h
Description: Link-Cut Tree. Given a function  $f(1 \dots N) \rightarrow 1 \dots N$ , eval-
uates  $f^b(a)$  for any  $a, b$ . sz is for path queries; sub, vsub are for subtree
queries. x->access() brings x to the top and propagates it; its left sub-
tree will be the path from x to the root and its right subtree will be empty.
Then sub will be the number of nodes in the connected component of x and
vsub will be the number of nodes under x. Use makeRoot for arbitrary path
queries.
Usage: FOR(i,1,N+1) LCT[i]=new snode(i); link(LCT[1],LCT[2],1);
Time:  $\mathcal{O}(\log N)$ 

```

```

e24bf7, 115 lines

typedef struct snode* sn;
struct snode { ////////// VARIABLES
    sn p, c[2]; // parent, children
    sn extra; // extra cycle node for "The Applicant"
    bool flip = 0; // subtree flipped or not
    int val, sz; // value in node, # nodes in current splay tree
    int sub, vsub = 0; // vsub stores sum of virtual children
    snode(int _val) : val(_val) {
        p = c[0] = c[1] = extra = NULL; calc(); }
    friend int getSz(sn x) { return x?x->sz:0; }
    friend int getSub(sn x) { return x?x->sub:0; }
    void prop() { // lazy prop
        if (!flip) return;
        swap(c[0],c[1]); flip = 0;
        FOR(i,2) if (c[i]) c[i]->flip ^= 1;
    }
    void calc() { // recalc vals
        FOR(i,2) if (c[i]) c[i]->prop();
        sz = 1+getS(c[0])+getS(c[1]);
        sub = 1+getSub(c[0])+getSub(c[1])+vsub;
    }
    ////////// SPLAY TREE OPERATIONS
    int dir() {
        if (!p) return -2;
        FOR(i,2) if (p->c[i] == this) return i;
        return -1; // p is path-parent pointer
    } // -> not in current splay tree
    // test if root of current splay tree
    bool isRoot() { return dir() < 0; }
    friend void setLink(sn x, sn y, int d) {
        if (y) y->p = x;
        if (d >= 0) x->c[d] = y; }
    void rot() { // assume p and p->p propagated
        assert(!isRoot()); int x = dir(); sn pa = p;
        setLink(pa->p, this, pa->dir());
        setLink(pa, c[x^1], x); setLink(this, pa, x^1);
        pa->calc();
    }
    void splay() {
        while (!isRoot() && !p->isRoot()) {
            p->p->prop(), p->prop(), prop();
            dir() = p->dir() ? p->rot() : rot();
            rot();
        }
        if (!isRoot()) p->prop(), prop(), rot();
        prop(); calc();
    }
    sn fbo(int b) { // find by order
        prop(); int z = getS(c[0]); // of splay tree
        if (b == z) { splay(); return this; }
        return b < z ? c[0]->fbo(b) : c[1] -> fbo(b-z-1);
    }
    ////////// BASE OPERATIONS
    void access() { // bring this to top of tree, propagate
        for (sn v = this, pre = NULL; v; v = v->p) {
            v->splay(); // now switch virtual children

```

```

            if (pre) v->vsub -= pre->sub;
            if (v->c[1]) v->vsub += v->c[1]->sub;
            v->c[1] = pre; v->calc(); pre = v;
        }
        splay(); assert(!c[1]); // right subtree is empty
    }
    void makeRoot() {
        access(); flip ^= 1; access(); assert(!c[0] && !c[1]); }
    ////////// QUERIES
    friend sn lca(sn x, sn y) {
        if (x == y) return x;
        x->access(), y->access(); if (!x->p) return NULL;
        x->splay(); return x->p?:x; // y was below x in latter case
    } // access at y did not affect x -> not connected
    friend bool connected(sn x, sn y) { return lca(x,y); }
    // # nodes above
    int distRoot() { access(); return getS(c[0]); }
    sn getRoot() { // get root of LCT component
        access(); sn a = this;
        while (a->c[0]) a = a->c[0], a->prop();
        a->access(); return a;
    }
    sn getPar(int b) { // get b-th parent on path to root
        access(); b = getS(c[0])-b; assert(b >= 0);
        return fbo(b);
    } // can also get min, max on path to root, etc
    ////////// MODIFICATIONS
    void set(int v) { access(); val = v; calc(); }
    friend void link(sn x, sn y, bool force = 0) {
        assert(!connected(x,y));
        if (force) y->makeRoot(); // make x par of y
        else { y->access(); assert(!y->c[0]); }
        x->access(); setLink(y,x,0); y->calc();
    }
    friend void cut(sn y) { // cut y from its parent
        y->access(); assert(y->c[0]);
        y->c[0]->p = NULL; y->c[0] = NULL; y->calc(); }
    friend void cut(sn x, sn y) { // if x, y adj in tree
        x->makeRoot(); y->access();
        assert(y->c[0] == x && !x->c[0] && !x->c[1]); cut(y); }
};

sn LCT[MX];

////////// THE APPLICANT SOLUTION
void setNex(sn a, sn b) { // set f[a] = b
    if (connected(a,b)) a->extra = b;
    else link(b,a); }
void delNex(sn a) { // set f[a] = NULL
    auto t = a->getRoot();
    if (t == a) { t->extra = NULL; return; }
    cut(a); assert(t->extra);
    if (!connected(t,t->extra))
        link(t->extra,t), t->extra = NULL;
}

sn getPar(sn a, int b) { // get f^b[a]
    int d = a->distRoot(); if (b <= d) return a->getPar(b);
    b -= d+1; auto r = a->getRoot()->extra; assert(r);
    d = r->distRoot()+1; return r->getPar(b%d);
}

```

Geometry (8)

8.1 Primitives

ComplexComp.h

Description: Allows you to sort complex numbers.

6f828b, 5 lines

```
#define x real()
```

```
#define y imag()
using P = complex<db>;
namespace std {
    bool operator<(P l,P r) { return mp(l.x,l.y)<mp(r.x,r.y); } }
```

PointShort.h
Description: Use in place of complex<T>.

```
26df5fc, 36 lines
using T = db; // or ll
const T EPS = 1e-9; // adjust as needed
using P = pair<T,T>; using vP = V<P>; using Line = pair<P,P>;
int sgn(T a) { return (a>EPS)-(a<-EPS); }
T sq(T a) { return a*a; }

T abs2(P p) { return sq(p.f)+sq(p.s); }
T abs(P p) { return sqrt(abs2(p)); }
T arg(P p) { return atan2(p.s,p.f); }
P conj(P p) { return P(p.f,-p.s); }
P perp(P p) { return P(-p.s,p.f); }
P dir(T ang) { return P(cos(ang),sin(ang)); }

P operator+(P l, P r) { return P(l.f+r.f,l.s+r.s); }
P operator-(P l, P r) { return P(l.f-r.f,l.s-r.s); }
P operator*(P l, T r) { return P(l.f*r,l.s*r); }
P operator/(P l, T r) { return P(l.f/r,l.s/r); }
P operator*(P l, P r) { // complex # multiplication
    return P(l.f*r.f-l.s*r.s,l.s*r.f+l.f*r.s); }
P operator/(P l, P r) { return l*conj(r)/abs2(r); }
```

```
P unit(const P& p) { return p/abs(p); }
T dot(const P& a, const P& b) { return a.f*b.f+a.s*b.s; }
T dot(const P& p, const P& a, const P& b) { return dot(a-p,b-p)
    ↪; }
T cross(const P& a, const P& b) { return a.f*b.s-a.s*b.f; }
T cross(const P& p, const P& a, const P& b) {
    return cross(a-p,b-p); }
P reflect(const P& p, const Line& l) {
    P a = l.f, d = l.s-l.f;
    return a+conj((p-a)/d)*d; }
P foot(const P& p, const Line& l) {
    return (p+reflect(p,l))/(T)2; }
bool onSeg(const P& p, const Line& l) {
    return sgn(cross(l.f,l.s,p)) == 0 && sgn(dot(p,l.f,l.s)) <= 0
    ↪; }
ostream& operator<<(ostream& os, P p) {
    return os << "(" << p.f << ", " << p.s << ")"; }
```

AngleCmp.h
Description: Sorts points in ccw order about origin in the same way as atan2, which returns real in $(-\pi,\pi]$ so points on negative x -axis come last.
Usage: vP v; sort(all(v),angleCmp);

```
"Point.h" 2df5fc, 8 lines
// WARNING: you will get unexpected results if you mistype this
    ↪ as bool instead of int
// -1 if lower half, 0 if origin, 1 if upper half
int half(P x) { return x.s != 0 ? sgn(x.s) : -sgn(x.f); }
bool angleCmp(P a, P b) { int A = half(a), B = half(b);
    return A == B ? cross(a,b) > 0 : A < B; }

// equivalent to: sort(all(v),[(P a, P b) {
//     return atan2(a.s,a.f) < atan2(b.s,b.f); }]);
```

SegDist.h
Description: computes distance between P and line (segment) AB

```
"Point.h" 0cb69a, 6 lines
T lineDist(const P& p, const Line& l) {
    return abs(cross(p,l.f,l.s))/abs(l.f-l.s); }
T segDist(const P& p, const Line& l) {
```

```
if (dot(l.f,p,l.s) <= 0) return abs(p-l.f);
if (dot(l.s,p,l.f) <= 0) return abs(p-l.s);
return lineDist(p,l); }
```

SegIsect.h
Description: computes the intersection point(s) of line (segments) a and b

```
"Point.h" 7f6ba0, 26 lines
// {unique intersection point} if it exists
// {b.f,b.s} if input lines are the same
// empty if lines do not intersect
vP lineIsect(const Line& a, const Line& b) {
    T a0 = cross(a.f,a.s,b.f), a1 = cross(a.f,a.s,b.s);
    if (a0 == a1) return a0 == 0 ? vP(b.f,b.s) : vP{};
    return {(b.s*a0-b.f*a1)/(a0-a1)}; }

// point in interior of both segments a and b, if it exists
vP strictIsect(const Line& a, const Line& b) {
    T a0 = cross(a.f,a.s,b.f), a1 = cross(a.f,a.s,b.s);
    T b0 = cross(b.f,b.s,a.f), b1 = cross(b.f,b.s,a.s);
    if (sgn(a0)*sgn(a1) < 0 && sgn(b0)*sgn(b1) < 0)
        return {(b.s*a0-b.f*a1)/(a0-a1)};
    return {}; }

// intersection of segments, a and b may be degenerate
vP segIsect(const Line& a, const Line& b) {
    vP v = strictIsect(a,b); if (sz(v)) return v;
    set<P> s;
    #define i(x,y) if (onSeg(x,y)) s.ins(x)
    i(a.f,b); i(a.s,b); i(b.f,a); i(b.s,a);
    return {all(s)}; } }
```

8.2 Polygons

PolygonCenArea.h
Description: centroid (center of mass) of a polygon with constant mass per unit area and SIGNED area
Time: $\mathcal{O}(N)$

```
"Point.h" 4ca221, 7 lines
pair<P,T> cenArea(const vP& v) { assert(sz(v) >= 3);
    P cen{}; T area{};
    F0R(i,sz(v)) {
        int j = (i+1)%sz(v); T a = cross(v[i],v[j]);
        cen += a*(v[i]+v[j]); area += a; }
    return {cen/area/(T)3,area/2}; // area is SIGNED } }
```

InPolygon.h
Description: Tests whether point is inside, on, or outside of a polygon (returns -1, 0, or 1). Both CW and CCW polygons are ok.
Time: $\mathcal{O}(N)$

```
"Point.h" 6ebff0, 9 lines
int inPoly(const P& p, const vP& poly) {
    int n = sz(poly), ans = 0;
    F0R(i,n) {
        P x = poly[i], y = poly[(i+1)%n]; if (x.s > y.s) swap(x,y);
        if (onSeg(p,{x,y})) return 0;
        ans ^= (x.s <= p.s && p.s < y.s && cross(p,x,y) > 0);
    }
    return ans ? -1 : 1; } }
```

ConvexHull.h
Description: top-bottom convex hull
Time: $\mathcal{O}(N \log N)$

```
"Point.h" 868655, 18 lines
```

```
pair<vi,vi> ulHull(const vP& v) {
    vi p(sz(v)), u, l; iota(all(p), 0);
    sort(all(p), [&v](int a, int b) { return v[a] < v[b]; });
    each(i,p) {
        #define ADDP(C, cmp) while (sz(C) > 1 && cross(\
            v[C[sz(C)-2]],v[C.bk],v[i]) cmp 0) C.pop_back(); C.pb(i);
        ADDP(u, >=); ADDP(l, <=);
    }
    return {u,l}; }
vi hullInd(const vP& v) { // returns indices in CCW order
    vi u,l; tie(u,l) = ulHull(v); if (sz(l) <= 1) return l;
    if (v[l[0]] == v[l[1]]) return {0};
    l.insert(end(l),l+rall(u)-1); return l; }
vP hull(const vP& v) {
    vi w = hullInd(v); vP res; each(t,w) res.pb(v[t]);
    return res; }
```

MinkowskiSum.h
Description: Minkowski sum of two convex polygons given in CCW order.
Time: $\mathcal{O}(N)$

```
"ConvexHull.h" e5d935, 29 lines
vP minkowski_sum(vP a, vP b) {
    if (sz(a) > sz(b)) swap(a, b);
    if (!sz(a)) return {};
    if (sz(a) == 1) {
        each(t, b) t += a.ft;
        return b;
    }
    rotate(begin(a), min_element(all(a)), end(a));
    rotate(begin(b), min_element(all(b)), end(b));
    a.pb(a[0]), a.pb(a[1]);
    b.pb(b[0]), b.pb(b[1]);
    vP result;
    int i = 0, j = 0;
    while (i < sz(a)-2 || j < sz(b)-2) {
        result.pb(a[i]+b[j]);
        T crs = cross(a[i+1]-a[i],b[j+1]-b[j]);
        i += (crs >= 0);
        j += (crs <= 0);
    }
    return result; } }
```

```
T diameter2(vP p) { // example application: squared diameter
    vP a = hull(p);
    vP b = a; each(t, b) t *= -1;
    vP c = minkowski_sum(a, b);
    T ret = 0; each(t, c) ckmax(ret, abs2(t));
    return ret; } }
```

HullDiameter.h
Description: Rotating calipers. Returns square of greatest distance between two points in P .
Time: $\mathcal{O}(N)$ given convex hull

```
"ConvexHull.h" 213a2a, 9 lines
db diameter2(vP P) {
    P = hull(P);
    int n = sz(P), ind = 1; T ans = 0;
    if (n > 1) F0R(i,n) for (int j = (i+1)%n;;ind = (ind+1)%n) {
        ckmax(ans, abs2(P[i]-P[ind]));
        if (cross(P[j]-P[i],P[(ind+1)%n]-P[ind]) <= 0) break;
    }
    return ans; } }
```

HullTangents.h

Description: Given convex polygon with no three points collinear and a point strictly outside of it, computes the lower and upper tangents.

Time: $\mathcal{O}(\log N)$

```
"Point.h" 85b807, 36 lines

bool lower;
bool better(P a, P b, P c) {
    T z = cross(a,b,c);
    return lower ? z < 0 : z > 0; }
int tangent(const vP& a, P b) {
    if (sz(a) == 1) return 0;
    int lo, hi;
    if (better(b,a[0],a[1])) {
        lo = 0, hi = sz(a)-1;
        while (lo < hi) {
            int mid = (lo+hi+1)/2;
            if (better(b,a[0],a[mid])) lo = mid;
            else hi = mid-1;
        }
        lo = 0;
    } else {
        lo = 1, hi = sz(a);
        while (lo < hi) {
            int mid = (lo+hi)/2;
            if (!better(b,a[0],a[mid])) lo = mid+1;
            else hi = mid;
        }
        hi = sz(a);
    }
    while (lo < hi) {
        int mid = (lo+hi)/2;
        if (better(b,a[mid],a[(mid+1)%sz(a)])) lo = mid+1;
        else hi = mid;
    }
    return lo%sz(a);
}
pi tangents(const vP& a, P b) {
    lower = 1; int x = tangent(a,b);
    lower = 0; int y = tangent(a,b);
    return {x,y};
}
```

LineHull.h

Description: lineHull accepts line and ccw convex polygon. If all vertices in poly lie to one side of the line, returns a vector of closest vertices to line as well as orientation of poly with respect to line (± 1 for above/below). Otherwise, returns the range of vertices that lie on or below the line. extrVertex returns the point of a hull with the max projection onto a line.

Time: $\mathcal{O}(\log N)$

```
"Point.h" 40e5a6, 41 lines

using Line = AR<P,2>;
#define cmp(i,j) sgn(-dot(dir,poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i+1,i) >= 0 && cmp(i,i-1+n) < 0
int extrVertex(const vP& poly, P dir) {
    int n = sz(poly), lo = 0, hi = n;
    if (extr(0)) return 0;
    while (lo+1 < hi) {
        int m = (lo+hi)/2;
        if (extr(m)) return m;
        int ls = cmp(lo+1,lo), ms = cmp(m+1,m);
        (ls < ms || (ls == ms && ls == cmp(lo,m))) ? hi : lo) = m;
    }
    return lo;
}
vi same(Line line, const vP& poly, int a) {
    // points on same parallel as a
    int n = sz(poly); P dir = perp(line[0]-line[1]);
    if (cmp(a+n-1,a) == 0) return {a+n-1)%n,a};
```

HullTangents LineHull HalfPlaneIsect HalfPlaneSet

```
    if (cmp(a,a+1) == 0) return {a,(a+1)%n};
    return {a};
}
#define cmpL(i) sgn(cross(line[0],line[1],poly[i]))
pair<int,vi> lineHull(Line line, const vP& poly) {
    int n = sz(poly); assert(n>1);
    int endA = extrVertex(poly,perp(line[0]-line[1])); // lowest
    if (cmpL(endA) >= 0) return {1,same(line,poly,endA)};
    int endB = extrVertex(poly,perp(line[1]-line[0])); // highest
    if (cmpL(endB) <= 0) return {-1,same(line,poly,endB)};
    AR<int,2> res;
    FOR(i,2) {
        int lo = endA, hi = endB; if (hi < lo) hi += n;
        while (lo < hi) {
            int m = (lo+hi+1)/2;
            if (cmpL(m%n) == cmpL(endA)) lo = m;
            else hi = m-1;
        }
        res[i] = lo%n; swap(endA,endB);
    }
    if (cmpL((res[0]+1)%n) == 0) res[0] = (res[0]+1)%n;
    return {0,{(res[1]+1)%n,res[0]}};
}
```

HalfPlaneIsect.h

Description: Returns vertices of half-plane intersection. A half-plane is the area to the left of a ray, which is defined by a point p and a direction dp. Area of intersection should be sufficiently precise when all inputs are integers with magnitude $\leq 10^5$. Intersection must be bounded. Probably works with floating point too (but EPS might need to be adjusted?).

Time: $\mathcal{O}(N \log N)$

```
"AngleCmp.h" 18f712, 52 lines

struct Ray {
    P p, dp; // origin, direction
    P isect(const Ray& L) const {
        return p+dp*(cross(L.dp,L.p-p)/cross(L.dp,dp)); }
    bool operator<(const Ray& L) const {
        return angleCmp(dp,L.dp); }
};

vP halfPlaneIsect(V<Ray> rays, bool add_bounds = false) {
    if (add_bounds) { // bound input by rectangle [0,DX] x [0,DY]
        int DX = 1e9, DY = 1e9;
        rays.pb({P{0,0},P{1,0}});
        rays.pb({P{DX,0},P{0,1}});
        rays.pb({P{DX,DY},P{-1,0}});
        rays.pb({P{0,DY},P{0,-1}});
    }
    sor(rays); // sort rays by angle
    { // remove parallel rays
        V<Ray> nrays;
        each(t,rays) {
            if (!sz(nrays) || cross(nrays.bk.dp,t.dp) > EPS) { nrays.
                ↳pb(t); continue; }
            // last two rays are parallel, keep only one
            if (cross(t.dp,t.p-nrays.bk.p) > 0) nrays.bk = t;
        }
        swap(rays, nrays);
    }
    auto bad = [&](const Ray& a, const Ray& b, const Ray& c) {
        P p1 = a.isect(b), p2 = b.isect(c);
        if (dot(p2-p1,b.dp) <= EPS) {
            if (cross(a.dp,c.dp) <= 0) return 2; // isect(a,b,c) =
                ↳empty
            return 1; // isect(a,c) == isect(a,b,c)
        }
        return 0; // all three rays matter
    };
    #define reduce(t) \
```

```
    while (sz(poly) > 1) { \
        int b = bad(poly.at(sz(poly)-2),poly.bk,t); \
        if (b == 2) return {}; \
        if (b == 1) poly.pop_back(); \
        else break; \
    }
deque<Ray> poly;
each(t,rays) { reduce(t); poly.pb(t); }
for(;;poly.pop_front()) {
    reduce(poly[0]);
    if (!bad(poly.bk,poly[0],poly[1])) break;
}
assert(sz(poly) >= 3); // expect nonzero area
vP poly_points; FOR(i,sz(poly))
    poly_points.pb(poly[i].isect(poly[(i+1)%sz(poly)]));
return poly_points;
}
```

HalfPlaneSet.h

Description: Online Half-Plane Intersection

d2027d, 77 lines

```
using T = int;
using T2 = long long;
using T4 = __int128_t;
const T2 INF = 2e9;

struct Line {
    T a, b;
    T2 c;
};

bool operator<(Line m, Line n) {
    auto half = [&](Line m) { return m.b < 0 || m.b == 0 && m.a <
        ↳0; };
    return make_tuple(half(m), (T2)m.b * n.a) <
        make_tuple(half(n), (T2)m.a * n.b);
}

tuple<T4, T4, T2> LineIntersection(Line m, Line n) {
    T2 d = (T2)m.a * n.b - (T2)m.b * n.a; // assert(d);
    T4 x = (T4)m.c * n.b - (T4)m.b * n.c;
    T4 y = (T4)m.a * n.c - (T4)m.c * n.a;
    return {x, y, d};
}

Line LineFromPoints(T x1, T y1, T x2, T y2) {
    // everything to the right of ray {x1, y1} -> {x2, y2}
    T a = y1 - y2, b = x2 - x1;
    T2 c = (T2)a * x1 + (T2)b * y1;
    return {a, b, c}; // ax + by <= c
}

ostream &operator<<(ostream &out, Line l) {
    out << "Line " << l.a << " " << l.b << " " << -l.c;
    // out << "(" << l.a << " * x + " << l.b << " * y <= " << l.c
        ↳<< ")";
    return out;
}

struct HalfplaneSet : multiset<Line> {
    HalfplaneSet() {
        insert({+1, 0, INF});
        insert({0, +1, INF});
        insert({-1, 0, INF});
        insert({0, -1, INF});
    };
    auto adv(auto it, int z) { // z = {-1, +1}
        return (z == -1 ? --(it == begin() ? end() : it)
            : (++it == end() ? begin() : it));
    }
    bool chk(auto it) {
        Line l = *it, pl = *adv(it, -1), nl = *adv(it, +1);
```

```
    auto [x, y, d] = LineIntersection(pl, nl);
    T4 sat = 1.a * x + 1.b * y - (T4)l.c * d;
    if (d < 0 && sat < 0) return clear(), 0; // unsat
    if ((d > 0 && sat <= 0) || (d == 0 && sat < 0)) return
        ↪erase(it), 1;
    return 0;
}
void Cut(Line l) { // add ax + by <= c
    if (empty()) return;
    auto it = insert(l);
    if (chk(it)) return;
    for (int z : {-1, +1})
        while (size() && chk(adv(it, z)))
            ;
}
double Maximize(T a, T b) { // max ax + by (UNTESTED)
    if (empty()) return -1 / 0.;
    auto it = lower_bound({a, b});
    if (it == end()) it = begin();
    auto [x, y, d] = LineIntersection(*adv(it, -1), *it);
    return (1.0 * a * x + 1.0 * b * y) / d;
}
double Area() {
    double total = 0.;
    for (auto it = begin(); it != end(); ++it) {
        auto [x1, y1, d1] = LineIntersection(*adv(it, -1), *it);
        auto [x2, y2, d2] = LineIntersection(*it, *adv(it, +1));
        total += (1.0 * x1 * y2 - 1.0 * x2 * y1) / d1 / d2;
    }
    return total * 0.5;
}
};
```

PolygonUnion.kt

Description: Compute union or intersection of two polygons and compute the area of the resulting figure.

Time: Runs quite quickly for two convex polygons with 10⁵ vertices each. 873515, 40 lines

```
import java.awt.geom.* // at beginning of file
fun loadPoly(): DoubleArray {
    val n = rInt() // read n points
    return rDbs(2*n).toDoubleArray()
}
fun makeArea(ps: DoubleArray): Area {
    val p = Path2D.Double()
    p.moveTo(ps[0],ps[1])
    for (i in 0..ps.size/2-1) p.lineTo(ps[2*i],ps[2*i+1])
    p.closePath()
    return Area(p)
}
fun computeArea(a: Area): Double {
    val iter = a.getPathIterator(null)
    val buf = DoubleArray(6,{0.0})
    var ret = 0.0
    val ps = ArrayList<Double>()
    while (!iter.isDone()) {
        when(iter.currentSegment(buf)) {
            PathIterator.SEG_MOVETO,PathIterator.SEG_LINETO->{
                ps.add(buf[0])
                ps.add(buf[1])
            }
            PathIterator.SEG_CLOSE -> {
                ps.add(ps[0])
                ps.add(ps[1])
                for (i in 0..ps.size/2-2) ret -= ps[2*i]*ps[2*i+1]
                    ↪+3]-ps[2*i+1]*ps[2*i+2]
                ps.clear()
            }
        }
    }
}
```

```
        iter.next()
    }
    assert(ret >= 0)
    return ret/2
}
fun intersectArea(a: DoubleArray, b: DoubleArray): Double {
    val ret = makeArea(a)
    ret.intersect(makeArea(b)) // or .add for union
    return computeArea(ret)
}
```

8.3 Circles

Circle.h

Description: represent circle as {center,radius}

"Point.h" 91f3fc, 6 lines

```
using Circ = pair<P,T>;
int in(const Circ& x, const P& y) { // -1 if inside, 0, 1
    return sgn(abs(y-x.f)-x.s); }
T arcLength(const Circ& x, P a, P b) {
    // precondition: a and b on x
    P d = (a-x.f)/(b-x.f); return x.s*acos(d.f); }
```

CircleIsect.h

Description: Circle intersection points and intersection area. Tangents will be returned twice.

"Circle.h" 21a173, 22 lines

```
vP isect(const Circ& x, const Circ& y) { // precondition: x!=y
    T d = abs(x.f-y.f), a = x.s, b = y.s;
    if (sgn(d) == 0) { assert(a != b); return {}; }
    T C = (a*a+d*d-b*b)/(2*a*d);
    if (abs(C) > 1+EPS) return {};
    T S = sqrt(max(1-C*C,(T)0)); P tmp = (y.f-x.f)/d*x.s;
    return {x.f+tmp*P(C,S),x.f+tmp*P(C,-S)};
}
vP isect(const Circ& x, const Line& y) {
    P c = foot(x.f,y); T sq_dist = sq(x.s)-abs2(x.f-c);
    if (sgn(sq_dist) < 0) return {};
    P offset = unit(y.s-y.f)*sqrt(max(sq_dist,T(0)));
    return {c+offset,c-offset};
}
T isect_area(Circ x, Circ y) { // not thoroughly tested
    T d = abs(x.f-y.f), a = x.s, b = y.s; if (a < b) swap(a,b);
    if (d >= a+b) return 0;
    if (d <= a-b) return PI*b*b;
    T ca = (a+a*d*d-b*b)/(2*a*d), cb = (b*b+d*d-a*a)/(2*b*d);
    T s = (a+b*d)/2, h = 2*sqrt(s*(s-a)*(s-b)*(s-d))/d;
    return a*a*acos(ca)+b*b*acos(cb)-d*h;
}
```

CircleTangents.h

Description: internal and external tangents between two circles

"Circle.h" d9a76f, 22 lines

```
P tangent(P x, Circ y, int t = 0) {
    y.s = abs(y.s); // abs needed because internal calls y.s < 0
    if (y.s == 0) return y.f;
    T d = abs(x-y.f);
    P a = pow(y.s/d,2)*(x-y.f)+y.f;
    P b = sqrt(d*d-y.s*y.s)/d*y.s*unit(x-y.f)*dir(PI/2);
    return t == 0 ? a+b : a-b;
}
V<pair<P,P>> external(Circ x, Circ y) {
    V<pair<P,P>> v;
    if (x.s == y.s) {
        P tmp = unit(x.f-y.f)*x.s*dir(PI/2);
        v.eb(x.f+tmp,y.f+tmp);
        v.eb(x.f-tmp,y.f-tmp);
    } else {
```

```
        P p = (y.s*x.f-x.s*y.f)/(y.s-x.s);
        F0R(i,2) v.eb(tangent(p,x,i),tangent(p,y,i));
    }
    return v;
}
V<pair<P,P>> internal(Circ x, Circ y) {
    return external({x.f,-x.s},y); }
```

Circumcenter.h

Description: returns {circumcenter,circumradius}

"Circle.h" a2c6a6, 5 lines

```
Circ ccCenter(P a, P b, P c) {
    b -= a; c -= a;
    P res = b*c*(conj(c)-conj(b))/(b*conj(c)-conj(b)*c);
    return {a+res,abs(res)};
}
```

MinEnclosingCirc.h

Description: minimum enclosing circle

Time: expected $\mathcal{O}(N)$

"Circumcenter.h" 53963d, 13 lines

```
circ mec(vP ps) {
    shuffle(all(ps), rng);
    P o = ps[0]; T r = 0, EPS = 1+1e-8;
    F0R(i,sz(ps)) if (abs(o-ps[i]) > r*EPS) {
        o = ps[i], r = 0; // point is on MEC
        F0R(j,i) if (abs(o-ps[j]) > r*EPS) {
            o = (ps[i]+ps[j])/2, r = abs(o-ps[i]);
            F0R(k,j) if (abs(o-ps[k]) > r*EPS)
                tie(o,r) = ccCenter(ps[i],ps[j],ps[k]);
        }
    }
    return {o,r};
}
```

8.4 Misc

ClosestPair.h

Description: Line sweep to find two closest points .

Time: $\mathcal{O}(N \log N)$

"Point.h" 2b60fa, 17 lines

```
pair<P,P> solve(vP v) {
    pair<db,pair<P,P>> bes; bes.f = INF;
    set<P> S; int ind = 0;
    sort(all(v));
    F0R(i,sz(v)) {
        if (i && v[i] == v[i-1]) return {v[i],v[i]};
        for (; v[i].f-v[ind].f >= bes.f; ++ind)
            S.erase({v[ind].s,v[ind].f});
        for (auto it = S.sub({v[i].s-bes.f,INF});
            it != end(S) && it->f < v[i].s+bes.f; ++it) {
            P t = {it->s,it->f};
            ckmin(bes,{abs(t-v[i]),{t,v[i]}});
        }
        S.insert({v[i].s,v[i].f});
    }
    return bes.s;
}
```

DelaunayFast.h

Description: Fast Delaunay triangulation assuming no duplicates and not all points collinear (in latter case, result will be empty). Should work for doubles as well, though there may be precision issues in 'circ'. Returns triangles in ccw order. Each circumcircle will contain none of the input points. If coordinates are ints at most B then T should be large enough to support ints on the order of B^4 .

Time: $\mathcal{O}(N \log N)$

"Point.h" 7ff542, 82 lines

```
// using l1l1 = ll; (if coords are < 2e4)
using l1l1 = __int128;
// returns true if p strictly within circumcircle(a,b,c)
bool inCircle(P p, P a, P b, P c) {
    a -= p, b -= p, c -= p; // assert(cross(a,b,c)>0);
    l1l1 x = (l1l1)abs2(a)*cross(b,c)+(l1l1)abs2(b)*cross(c,a)
        +(l1l1)abs2(c)*cross(a,b);
    return x*(cross(a,b,c)>0?-1:-1) > 0;
}

P arb(LLONG_MAX,LLONG_MAX); // not equal to any other point
using Q = struct Quad*;
struct Quad {
    bool mark; Q o, rot; P p;
    P F() { return r()->p; }
    Q r() { return rot->rot; }
    Q prev() { return rot->o->rot; }
    Q next() { return r()->prev(); }
};
Q makeEdge(P orig, P dest) {
    Q q[] {new Quad{0,0,0,orig}, new Quad{0,0,0,arb},
        new Quad{0,0,0,dest}, new Quad{0,0,0,arb}};
    F0R(i,4) q[i]->o = q[-i & 3], q[i]->rot = q[(i+1) & 3];
    return *q;
}
void splice(Q a, Q b) { swap(a->o->rot->o, b->o->rot->o); swap(
    a->o, b->o); }
Q connect(Q a, Q b) {
    Q q = makeEdge(a->F(), b->p);
    splice(q, a->next()); splice(q->r(), b);
    return q;
}
pair<Q,Q> rec(const vP& s) {
    if (sz(s) <= 3) {
        Q a = makeEdge(s[0], s[1]), b = makeEdge(s[1], s.bk);
        if (sz(s) == 2) return { a, a->r() };
        splice(a->r(), b);
        auto side = cross(s[0], s[1], s[2]);
        Q c = side ? connect(b, a) : 0;
        return {side < 0 ? c->r() : a, side < 0 ? c : b->r() };
    }

#define H(e) e->F(), e->p
#define valid(e) (cross(e->F(),H(base)) > 0)
    Q A, B, ra, rb;
    int half = sz(s) / 2;
    tie(ra, A) = rec({all(s)-half});
    tie(B, rb) = rec({sz(s)-half+all(s)});
    while ((cross(B->p,H(A)) < 0 && (A = A->next())) ||
        (cross(A->p,H(B)) > 0 && (B = B->r()->o)));
    Q base = connect(B->r(), A);
    if (A->p == ra->p) ra = base->r();
    if (B->p == rb->p) rb = base;

#define DEL(e, init, dir) Q e = init->dir; if (valid(e)) \
    while (inCircle(e->dir->F(), H(base), e->F())) { \
        Q t = e->dir; \
        splice(e, e->prev()); \
        splice(e->r(), e->r()->prev()); \
        e = t; \
    }
    while (1) {
        DEL(LC, base->r(), o); DEL(RC, base, prev());
        if (!valid(LC) && !valid(RC)) break;
        if (!valid(LC) || (valid(RC) && inCircle(H(RC), H(LC))))
            base = connect(RC, base->r());
        else base = connect(base->r(), LC->r());
    }
}
```

```
return {ra, rb};
}
V<AR<P,3>> triangulate(vP pts) {
    sor(pts); assert(unique(all(pts)) == end(pts)); // no
        ↳duplicates
    if (sz(pts) < 2) return {};
    Q e = rec(pts).f; V<Q> q = {e};
    while (cross(e->o->F(), e->F(), e->p) < 0) e = e->o;
#define ADD { Q c = e; do { c->mark = 1; pts.pb(c->p); \
    q.pb(c->r()); c = c->next(); } while (c != e); }
    ADD; pts.clear();
    int qi = 0; while (qi < sz(q)) if (!(e = q[qi++])->mark) ADD;
    V<AR<P,3>> ret (sz(pts)/3);
    F0R(i,sz(pts)) ret[i/3][i%3] = pts[i];
    return ret;
}
```

ManhattanMST.h

Description: Given N points, returns up to $4N$ edges which are guaranteed to contain a MST for graph with edge weights $w(p,q) = |p.x-q.x|+|p.y-q.y|$. Edges are in the form {dist, {src, dst}}.

Time: $\mathcal{O}(N \log N)$

```
"DSU.h" b7a3bd, 24 lines
// use standard MST algorithm on result to find final MST
V<pair<int,pi>> manhattanMst(vpi v) {
    vi id(sz(v)); iota(all(id),0);
    V<pair<int,pi>> ed;
    F0R(k,4) {
        sort(all(id), [&](int i, int j) {
            return v[i].f+v[i].s < v[j].f+v[j].s; });
        map<int,int> sweep; // find first octant neighbors
        each(i,id) { // those in sweep haven't found neighbor yet
            for (auto it = sweep.lb(-v[i].s);
                it != end(sweep); sweep.erase(it++)) {
                int j = it->s;
                pi d{v[i].f-v[j].f,v[i].s-v[j].s}; if (d.s>d.f) break;
                ed.pb({d.f+d.s,{i,j}});
            }
            sweep[-v[i].s] = i;
        }
        each(p,v) {
            if (k%4) p.f *= -1;
            else swap(p.f,p.s);
        }
    }
    return ed;
}
```

8.5 3D

Point3D.h

Description: Basic 3D geometry.

```
"Point.h" d7d537, 82 lines
using P3 = AR<T,3>; using Tri = AR<P3,3>; using vP3 = V<P3>;
T abs2(const P3& x) {
    T sum = 0; F0R(i,3) sum += sq(x[i]);
    return sum; }
T abs(const P3& x) { return sqrt(abs2(x)); }

P3& operator+=(P3& l, const P3& r) { F0R(i,3) l[i] += r[i];
    return l; }
P3& operator-=(P3& l, const P3& r) { F0R(i,3) l[i] -= r[i];
    return l; }
P3& operator*=(P3& l, const T& r) { F0R(i,3) l[i] *= r;
    return l; }
P3& operator/=(P3& l, const T& r) { F0R(i,3) l[i] /= r;
    return l; }
P3 operator-(P3 l) { l *= -1; return l; }
P3 operator+(P3 l, const P3& r) { return l += r; }
```

```
P3 operator-(P3 l, const P3& r) { return l -= r; }
P3 operator*(P3 l, const T& r) { return l *= r; }
P3 operator*(const T& r, const P3& l) { return l*r; }
P3 operator/(P3 l, const T& r) { return l /= r; }
```

```
P3 unit(const P3& x) { return x/abs(x); }
T dot(const P3& a, const P3& b) {
    T sum = 0; F0R(i,3) sum += a[i]*b[i];
    return sum; }
P3 cross(const P3& a, const P3& b) {
    return {a[1]*b[2]-a[2]*b[1],a[2]*b[0]-a[0]*b[2],
        a[0]*b[1]-a[1]*b[0]}; }
P3 cross(const P3& a, const P3& b, const P3& c) {
    return cross(b-a,c-a); }
P3 perp(const P3& a, const P3& b, const P3& c) {
    return unit(cross(a,b,c)); }
```

```
bool isMult(const P3& a, const P3& b) { // for long longs
    P3 c = cross(a,b); F0R(i,sz(c)) if (c[i] != 0) return 0;
    return 1; }
bool collinear(const P3& a, const P3& b, const P3& c) {
    return isMult(b-a,c-a); }
```

```
T DC(const P3&a,const P3&b,const P3&c,const P3&p) {
    return dot(cross(a,b,c),p-a); }
bool coplanar(const P3&a,const P3&b,const P3&c,const P3&p) {
    return DC(a,b,c,p) == 0; }
bool op(const P3& a, const P3& b) {
    int ind = 0; // going in opposite directions?
    F0R(i,1,3) if (std::abs(a[i]*b[i])>std::abs(a[ind]*b[ind]))
        ind = i;
    return a[ind]*b[ind] < 0;
}
// coplanar points, b0 and b1 on opposite sides of a0-a1?
bool opSide(const P3&a,const P3&b,const P3&c,const P3&d) {
    return op(cross(a,b,c),cross(a,b,d)); }
// coplanar points, is a in Triangle b
bool inTri(const P3& a, const Tri& b) {
    F0R(i,3) if (opSide(b[i],b[(i+1)%3],b[(i+2)%3],a)) return 0;
    return 1; }
```

```
// point-seg dist
T psDist(const P3&p,const P3&a,const P3&b) {
    if (dot(a-p,a-b) <= 0) return abs(a-p);
    if (dot(b-p,b-a) <= 0) return abs(b-p);
    return abs(cross(p,a,b))/abs(a-b);
}
```

```
// projection onto line
P3 foot(const P3& p, const P3& a, const P3& b) {
    P3 d = unit(b-a); return a+dot(p-a,d)*d; }
// rotate p about axis
P3 rotAxis(const P3& p, const P3& a, const P3& b, T theta) {
    P3 dz = unit(b-a), f = foot(p,a,b);
    P3 dx = p-f, dy = cross(dz,dx);
    return f+cos(theta)*dx+sin(theta)*dy;
}
// projection onto plane
P3 foot(const P3& a, const Tri& b) {
    P3 c = perp(b[0],b[1],b[2]);
    return a-c*(dot(a,c)-dot(b[0],c)); }
// line-plane intersection
P3 lpIntersect(const P3&a0,const P3&a1,const Tri&b) {
    P3 c = unit(cross(b[2]-b[0],b[1]-b[0]));
    T x = dot(a0,c)-dot(b[0],c), y = dot(a1,c)-dot(b[0],c);
    return (y*a0-x*a1)/(y-x);
}
```

Hull3D.h

Description: Incremental 3D convex hull where not all points are coplanar. Normals to returned faces point outwards. If coordinates are ints at most B then T should be large enough to support ints on the order of B^3 . Changes order of points. The number of returned faces may depend on the random seed, because points that are on the boundary of the convex hull may or may not be included in the output.

Time: $\mathcal{O}(N^2)$, $\mathcal{O}(N \log N)$

```
5f4f0e, 91 lines
// using T = 1l;
bool above(const P3&a,const P3&b,const P3&c,const P3&p) {
    return DC(a,b,c,p) > 0; } // is p strictly above plane
void prep(vP3& p) { // rearrange points such that
    shuffle(all(p),rng); // first four are not coplanar
    int dim = 1;
    FOR(i,1,sz(p))
        if (dim == 1) {
            if (p[0] != p[i]) swap(p[1],p[i]), ++dim;
        } else if (dim == 2) {
            if (!collinear(p[0],p[1],p[i]))
                swap(p[2],p[i]), ++dim;
        } else if (dim == 3) {
            if (!coplanar(p[0],p[1],p[2],p[i]))
                swap(p[3],p[i]), ++dim;
        }
    }
    assert(dim == 4);
}
```

```
using F = AR<int,3>; // face
V<F> hull3d(vP3& p) {
    // s.t. first four points form tetra
    prep(p); int N = sz(p); V<F> hull; // triangle for each face
    auto ad = [&](int a, int b, int c) { hull.pb({a,b,c}); };
    // +new face to hull
    ad(0,1,2), ad(0,2,1); // initialize hull as first 3 points
    V<vb> in(N,vb(N)); // is zero before each iteration
    FOR(i,3,N) { // incremental construction
        V<F> def, HULL; swap(hull,HULL);
        // HULL now contains old hull
        auto ins = [&](int a, int b, int c) {
            if (in[b][a]) in[b][a] = 0; // kill reverse face
            else in[a][b] = 1, ad(a,b,c);
        };
        each(f,HULL) {
            if (above(p[f[0]],p[f[1]],p[f[2]],p[i]))
                F0R(j,3) ins(f[j],f[(j+1)%3],i);
            // recalc all faces s.t. point is above face
            else def.pb(f);
        }
        each(t,hull) if (in[t[0]][t[1]]) // edge exposed,
            in[t[0]][t[1]] = 0, def.pb(t); // add a new face
        swap(hull,def);
    }
    return hull;
}
V<F> hull3dFast(vP3& p) {
    prep(p); int N = sz(p); V<F> hull;
    vb active; // whether face is active
    V<vi> rvis; // points visible from each face
    V<AR<pi,3>> other; // other face adjacent to each edge of
        ↪face
    V<vi> vis(N); // faces visible from each point
    auto ad = [&](int a, int b, int c) {
        hull.pb({a,b,c}); active.pb(1); rvis.eb(); other.eb(); };
    auto ae = [&](int a, int b) { vis[b].pb(a), rvis[a].pb(b); };
    auto abv = [&](int a, int b) {
        F f=hull[a]; return above(p[f[0]],p[f[1]],p[f[2]],p[b]); };
    auto edge = [&](pi e) -> pi {
        return {hull[e.f][e.s],hull[e.f][(e.s+1)%3]}; };
}
```

Hull3D PolySaVol KMP Z Manacher LyndonFactor

```
auto glue = [&](pi a, pi b) { // link two faces by an edge
    pi x = edge(a); assert(edge(b) == mp(x.s,x.f));
    other[a.f][a.s] = b, other[b.f][b.s] = a;
}; // ensure face 0 is removed when i=3
ad(0,1,2), ad(0,2,1); if (abv(1,3)) swap(p[1],p[2]);
F0R(i,3) glue({0,i},{1,2-i});
FOR(i,3,N) ae(abv(1,i),i); // coplanar points go in rvis[0]
vi label(N,-1);
FOR(i,3,N) { // incremental construction
    vi rem; each(t,vis[i]) if (active[t]) active[t]=0, rem.pb(t
        ↪);
    if (!sz(rem)) continue; // hull unchanged
    int st = -1;
    each(r,rem) F0R(j,3) {
        int o = other[r][j].f;
        if (active[o]) { // create new face!
            int a,b; tie(a,b) = edge({r,j}); ad(a,b,i); st = a;
            int cur = sz(rvis)-1; label[a] = cur;
            vi tmp; set_union(all(rvis[r]),all(rvis[o]),
                back_inserter(tmp));
            // merge sorted vectors ignoring duplicates
            each(x,tmp) if (abv(cur,x)) ae(cur,x);
            glue({cur,0},other[r][j]); // glue old w/ new face
        }
    }
    for (int x = st, y; ; x = y) { // glue new faces together
        int X = label[x]; glue({X,1},{label[y=hull[X][1]],2});
        if (y == st) break;
    }
}
V<F> ans; F0R(i,sz(hull)) if (active[i]) ans.pb(hull[i]);
return ans;
}
```

PolySaVol.h

Description: surface area and volume of polyhedron, normals to faces must point outwards

```
"Hull3D.h" 52fc2b, 8 lines
pair<T,T> SaVol(vP3 p, V<F> faces) {
    T s = 0, v = 0;
    each(i,faces) {
        P3 a = p[i[0]], b = p[i[1]], c = p[i[2]];
        s += abs(cross(a,b,c)); v += dot(cross(a,b),c);
    }
    return {s/2,v/6};
}
```

Strings (9)

9.1 Light

KMP.h

Description: $f[i]$ is length of the longest proper suffix of the i -th prefix of s that is a prefix of s

Time: $\mathcal{O}(N)$

```
4538e4, 13 lines
vi kmp(str s) {
    int N = sz(s); vi f(N+1); f[0] = -1;
    FOR(i,1,N+1) {
        for (f[i]=f[i-1];f[i]!==-1&&s[f[i]]!=s[i-1];)f[i]=f[f[i]];
        ++f[i]; }
    return f;
}
vi getOc(str a, str b) { // find occurrences of a in b
    vi f = kmp(a+"@"+b), ret;
    FOR(i,sz(a),sz(b)+1) if (f[i+sz(a)+1] == sz(a))
        ret.pb(i-sz(a));
    return ret;
}
```

}

Z.h

Description: $f[i]$ is the max len such that $s.substr(0,len) == s.substr(i,len)$

Time: $\mathcal{O}(N)$

```
566170, 15 lines
vi z(str s) {
    int N = sz(s), L = 1, R = 0; s += '#';
    vi ans(N); ans[0] = N;
    FOR(i,1,N) {
        if (i <= R) ans[i] = min(R-i+1,ans[i-L]);
        while (s[i+ans[i]] == s[ans[i]]) ++ans[i];
        if (i+ans[i]-1 > R) L = i, R = i+ans[i]-1;
    }
    return ans;
}
vi getPrefix(str a, str b) { // find prefixes of a in b
    vi t = z(a+b); t = vi(sz(a)+all(t));
    each(u,t) ckmin(u,sz(a));
    return t;
}
```

Manacher.h

Description: length of largest palindrome centered at each character of string and between every consecutive pair

Time: $\mathcal{O}(N)$

```
fcc3f7, 13 lines
vi manacher(str _S) {
    str S = "@"; each(c,_S) S += c, S += "#";
    S.bk = '&#39;';
    vi ans(sz(S)-1); int lo = 0, hi = 0;
    FOR(i,1,sz(S)-1) {
        if (i != 1) ans[i] = min(hi-i,ans[hi-i+lo]);
        while (S[i-ans[i]-1] == S[i+ans[i]+1]) ++ans[i];
        if (i+ans[i] > hi) lo = i-ans[i], hi = i+ans[i];
    }
    ans.erase(begin(ans));
    F0R(i,sz(ans)) if (i%2 == ans[i]%2) ++ans[i];
    return ans;
}
```

LyndonFactor.h

Description: A string is "simple" if it is strictly smaller than any of its own nontrivial suffixes. The Lyndon factorization of the string s is a factorization $s = w_1w_2 \dots w_k$ where all strings w_i are simple and $w_1 \geq w_2 \geq \dots \geq w_k$. Min rotation gets min index i such that cyclic shift of s starting at i is minimum.

Time: $\mathcal{O}(N)$

```
af38ba, 19 lines
vs duval(str s) {
    int N = sz(s); vs factors;
    for (int i = 0; i < N; ) {
        int j = i+1, k = i;
        for (; j < N && s[k] <= s[j]; ++j) {
            if (s[k] < s[j]) k = i;
            else ++k;
        }
        for (; i <= k; i += j-k) factors.pb(s.substr(i,j-k));
    }
    return factors;
}
int minRotation(str s) {
    int N = sz(s); s += s;
    vs d = duval(s); int ind = 0, ans = 0;
    while (ans+sz(d[ind]) < N) ans += sz(d[ind++]);
    while (ind && d[ind] == d[ind-1]) ans -= sz(d[ind--]);
    return ans;
}
```

HashRange.h

Description: Polynomial hash for substrings with two bases.

fc0b90, 24 lines

```
using H = AR<int,2>; // bases not too close to ends
H makeH(char c) { return {c,c}; }
uniform_int_distribution<int> BDIST(0.1*MOD,0.9*MOD);
const H base(BDIST(rng),BDIST(rng));
H operator+(H l, H r) {
    FOR(i,2) if ((l[i] += r[i]) >= MOD) l[i] -= MOD;
    return l; }
H operator-(H l, H r) {
    FOR(i,2) if ((l[i] -= r[i]) < 0) l[i] += MOD;
    return l; }
H operator*(H l, H r) {
    FOR(i,2) l[i] = (ll)l[i]*r[i]%MOD;
    return l; }

V<H> pows{{1,1}};
struct HashRange {
    str S; V<H> cum{};
    void add(char c) { S += c; cum.pb(base*cum.bk+makeH(c)); }
    void add(str s) { each(c,s) add(c); }
    void extend(int len) { while (sz(pows) <= len)
        pows.pb(base*pows.bk); }
    H hash(int l, int r) { int len = r+1-l; extend(len);
        return cum[r+1]-pows[len]*cum[l]; }
};
```

ReverseBW.h

Description: Used only once. Burrows-Wheeler Transform appends # to a string, sorts the rotations of the string in increasing order, and constructs a new string that contains the last character of each rotation. This function reverses the transform.

Time: $\mathcal{O}(N \log N)$

e400d8, 7 lines

```
str reverseBW(str t) {
    vi nex(sz(t)); iota(all(nex),0);
    stable_sort(all(nex), [&t](int a,int b){return t[a]<t[b];});
    str ret; for (int i = nex[0]; i; )
        ret += t[i = nex[i]];
    return ret;
}
```

AhoCorasickFixed.h

Description: Aho-Corasick for fixed alphabet. For each prefix, stores link to max length suffix which is also a prefix.

Time: $\mathcal{O}(N \sum)$

96dfcc, 27 lines

```
template<size_t ASZ> struct ACfixed {
    struct Node { AR<int, ASZ> to; int link; };
    V<Node> d{{{}};
    int add(str s) { // add word
        int v = 0;
        each(C,s) {
            int c = C-'a';
            if (!d[v].to[c]) d[v].to[c] = sz(d), d.eb();
            v = d[v].to[c];
        }
        return v;
    }
    void init() { // generate links
        d[0].link = -1;
        queue<int> q; q.push(0);
        while (sz(q)) {
            int v = q.ft; q.pop();
            FOR(c,ASZ) {
                int u = d[v].to[c]; if (!u) continue;
                d[u].link = d[v].link == -1 ? 0 : d[d[v].link].to[c];
                q.push(u);
            }
        }
    }
};
```

```
        if (v) FOR(c,ASZ) if (!d[v].to[c])
            d[v].to[c] = d[d[v].link].to[c];
    }
};
```

SuffixArray.h

Description: Sort suffixes. First element of sa is sz(S), isa is the inverse of sa, and lcp stores the longest common prefix between every two consecutive elements of sa.

Time: $\mathcal{O}(N \log N)$

"RMQ.h"

27a566, 30 lines

```
struct SuffixArray {
    str S; int N; vi sa, isa, lcp;
    void init(str _S) { N = sz(S = _S)+1; genSa(); genLcp(); }
    void genSa() { // sa has size sz(S)+1, starts with sz(S)
        sa = isa = vi(N); sa[0] = N-1; iota(1+all(sa),0);
        sort(1+all(sa), [&](int a, int b) { return S[a] < S[b]; });
        FOR(i,1,N) { int a = sa[i-1], b = sa[i];
            isa[b] = i > 1 && S[a] == S[b] ? isa[a] : i; }
        for (int len = 1; len < N; len *= 2) { // currently sorted
            // by first len chars
            vi s(sa), is(isa), pos(N); iota(all(pos),0);
            each(t,s) {int T=t-len;if (T>=0) sa[pos[isa[T]]++] = T;}
            FOR(i,1,N) { int a = sa[i-1], b = sa[i];
                isa[b] = is[a]==is[b]&&is[a+len]==is[b+len]?isa[a]:i; }
            }
        }
    }
    void genLcp() { // Kasai's Algo
        lcp = vi(N-1); int h = 0;
        FOR(b,N-1) { int a = sa[isa[b]-1];
            while (a+h < sz(S) && S[a+h] == S[b+h]) ++h;
            lcp[isa[b]-1] = h; if (h) h--; }
        R.init(lcp);
    }
    RMQ<int> R;
    int getLCP(int a, int b) { // lcp of suffixes starting at a,b
        if (a == b) return sz(S)-a;
        int l = isa[a], r = isa[b]; if (l > r) swap(l,r);
        return R.query(l,r-1);
    }
};
```

SuffixArrayLinear.h

Description: Linear-time suffix array.

Usage: sa.is(s, 26) // all entries must be in [0, 26)

Time: $\mathcal{O}(N)$, ~100ms for $N = 5 \cdot 10^5$

ed0bb4, 46 lines

```
vi sa_is(const vi& s, int upper) {
    int n = sz(s); if (!n) return {};
    vi sa(n); vb ls(n);
    R0F(i,n-1) ls[i] = s[i] == s[i+1] ? ls[i+1] : s[i] < s[i+1];
    vi sum_l(upper), sum_s(upper);
    FOR(i,n) (ls[i] ? sum_l[s[i]+1] : sum_s[s[i]]++)++;
    FOR(i,upper) {
        if (i) sum_l[i] += sum_s[i-1];
        sum_s[i] += sum_l[i];
    }
    auto induce = [&](const vi& lms) {
        fill(all(sa),-1);
        vi buf = sum_s;
        for (int d: lms) if (d != n) sa[buf[s[d]]++] = d;
        buf = sum_l; sa[buf[s[n-1]]++] = n-1;
        FOR(i,n) {
            int v = sa[i]-1;
            if (v >= 0 && !ls[v]) sa[buf[s[v]]++] = v;
        }
        buf = sum_l;
        R0F(i,n) {
```

```
            int v = sa[i]-1;
            if (v >= 0 && ls[v]) sa[--buf[s[v]+1]] = v;
        }
    };
    vi lms_map(n+1,-1), lms; int m = 0;
    FOR(i,1,n) if (!ls[i-1] && ls[i]) lms_map[i]=m++, lms.pb(i);
    induce(lms); // sorts LMS prefixes
    vi sorted_lms;each(v,sa)if (lms_map[v]!==-1)sorted_lms.pb(v);
    vi rec_s(m); int rec_upper = 0; // smaller subproblem
    FOR(i,1,m) { // compare two lms substrings in sorted order
        int l = sorted_lms[i-1], r = sorted_lms[i];
        int end_l = lms_map[l]+1 < m ? lms[lms_map[l]+1] : n;
        int end_r = lms_map[r]+1 < m ? lms[lms_map[r]+1] : n;
        bool same = 0; // whether lms substrings are same
        if (end_l-1 == end_r-r) {
            for (;l < end_l && s[l] == s[r]; ++l,++r);
            if (l != n && s[l] == s[r]) same = 1;
        }
        rec_s[lms_map[sorted_lms[i]]] = (rec_upper += !same);
    }
    vi rec_sa = sa_is(rec_s,rec_upper+1);
    FOR(i,m) sorted_lms[i] = lms[rec_sa[i]];
    induce(sorted_lms); // sorts LMS suffixes
    return sa;
}
```

TandemRepeats.h

Description: Find all (i,p) such that $s.substr(i,p) == s.substr(i+p,p)$. No two intervals with the same period intersect or touch.

Usage: solve("aaabababa") // {{0, 1, 1}, {2, 5, 2}}

Time: $\mathcal{O}(N \log N)$

"SuffixArray.h"

661326, 13 lines

```
V<AR<int,3>> solve(str s) {
    int N = sz(s); SuffixArray A,B;
    A.init(s); reverse(all(s)); B.init(s);
    V<AR<int,3>> runs;
    for (int p = 1; 2*p <= N; ++p) { // do in  $\mathcal{O}(N/p)$  for period p
        for (int i = 0, lst = -1; i+p <= N; i += p) {
            int l = i-B.getLCP(N-i-p,N-i), r = i-p+A.getLCP(i,i+p);
            if (l > r || l == lst) continue;
            runs.pb({lst = l,r,p}); // for each i in [l,r],
        } // s.substr(i,p) == s.substr(i+p,p)
    }
    return runs;
}
```

9.2 Heavy

PalTree.h

Description: Used infrequently. Palindromic tree computes number of occurrences of each palindrome within string. $ans[i][0]$ stores min even x such that the prefix $s[1..i]$ can be split into exactly x palindromes, $ans[i][1]$ does the same for odd x .

Time: $\mathcal{O}(N \sum)$ for addChar, $\mathcal{O}(N \log N)$ for updAns

8a7d31, 41 lines

```
struct PalTree {
    static const int ASZ = 26;
    struct node {
        AR<int,ASZ> to = AR<int,ASZ>();
        int len, link, oc = 0; // # occurrences of pal
        int slink = 0, diff = 0;
        AR<int,2> seriesAns;
        node(int _len, int _link) : len(_len), link(_link) {}
    };
    str s = "@"; V<AR<int,2>> ans = {{0,MOD}};
    V<node> d = {{0,1},{-1,0}}; // dummy pals of len 0,-1
    int last = 1;
    int getLink(int v) {
```

```

    while (s[sz(s)-d[v].len-2] != s.bk) v = d[v].link;
    return v;
}
void updAns() { // serial path has O(log n) vertices
    ans.pb({MOD,MOD});
    for (int v = last; d[v].len > 0; v = d[v].slink) {
        d[v].seriesAns=ans[sz(s)-1-d[d[v].slink].len-d[v].diff];
        if (d[v].diff == d[d[v].link].diff)
            FOR(i,2) ckmin(d[v].seriesAns[i],
                d[d[v].link].seriesAns[i]);
        // start of previous oc of link[v]=start of last oc of v
        FOR(i,2) ckmin(ans.bk[i],d[v].seriesAns[i^1]+1);
    }
}
void addChar(char C) {
    s += C; int c = C-'a'; last = getLink(last);
    if (!d[last].to[c]) {
        d.pb(d[last].len+2,d[getLink(d[last].link)].to[c]);
        d[last].to[c] = sz(d)-1;
        auto& z = d.bk; z.diff = z.len-d[z.link].len;
        z.slink = z.diff == d[z.link].diff
            ? d[z.link].slink : z.link;
    } // max suf with different dif
    last = d[last].to[c]; ++d[last].oc;
    updAns();
}
void numOc() { ROF(i,2,sz(d)) d[d[i].link].oc += d[i].oc; }
};

```

SuffixAutomaton.h

Description: Used infrequently. Constructs minimal deterministic finite automaton (DFA) that recognizes all suffixes of a string. len corresponds to the maximum length of a string in the equivalence class, pos corresponds to the first ending position of such a string, lnk corresponds to the longest suffix that is in a different class. Suffix links correspond to suffix tree of the reversed string!

Time: $\mathcal{O}(N \log \Sigma)$

a99c6d, 67 lines

```

struct SuffixAutomaton {
    int N = 1; vi lnk{-1}, len{0}, pos{-1}; // suffix link,
    // max length of state, last pos of first occurrence of state
    V<map<char,int>> nex{1}; V<bool> isClone{0};
    // transitions, cloned -> not terminal state
    V<vi> iLnk; // inverse links
    int add(int p, char c) { // ~p nonzero if p != -1
        auto getNex = [&]() {
            if (p == -1) return 0;
            int q = nex[p][c]; if (len[p]+1 == len[q]) return q;
            int clone = N++; lnk.pb(lnk[q]); lnk[q] = clone;
            len.pb(len[p]+1), nex.pb(nex[q]),
            pos.pb(pos[q]), isClone.pb(1);
            for (; ~p && nex[p][c] == q; p = lnk[p]) nex[p][c]=clone;
            return clone;
        };
        // if (nex[p].count(c)) return getNex();
        // ^ need if adding > 1 string
        int cur = N++; // make new state
        lnk.pb(), len.pb(len[p]+1), nex.pb(),
        pos.pb(pos[p]+1), isClone.pb(0);
        for (; ~p && !nex[p].count(c); p = lnk[p]) nex[p][c] = cur;
        int x = getNex(); lnk[cur] = x; return cur;
    }
}
void init(str s) { int p = 0; each(x,s) p = add(p,x); }
// inverse links
void genIlnk() { iLnk.rsz(N); FOR(v,1,N) iLnk[lnk[v]].pb(v); }
// APPLICATIONS
void getAllOccur(vi& oc, int v) {
    if (!isClone[v]) oc.pb(pos[v]); // terminal position
    each(u,iLnk[v]) getAllOccur(oc,u); }

```

```

vi allOccur(str s) { // get all occurrences of s in automaton
    int cur = 0;
    each(x,s) {
        if (!nex[cur].count(x)) return {};
        cur = nex[cur][x]; }
    // convert end pos -> start pos
    vi oc; getAllOccur(oc,cur); each(t,oc) t += 1-sz(s);
    sort(all(oc)); return oc;
}
vl distinct;
ll getDistinct(int x) {
    // # distinct strings starting at state x
    if (distinct[x]) return distinct[x];
    distinct[x]=1;each(y,nex[x]) distinct[x]+=getDistinct(y.s);
    return distinct[x]; }
ll numDistinct() { // # distinct substrings including empty
    distinct.rsz(N); return getDistinct(0); }
ll numDistinct2() { // assert(numDistinct()==numDistinct2());
    ll ans = 1; FOR(i,1,N) ans += len[i]-len[lnk[i]];
    return ans; }
};

```

```

SuffixAutomaton S;
vi sa; str s;
void dfs(int x) {
    if (!S.isClone[x]) sa.pb(sz(s)-1-S.pos[x]);
    V<pair<char,int>> chr;
    each(t,S.iLnk[x]) chr.pb({s[S.pos[t]-S.len[x]],t});
    sort(all(chr)); each(t,chr) dfs(t.s);
}

```

```

int main() {
    re(s); reverse(all(s));
    S.init(s); S.genIlnk();
    dfs(0); ps(sa); // generating suffix array for s
}

```

SuffixTree.h

Description: Used infrequently. Ukkonen's algorithm for suffix tree. Longest non-unique suffix of s has length len[p]+lef after each call to add terminates. Each iteration of loop within add decreases this quantity by one.

Time: $\mathcal{O}(N \log \Sigma)$

39751c, 51 lines

```

struct SuffixTree {
    str s; int N = 0;
    vi pos, len, lnk; V<map<char,int>> to;
    int make(int POS, int LEN) { // lnk[x] is meaningful when
        // x!=0 and len[x] != MOD
        pos.pb(POS);len.pb(LEN);lnk.pb(-1);to.pb();return N++; }
    void add(int& p, int& lef, char c) { // longest
        // non-unique suffix is at node p with lef extra chars
        s += c; ++lef; int lst = 0;
        for (;lef;p=lnk[p]:lef--) { // if p != root then lnk[p]
            // must be defined
            while (lef>1 && lef>len[to[p][s[sz(s)-lef]]])
                p = to[p][s[sz(s)-lef]], lef -= len[p];
            // traverse edges of suffix tree while you can
            char e = s[sz(s)-lef]; int& q = to[p][e];
            // next edge of suffix tree
            if (!q) q = make(sz(s)-lef,MOD); lnk[lst] = p, lst = 0;
            // make new edge
            else {
                char t = s[pos[q]+lef-1];
                if (t == c) { lnk[lst] = p; return; } // suffix not
                ↪ unique
                int u = make(pos[q],lef-1);
                // new node for current suffix-1, define its link
                to[u][c] = make(sz(s)-1,MOD); to[u][t] = q;
                // new, old nodes
            }
        }
    }
}

```

```

        pos[q] += lef-1; if (len[q] != MOD) len[q] -= lef-1;
        q = u, lnk[lst] = u, lst = u;
    }
}
void init(str _s) {
    make(-1,0); int p = 0, lef = 0;
    each(c,_s) add(p,lef,c);
    add(p,lef,'$'); s.pop_back(); // terminal char
}
int maxPre(str x) { // max prefix of x which is substring
    for (int p = 0, ind = 0;;) {
        if (ind == sz(x) || !to[p].count(x[ind])) return ind;
        p = to[p][x[ind]];
        FOR(i,len[p]) {
            if (ind == sz(x) || x[ind] != s[pos[p]+i]) return ind;
            ind ++;
        }
    }
}
vi sa; // generate suffix array
void genSa(int x = 0, int Len = 0) {
    if (!sz(to[x])) sa.pb(pos[x]-Len); // found terminal node
    else each(t,to[x]) genSa(t.s,Len+len[x]);
}
};

```

Various (10)

10.1 Dynamic programming

When doing DP on intervals:

$a[i][j] = \min_{i < k < j} (a[i][k] + a[k][j]) + f(i, j)$, where the (minimal) optimal k increases with both i and j ,

- one can solve intervals in increasing order of length, and search $k = p[i][j]$ for $a[i][j]$ only between $p[i][j-1]$ and $p[i+1][j]$.

- This is known as Knuth DP. Sufficient criteria for this are if $f(b, c) \leq f(a, d)$ and $f(a, c) + f(b, d) \leq f(a, d) + f(b, c)$ for all $a \leq b \leq c \leq d$.

- Consider also: LineContainer (ch. Data structures), monotone queues, ternary search.

CircularLCS.h

Description: Used only twice. For strs A, B calculates longest common subsequence of A with all rotations of B

Time: $\mathcal{O}(|A| \cdot |B|)$

db21cf, 26 lines

```

int circular_lcs(str A, str B) {
    B += B;
    int max_lcs = 0;
    V<vb> dif_left(sz(A)+1,vb(sz(B)+1)), dif_up(sz(A)+1,vb(sz(B)
        ↪ +1));
    auto recalc = [&](int x, int y) { assert(x && y);
        int res = (A.at(x-1) == B.at(y-1)) |
            dif_up[x][y-1] | dif_left[x-1][y];
        dif_left[x][y] = res-dif_up[x][y-1];
        dif_up[x][y] = res-dif_left[x-1][y];
    };
    FOR(i,1,sz(A)+1) FOR(j,1,sz(B)+1) recalc(i,j);
    FOR(j,sz(B)/2) {
        // 1. zero out dp[.][j], update dif_left and dif_right
    }
}

```

```
if (j) for (int x = 1, y = j; x <= sz(A) && y <= sz(B); ) {
    int pre_up = dif_up[x][y];
    if (y == j) dif_up[x][y] = 0;
    else recal(x,y);
    (pre_up == dif_up[x][y]) ? ++x : ++y;
}
// 2. calculate LCS(A[0:sz(A)),B[j:sz(B)/2])
int cur_lcs = 0;
FOR(x,1,sz(A)+1) cur_lcs += dif_up[x][j+sz(B)/2];
ckmax(max_lcs,cur_lcs);
}
return max_lcs;
}
```

SMAWK.h
Description: Given negation of totally monotone matrix with entries of type D, find indices of row maxima (their indices increase for every submatrix). If tie, take lesser index. f returns matrix entry at (r,c) in $O(1)$. Use in place of divide & conquer to remove a log factor.
Time: $O(R+C)$, can be reduced to $O(C(1+\log R/C))$ evaluations of f

```
template<class F, class D=ll> vi smawk (F f, vi x, vi y) {
    vi ans(sz(x),-1); // x = rows, y = cols
    #define upd() if (ans[i] == -1 || w > mx) ans[i] = c, mx = w
    if (min(sz(x),sz(y)) <= 8) {
        FOR(i,sz(x)) { int r = x[i]; D mx;
            each(c,y) { D w = f(r,c); upd(); } }
        return ans;
    }
    if (sz(x) < sz(y)) { // reduce subset of cols to consider
        vi Y; each(c,y) {
            for (;sz(Y);Y.pop_back()) { int X = x[sz(Y)-1];
                if (f(X,Y.bk) >= f(X,c)) break; }
            if (sz(Y) < sz(x)) Y.pb(c);
        } y = Y;
    } // recurse on half the rows
    vi X; for (int i = 1; i < sz(x); i += 2) X.pb(x[i]);
    vi ANS = smawk(f,X,y); FOR(i,sz(ANS)) ans[2*i+1] = ANS[i];
    for (int i = 0, k = 0; i < sz(x); i += 2){
        int to = i+1 < sz(ans) ? ans[i+1] : y.bk; D mx;
        for(int r = x[i]; ++k) {
            int c = y[k]; D w = f(r,c); upd();
            if (c == to) break; }
    }
    return ans;
};
```

10.2 Debugging tricks

- signal(SIGSEGV, [](int) { _Exit(0); }); converts segfaults into Wrong Answers. Similarly one can catch SIGABRT (assertion failures) and SIGFPE (zero divisions). _GLIBCXX_DEBUG violations generate SIGABRT (or SIGSEGV on gcc 5.4.0 apparently).
- feenableexcept (29); kills the program on NaNs (1), 0-divs (4), infinities (8) and denormals (16).

10.3 Optimization tricks

10.3.1 Bit hacks

- x & -x is the least bit in x.
- for (int x = m; x;) { --x &= m; ... } loops over all subset masks of m (except m itself).

- c = x&-x, r = x+c; (((r^x) >> 2)/c) | r is the next number after x with the same number of bits set.
- FOR(b,k) FOR(i,1<<K) if (i&1<<b) D[i] += D[i^1<<b]; computes all sums of subsets.
- #pragma GCC optimize ("Ofast") will make GCC auto-vectorize for loops and optimizes floating points better (assumes associativity and turns off denormals).
- #pragma GCC target ("avx,avx2") can double performance of vectorized code, but causes crashes on old machines. Also consider older #pragma GCC target ("sse4").
- #pragma GCC optimize ("trapv") kills the program on integer overflows (but is really slow).

FastIO.h
Description: Fast input and output for integers and strings. For doubles, read them as strings and convert them to double using stod.
Usage: initO(); int a,b; ri(a,b); wi(b,'n'); wi(a,'n');
Time: input is ~300ms faster for 10⁶ long longs on CF.

```
inline namespace FastIO {
const int BSZ = 1<<15; INPUT
char ibuf[BSZ]; int ipos, ilen;
char nc() { // next char
    if (ipos == ilen) {
        ipos = 0; ilen = fread(ibuf,1,BSZ,stdin);
        if (!ilen) return EOF;
    }
    return ibuf[ipos++];
}
void rs(str& x) { // read str
    char ch; while (isspace(ch = nc()));
    do { x += ch; } while (!isspace(ch = nc()) && ch != EOF);
}
tcT> void ri(T& x) { // read int or ll
    char ch; int sgn = 1;
    while (!isdigit(ch = nc())) if (ch == '-') sgn *= -1;
    x = ch-'0'; while (isdigit(ch = nc())) x = x*10+(ch-'0');
    x *= sgn;
}
tcT, class... Ts> void ri(T& t, Ts&... ts) {
    ri(t); ri(ts...); } // read ints
//////// OUTPUT (call initO() at start)
char obuf[BSZ], numBuf[100]; int opos;
void flushOut() { fwrite(obuf,1,opos,stdout); opos = 0; }
void wc(char c) { // write char
    if (opos == BSZ) flushOut();
    obuf[opos++] = c; }
void ws(str s) { each(c,s) wc(c); } // write str
tcT> void wi(T x, char after = '\0') {
    if (x < 0) wc('-'), x *= -1;
    int len = 0; for (;x>=10;x/=10) numBuf[len++] = '0'+(x%10);
    wc('0'+x); R0F(i,len) wc(numBuf[i]);
    if (after) wc(after);
}
void initO() { assert(atexit(flushOut) == 0); }
```

10.4 Other languages

Python3.py

Description: Python review.

40 lines

```
from math import *
import sys, random
def nextInt():
    return int(input())
def nextStrs():
    return input().split()
def nextInts():
    return list(map(int,nextStrs()))

n = nextInt()
v = [n]
def process(x):
    global v
    x = abs(x)
    V = []
    for t in v:
        g = gcd(t,x)
        if g != 1:
            V.append(g)
        if g != t:
            V.append(t//g)
    v = V
for i in range(50):
    x = random.randint(0,n-1)
    if gcd(x,n) != 1:
        process(x)
    else:
        sx = x*x%n # assert(gcd(sx,n) == 1)
        print(f"sqrt {sx}")
        sys.stdout.flush()
        X = nextInt()
        process(x+X)
        process(x-X)
print(f'! {len(v)}',end='')
for i in v:
    print(f' {i}',end='')
print()
sys.stdout.flush() # sys.exit(0) -> exit
# sys.setrecursionlimit(int(1e9)) -> stack size
# print(f'{ans:6f}') -> print ans to 6 decimal places
```

Decimal.py

Description: Arbitrary-precision decimals

5 lines

```
from decimal import *

getcontext().prec = 100 # how many digits of precision
print(Decimal(1) / Decimal(7)) # 0.142857142857...
print(Decimal(10) ** -100) # 1E-100
```